SHADE: Trace-Based Humanization for Robust and Composable AI-Text Detector Evasion

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Abstract

Classifiers detect text generated by large language models using statistical differences from human writing. Attacks often optimize a detector's score without identifying which differences they remove or whether their effects survive retraining.

We present **SHADE** (**S**tatistical **H**uman-**A**lignment for **D**etector **E**vasion), a trace-based humanization method: it treats measurable statistical properties of text as traces to target explicitly. SHADE combines controlled generation with local editing to bring selected feature values into ranges typically observed in human texts of the same genre, while preserving previously attained targets. These features capture surface properties and how predictive information changes with the amount of preceding context.

On 74 held-out source documents, SHADE reduces detection performance substantially across five detectors, four of them unused during development, in several cases to near chance. After retraining on attacked examples, three detectors recover little or nothing while another recovers almost fully, revealing a remaining coverage gap. On a separate cohort, adding a complementary editing attack improves evasion over either component alone across all five detectors.

These results support organizing humanization around measurable targets, using adaptation tests to diagnose residual weaknesses and complementary modules to extend coverage.

1 Introduction

A text detector is a classifier that estimates whether writing is human- or machine-generated. Such detectors are vulnerable to paraphrasing, rewriting and style manipulation (Sadasivan et al., 2025; Krishna et al., 2023; Wang et al., 2024b; Macko et al., 2024). A frozen detector is held unchanged during attack evaluation; evading it need not remove the evidence that made detection possible. Discriminative features can invert under distribution shift (Gao et al., 2026); attack-aware adaptation or augmentation can restore discrimination (Koike et al., 2024; Masrour et al., 2025).

A humanizer generates or edits text to evade detection; unlike a paraphraser, which reformulates content, it explicitly targets detectability. We ask whether humanization can target measurable machine–human differences and stop within ranges typical of human writing. A *feature* is a numerical property of text; we call it a *trace* when examining it as potential evidence of machine generation. **Trace-based humanization** targets these properties explicitly rather than only a detector score. SHADE combines this objective with staged construction and preservation: address remaining differences associated with AI text, then preserve previously attained targets. Its features describe vocabulary, sentences, punctuation, parts of speech, named entities and predictability with varying preceding context.

The evaluation follows three questions. Does SHADE transfer beyond its development detectors? Does frozen evasion survive limited labeled exposure? Can a residual attack complement the pipeline where its trace coverage is incomplete? The disjoint 74-source result supports transfer; retraining each detector distinguishes reductions that persist from those it can reverse. RoBERTa (Liu et al., 2019), a pretrained language encoder used here as an AI-text classifier (Dugan et al., 2024), supplies a natural negative-control contrast: SHADE lowers its frozen AUROC (area under the receiver operating characteristic curve), but exposure restores strong discrimination. This remain-

ing weakness motivates **SHADE+RAFT**, the composition of SHADE with RAFT (Wang et al., 2024a), a detector-guided local substitution attack using RoBERTa feedback here, tested against both constituents and a matched composition with StealthRL (Ranganath and Ramesh, 2026), a reinforcement-learning humanization attack.

Three findings organize the argument. First, frozen SHADE transfers to representations that did not guide its construction and reproduces the same detector-specific pattern on new source identities. Second, limited exposure separates detectors whose discrimination remains attenuated from those that relearn, despite substantial frozen gains in both groups. Third, a local residual module can outperform each method alone while preserving much of the parent's measured surface neutralization. These are complementary observations: transfer alone does not establish resistance, and composition alone does not establish feature preservation.

Our contribution is the human-reference trace-neutralization objective together with attainment, recovery and composition audits. We do not claim the first feature-directed, modular or adaptation-tested attack. The construction record and earlier dependency-trace experiments remain in the appendices; the main paper focuses on what this objective makes measurable and actionable.

2 Related Work

Detection and adversarial robustness. Probability- and rank-based approaches include GLTR (Gehrmann et al., 2019), a token-rank visualization framework used here to build a classifier; DetectGPT (Mitchell et al., 2023) and Fast-DetectGPT (Bao et al., 2024), likelihood-based detectors; Binoculars (Hans et al., 2024), a two-model likelihood-ratio detector; and WaveDetect (Liu et al., 2026c), a detector of patterns in token probabilities. Ghostbuster (Verma et al., 2024) classifies language-model probability features. The feature-based detector of Fröhling and Zubiaga (2021), abbreviated FZ here, exposes surface statistics. We use three feature-count variants during development: FZ118, FZ186 and FZ301, named for their 118-, 186- and 301-feature extractors; individual classifier configurations may exclude features. Related systems add syntactic and discourse signals (Darwinkel et al., 2024). RoBERTa-based detectors provide a less explicitly factorized alternative. Attack-aware adaptation or augmentation can restore discrimination (Koike et al., 2024; Masrour et al., 2025; Zhou et al., 2024), motivating separate frozen and post-exposure evaluation.

Humanization objectives and local composition. Anonymouth (McDonald et al., 2012) and StyleRemix (Fisher et al., 2024) are authorship-obfuscation systems controlling style. Humanization attacks include TempParaphraser (Huang et al., 2025), which exploits sampling weaknesses; HIP (Xu et al., 2026), a model-based humanizer; MASH (Gu et al., 2026), a multistage attack; and UTS (Galat and Rizoiu, 2026), a structural rewriting attack. StealthRL optimizes detector rewards; related approaches use detector-guided off-the-shelf LLM paraphrasing (Cheng et al., 2025) or RL/preference optimization (Nicks et al., 2024). SHADE targets ranges typical of human writing; RAFT supplies local residual edits. We require composition to outperform both constituents.

3 Trace-Based Humanization

3.1 Human-reference trace neutralization

For each selected feature, we estimate a range typical of human texts in the same genre, called a *human reference band*. Values outside it incur distance to the nearest boundary; values inside incur none. *AI-ward* denotes the direction of the original AI–human difference for that feature; *human-ward* denotes movement toward the human reference. For text x, feature index j and genre g, $\phi_j(x)$ is a real-valued measurement and $L_{j,g}, U_{j,g}$ are real-valued band bounds. The distance is

$$d_{j,g}(x) = \begin{cases} L_{j,g} - \phi_j(x), & \phi_j(x) < L_{j,g}, \\ 0, & L_{j,g} \le \phi_j(x) \le U_{j,g}, \\ \phi_j(x) - U_{j,g}, & \phi_j(x) > U_{j,g}. \end{cases} \tag{1}$$

Distance grows on both sides of the band, so moving past the human range is penalized rather than rewarded, unlike minimizing a detector score. With robust human scale $s_{j,g} > 0$ making distances comparable across features,

selected-feature set $\mathcal{J}$, and relevance weight $w_j \geq 0$ reflecting feature j's contribution to detector scores, the normalized trace loss is

$$D_g(x) = \sum_{j \in \mathcal{J}} w_j \frac{d_{j,g}(x)}{s_{j,g}}. \tag{2}$$

In equation (2), $\mathcal{J}$ and w_j select residual AI-ward discrepancies and their relevance: uniform coverage of every extracted feature is not the objective. Preservation concerns features deliberately neutralized or attained by earlier stages, not every feature initially inside a human band. The equation expresses the shared design principle, not one jointly fitted loss used identically at every stage. Historical controls and retrospective weighted diagnostics are reported separately. In tree-based FZ stages, path contributions identify residual features associated with the classifier's AI score (Saabas, 2014); they do not define the human target or establish causal feature effects. Contextual controls similarly use human surprisal/context behavior rather than an unseen detector's score. A detector can therefore identify a useful target without becoming the scalar objective of every intervention. The operative sequence is *measure discrepancy, target the human region, preserve prior gains.*

3.2 Staged feature control

NS*i* denotes successive frozen configurations of SHADE; **NS10** is the endpoint evaluated throughout and referred to as SHADE in the results, with the designation retained for correspondence with the released reproduction package. The composed pipeline, SHADE+RAFT, has configuration designation **NS11**.

Figure 1 shows document flow, human targets and preservation checks within SHADE.

For tokenized text $x_{1:n}$, a fixed autoregressive scorer q measures *absolute surprisal* L_k: mean prediction loss with k preceding tokens. *Context utility* $U_x(k)$ is the loss reduction relative to eight preceding tokens:

$$L_k(x) = -\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \log q(x_t \mid x_{t-k:t-1}),$$
$$U_x(k) = L_8(x) - L_k(x).$$

The target positions $\mathcal{T}$ are fixed across context lengths and have sufficient preceding text.

Algorithm 1 SHADE trace-neutralization scaffold

Require: source text x, genre g, human references H_g, trace families $\Phi_1, \ldots, \Phi_R$, edit/generation operators $\mathcal{O}_1, \ldots, \mathcal{O}_R$

$y \leftarrow$ controlled generation or x
$\mathcal{L} \leftarrow \varnothing$ ▷ locked/accepted features
for $r = 1, \ldots, R$ **do**
  estimate human bands and residual AI-ward discrepancies for Φ_r
  select target features $\mathcal{J}_r$ and admissible operators
  for candidate $y' \in \mathcal{O}_r(y)$ **do**
    if $D_g(y') < D_g(y)$ on $\mathcal{J}_r$ **and** locked-feature/non-degradation guards pass **then**
      $y \leftarrow y'$
    end if
  end for
  add attained features to $\mathcal{L}$
end for
return y

Positive $U_x(k)$ means longer context improves prediction under q; it does not measure the source model's attention. We omit x when clear and write U32 and L32 for $U_x(32)$ and $L_{32}(x)$, analogously at other lengths.

SHADE combines its context-controlled generator (the U-series) targeting lexical diversity and surprisal/context behavior with sentence, punctuation, long-word, part-of-speech (POS), preposition/number, lexical-substitution and named-entity edits. Human bands supply the targets; likelihood/grammar guards and locked-feature checks constrain edits. Algorithm 1 summarizes the scaffold. Operators can affect several features and some targets remain unattained; the design makes these interactions auditable rather than assuming independence. The local stages use ordinary-language operations rather than encoding perturbations. Running human references guide lexical control during generation; later sentence, punctuation, POS and entity operations address remaining surface discrepancies. Stage-specific guards and partial-attainment outcomes matter because a single edit may improve its target while moving another feature. Full procedures and stagewise results are in Appendix F.

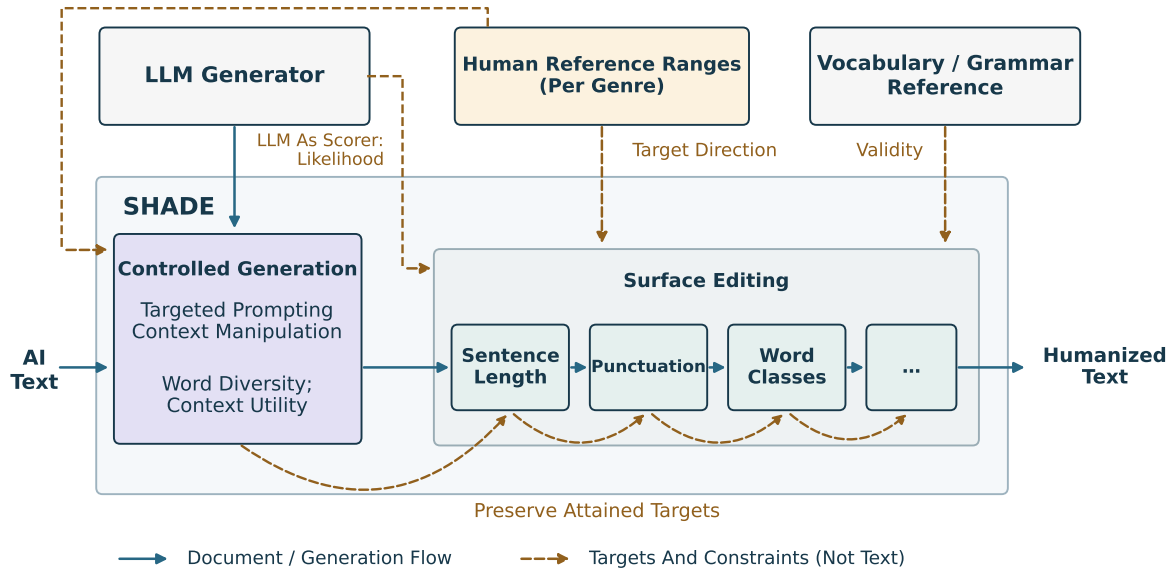


Figure 1: **SHADE applies feature controls in order.** The enclosing region is the method: purple marks generator-stage intervention, and green marks successive surface-editing stages (ellipsis: additional feature families). Solid arrows carry document/generation flow. Dashed arrows carry targets and constraints, not text. Human ranges guide generation and supply surface targets; the LLM supplies likelihood scores, while vocabulary/grammar references constrain valid edits. The shared band supplies this information to surface stages without separate source-to-stage arrows. Each stage's attained targets become preservation constraints for later stages. Checks are stage-specific and attainment is incomplete.

3.3 Checks with distinct evidential roles

Canonical Integrity is mechanically checkable under declared normalization: evasion must not rely on invisible characters or homoglyphs. **Linguistic Integrity / Non-degradation** is a modest source-relative admissibility criterion excluding deliberate misspellings, grammatical breakage, incoherent filler and major content deletion or distortion. Local guards and author inspection support it; no truthfulness, human-equivalent quality, formally verified semantic equivalence or blinded multi-rater study is claimed. **Adaptive Camouflage** is a quantitative experiment: expose a defender to labeled attack examples and measure recovery on held-out attacked texts under a specified budget.

3.4 Modular residual attacks

For frozen AUROC a_d in the original AI-positive orientation, define ordered-composition gain

$$G_d(A, B) = \min\{a_d(A), a_d(B)\} - a_d(B \circ A). \quad (3)$$

Positive gain means beating both standalone routes. It does not imply independent features, commutativity or causal superadditivity. We compare SHADE+RAFT with RAFT(OG), where OG denotes the original generated text, SHADE and StealthRL→RAFT under the same local procedure.

4 Experimental Setup

Cohort and detector roles. The primary evaluation uses 74 disjoint sources from RAID (Dugan et al., 2024), a benchmark corpus for AI-text detection, generated after freezing NS10, the configuration of SHADE applied unchanged to this held-out complement; the original 54 sources were repeatedly inspected during development and support the competitor/composition study. Both cohorts come from the same RAID test population; pooled 128-source results are secondary (Appendix Q). FZ and GHOSTBUSTER guide SHADE development. The five evaluation detectors are Ghostbuster, Binoculars, RoBERTa, GLTR and WaveDetect. The latter four—using Binoculars with NF4 (4-bit NormalFloat) quantization (Dettmers et al., 2023), RAID-trained

RoBERTa, a four-feature GLTR classifier and RAID-trained WaveDetect—do not guide SHADE. The composed pipeline's RAFT stage uses RoBERTa, making its RoBERTa evaluation detector-aware; the other four detectors are evaluation-only for that stage.

Adaptation. One permutation (seed 20260923) defines nested 20/30/40-example SHADE exposure sets on the 74-source complement, leaving 54/44/34 held out. Exposed positives augment original RAID training data. Ghostbuster refits its calibrated classifier; Binoculars a two-component readout with frozen Falcon language models (Almazrouei et al., 2023); GLTR its logistic head; RoBERTa and WaveDetect continue full-model training. Original RAID validation selects neural checkpoints. Human references are rescored and genre-weighted to each held-out cohort. Appendix N also varies the 30/44 partition across five seeds with training seed fixed.

Uncertainty and scoring. AUROC retains the original AI-positive orientation; values below .5 can reflect ranking reversal, not distributional equality. Confidence intervals are pointwise paired case-bootstrap intervals over evaluation source documents and reflect resampling of that cohort only. They do not include generation randomness, human reference pool composition or detector training variation. Nested-budget estimates condition on one partition. Competitor provenance, normalization, low-FPR results and protocol details are retained in the appendices. The competitor pool contains limited local reproductions of HIP, MASH, the synthetic and streaming UTS variants (UTS-SYNTH and UTS-STREAM), and released 4B HIP and StealthRL checkpoints. "Limited" and "released" denote provenance rather than performance tiers. Composition uses the same local RAFT procedure across starting texts; matched source identities and human weighting allow paired comparisons, without implying equal realized edits or quality across outputs.

5 Results

5.1 Progressive neutralization constructs the evaluated configuration

Steps 1–8 reduce FZ186 AUROC from .876 to .651. Richer FZ301 features expose residual entity structure; Organization-name (ORG) and number-entity (CARDINAL) controls reduce post-Step-8 AUROC from .700 to .671. Combining the selected U-series generator with Steps 2–10 produces SHADE, reducing Ghostbuster AUROC from .917 to .459 on the 54 development sources. These stagewise findings document construction (Appendix F); the frozen replication is the primary endpoint.

5.2 Frozen cross-detector transfer

On the disjoint 74-source complement, SHADE lowers AUROC on all five detectors by .199–.421, with all paired reduction intervals excluding zero (table 1). SHADE AUROCs differ from the original 54-source results by at most .031, supporting within-RAID replication of the detector-specific pattern. This is source-disjoint replication, not an external-dataset claim.

The matched 54-source competitor matrix is in Appendix J (table 2). SHADE has lower AUROC than every evaluated HIP, MASH and UTS configuration across all five detectors. StealthRL is stronger on Binoculars and GLTR, but weaker on Ghostbuster, RoBERTa and WaveDetect. No standalone method dominates the panel. On these 54 sources, SHADE lowers WaveDetect from .797 to .549 and GLTR from .816 to .506 without either detector guiding construction. Binoculars falls from .905 to .722 and RoBERTa from .992 to .733, leaving meaningful residual discrimination. Thus cross-detector transfer identifies substantial shared vulnerability while preserving the differences that the exposure experiments test. These comparisons concern the evaluated limited reproductions and released checkpoints, not complete method families or equalized compute budgets.

5.3 Adaptive Camouflage distinguishes evasion from resistance

Exposure produces near-zero recovery on GLTR and Binoculars, negative recovery on WaveDetect, partial recovery on Ghostbuster, and

Detector	Original AI	SHADE	Δ AUROC	95% paired CI
Ghostbuster	.910	.489	-.421	[−.503, −.341]
Binoculars NF4	.906	.706	-.199	[−.289, −.114]
RoBERTa	.985	.732	-.253	[−.321, −.191]
GLTR	.835	.537	-.298	[−.376, −.222]
WaveDetect	.775	.550	-.225	[−.315, −.134]

Table 1: **Primary frozen evaluation on the disjoint 74-source complement.** These sources were generated after freezing SHADE and exclude the 54-source development cohort. All five paired SHADE-minus-OG intervals exclude zero; interval scope is defined in Section 4. This is a within-RAID source replication; secondary pooled estimates appear in table 40.

	20 exposed / 54 held out			30 / 44			40 / 34		
Detector	Frozen	Adapted	RF (%)	Frozen	Adapted	RF (%)	Frozen	Adapted	RF (%)
Ghostbuster	.452	.491	+8.6	.438	.520	+17.7	.457	.582	+28.9
Binoculars NF4	.694	.698	+2.2†	.672	.681	+4.2†	.661	.674	+6.4†
RoBERTa	.740	.907	+68.0	.742	.744	+1.0	.767	.940	+80.8
GLTR	.518	.519	+0.4	.520	.521	+0.3	.533	.535	+0.6
WaveDetect	.513	.414	-39.6	.508	.456	-21.9	.486	.399	-31.2

Table 3: **Nested exposure on the disjoint 74-source complement.** RF is adapted-minus-frozen AUROC divided by the matched OG–SHADE gap. † Binoculars instead uses the saved raw-ratio gap to normalize readout recovery. These are point estimates, not confidence intervals, from one nested partition (Section 4); definitions and the baseline distinction appear in Appendix N.

strong recovery on RoBERTa at twenty and forty examples (table 3). Recovery fraction (RF) expresses the adapted-minus-frozen increase as a proportion of the matched original-AI–SHADE gap. At twenty examples, Ghostbuster recovers 8.6%, GLTR 0.4%, and RoBERTa 68.0%, while WaveDetect moves farther in the reverse-ranking direction. Binoculars' raw-ratio gap (.203) is comparable to RoBERTa's (.246), yet its readout update is only .0044: the small update does not merely accompany a negligible attack gap. Exact definitions and the Binoculars reference-gap distinction are in Appendix N.

Adaptation resistance is measured per detector–attack pair. This resolution identifies RoBERTa as a remaining weakness and motivates composition; an aggregate resistance figure would conceal that weakness. As a mechanistic interpretation, Binoculars' perplexity-based readout is adjacent to targeted surprisal/context traces, whereas RoBERTa's residual learned structure has no explicit counterpart in the trace set. SHADE's frozen RoBERTa gain is therefore plausibly incidental to those controls. The measured contrast is direct; this account of its cause is interpretation. The twenty- and forty-example RoBERTa fractions recover 68.0% and 80.8% of the opened gap, whereas GLTR remains below 1% across all three budgets. Ghostbuster's increasing recovery, from 8.6% to 28.9%, supplies an intermediate case rather than a binary success/failure split. The normalization makes these outcomes comparable relative to the frozen gap rather than only in absolute AUROC units. Variation aligned with interpretable coverage supplies evidence that uniform resistance would not provide.

RoBERTa's thirty-example recovery is partition-sensitive: the C09a sweep finds the original partition has the smallest recovery among five tested partitions, with training seed fixed. WaveDetect's negative recovery reflects reversed ranking under the specified continuation protocol, not absence of all exploitable signal. Both diagnostics and their limits are detailed in Appendix N.

5.4 Residual failure becomes a modular target

The residual module must edit existing text as a suffix. We selected RAFT through preliminary standalone RoBERTa screening; only RAFT was composed with SHADE (Appendix H). This establishes an available complementary module, not an optimal one. SHADE+RAFT improves on both SHADE and RAFT alone

Detector	SHADE+RAFT G_d	SHADE+RAFT 95% paired CI	StealthRL → RAFT G_d	StealthRL → RAFT 95% paired CI
GHOSTBUSTER	+.108	[+.020, +.197]	-.239	[−.372, −.104]
Binoculars NF4	+.127	[+.009, +.205]	+.226	[+.149, +.313]
RoBERTa	+.332	[+.255, +.410]	-.108	[−.174, −.042]
GLTR	+.223	[+.161, +.289]	+.210	[+.146, +.255]
WAVEDETECT	+.295	[+.203, +.369]	-.106	[−.209, −.003]

Table 4: **Operational complementarity with uncertainty.** $G_d = \min\{A_d(A), A_d(\text{RAFT}(OG))\} - A_d(\text{RAFT}(A))$. The minimum is recomputed in each of 10,000 paired source-bootstrap replicates, with fixed detectors and human references. SHADE composition has positive gains and pointwise intervals excluding zero on all five detectors; StealthRL composition does so on two. Cohort-only interval scope is defined in Section 4; intervals are not adjusted for simultaneous inference across detectors.

across all five detectors, with all five pointwise paired gain intervals excluding zero (table 4). The largest gain is on RoBERTa: .733 for SHADE and .591 for RAFT become .259 for SHADE+RAFT. Adaptation nevertheless restores SHADE+RAFT RoBERTa AUROC to .803: composition strongly improves frozen evasion without eliminating recoverable structure. The same RAFT module after StealthRL is complementary on only GLTR and Binoculars, and is stronger than SHADE+RAFT on Binoculars. Standalone strength and composability therefore differ; neither pipeline is universally superior. In particular, reducing StealthRL's own score after adding RAFT is insufficient for complementarity if the result remains worse than RAFT applied directly to OG. The minimum over the two constituents in equation (3) enforces this distinction, and is recomputed within each paired bootstrap replicate. The five SHADE intervals support complementarity in this tested order; they are pointwise, conditional intervals rather than evidence of universal or order-independent composition.

5.5 Feature-level evidence for compatibility

SHADE reduces FZ301 mean human-adjusted AI-score excess from .195 to .037, and SHADE+RAFT further reduces it to .021; RAFT alone leaves .174. This provides positive evidence that composition preserves and extends the dominant targeted surface gains supplied by SHADE.

As a secondary all-feature diagnostic, unweighted central-80% band coverage changes from 84.67% to 81.01% under OG→SHADE, while retrospective relevance-weighted coverage changes from 77.33% to 78.31%. Global coverage is not SHADE's objective: it targets selected residual AI-ward features and weights their relevance (equation (2)), rather than matching all 301 features. Appendix P gives the weight construction and full per-arm results; this binary coverage diagnostic is distinct from the weighted distance objective.

On 39 jointly eligible sources, SHADE →SHADE+RAFT improves U32/U128/extra-context band attainment but reduces L32/L128 absolute-surprisal attainment from 87.2/74.4% to 51.3/48.7%: utility differences and loss levels expose distinct effects. Cross-adaptation further identifies shared attack-sensitive signals; the exposure-by-evaluation interaction is positive with conditional intervals excluding zero for Ghostbuster and GLTR. Detailed weighting, band-distance and interaction analyses remain in Appendices O–P. Together these findings support surface preservation, partial contextual overlap and measurable composition costs, without isolating a causal RAFT fingerprint.

6 Discussion

Trace coverage explains the question adaptation asks. Frozen evasion and relearning resistance are separate outcomes. RoBERTa offers a natural negative-control contrast: SHADE lowers its frozen discrimination without targeting its learned representation, then exposure restores strong discrimination. Several representations closer to the targeted traces recover much less; competitor attacks also exhibit frozen evasion followed by relearning. This pattern supports the trace-coverage interpretation rather than equating score suppression with neutralization. It does not causally isolate the objective or imply

information-theoretic impossibility of recovery. Protocol capacity matters, but cannot alone explain low recovery: WaveDetect has full-model adaptation and still recovers negatively, unlike RoBERTa.

Measurable traces make composition auditable. Human-band targets expose what later stages should preserve. RAFT supplies complementary gains after SHADE while retaining dominant surface reductions, but also introduces absolute-surprisal and observed coherence costs. The non-surface overlap is observable because U supplies a quantifiable trace where learned structure is often opaque. The present panel cannot separate residual recoverability into re-exposed contextual structure and a RAFT-specific signature; that attribution requires controlled interventions. Thus SHADE remains the primary endpoint and SHADE+RAFT a composition demonstration. Modularity follows from explicit targets and preservation checks, rather than constituting a separate priority claim. A future residual module can use the audit to preserve dominant surface gains while constraining both context-utility differences and absolute loss levels. This is more informative than requiring its final detector score merely to decrease: the audit identifies which earlier changes survive and where composition introduces interference. Non-degradation remains a separate source-relative admissibility question, supported here by guards and author inspection rather than a quantitative quality study.

7 Conclusion

SHADE targets selected machine–human discrepancies with human-reference controls and audits their preservation, transfer and recoverability. Frozen SHADE transfers across five detector families on a disjoint 74-source complement. Under the evaluated protocols it exhibits substantial adaptation resistance for GLTR, Binoculars and WaveDetect, partial Ghostbuster recovery, and strong residual RoBERTa recoverability. The mechanistic interpretation is scoped to coverage of targeted traces, not all learned features. RoBERTa's boundary motivates RAFT composition: SHADE+RAFT outperforms each method alone on all five detectors, while matched StealthRL composition is less broadly complementary. Trace-based humanization therefore provides a practical framework for measuring why evasion persists, locating residual gaps, and testing complementary modules.

Limitations

The pooled 128-source frozen evaluation contains two different source roles: the original 54-source development/evaluation cohort and a disjoint 74-source complement generated only after the SHADE pipeline was frozen. The latter is a useful within-RAID replication but not an external-dataset confirmatory sample; both partitions come from the same RAID test population. FZ and Ghostbuster were repeatedly examined while SHADE was constructed, whereas Binoculars, RoBERTa, GLTR, and WaveDetect did not guide SHADE. Competitor and composition results remain limited to the original 54 sources.

Adaptive Camouflage is evaluated at three nested exposure budgets (20, 30, and 40 examples) on one fixed 74-source permutation and one training seed per budget. This is stronger than a single-budget check but still does not include repeated exposure samples, repeated training seeds, alternative optimizers, or unrestricted defender redesign. RoBERTa's recovery fractions of 68.0%, 1.0%, and 80.8% underscore partition sensitivity: C09a finds the original thirty-example partition has the smallest recovery among five tested partitions (Appendix N), while the twenty- and forty-example rebounds demonstrate strong residual recoverability.

Adaptation capacity and trace coverage. Adaptation protocols necessarily differ in capacity: Ghostbuster refits a calibrated classifier over frozen features, Binoculars updates a two-feature readout, and GLTR refits a logistic head, whereas RoBERTa and WaveDetect continue full-model training. Consequently, table 3 measures recovery under these specified protocols rather than isolating trace coverage from adaptation capacity. However, limited update capacity alone cannot explain the observed low recovery: WaveDetect belongs to the full-model continuation group yet has negative recovery at all three exposure budgets ($-.100, -.052, -.086$ AUROC), while

RoBERTa recovers strongly at two budgets under full-model continuation. Thus low recovery is not confined to restricted readouts. This contrast strengthens the case for representation-specific responses to the intervention, while leaving optimization, training objectives and residual feature coverage as possible contributors; full-model access does not guarantee an equally effective adaptation procedure across architectures.

The competitor study mixes limited local reproductions with released checkpoints, does not equalize compute budgets, and does not establish implementation equivalence to every published configuration. HIP training-data overlap with this cohort has not been fully audited. Our comparisons therefore describe the evaluated configurations, not complete method families.

Linguistic Integrity / Non-degradation is supported by local likelihood/grammar guards and author inspection, without a blinded multi-rater study. Its reference is the original generated text: it neither establishes truthfulness or human-equivalent writing quality nor formally verifies semantic equivalence. It is an exclusion criterion for corruption-based evasion, not a standalone quality result. SHADE+RAFT shows visible coherence degradation after RAFT and is therefore treated as a composition proof of concept rather than the preferred endpoint. No matched quality comparison is available for every competitor configuration. These qualitative limits are distinct from mechanically checkable Canonical Integrity and the quantitative Adaptive Camouflage experiments.

Human-reference band attainment is feature-specific and incomplete. Matching marginal bands, means, or one detector representation does not imply equality of the joint human and attacked distributions. Tails, feature dependencies, conditional patterns, and new learned representations may remain discriminative. RoBERTa's adaptive rebound is direct evidence of this limitation.

The detector suite covers five research implementations and does not include every contemporary commercial or semantic-structural detector. Precision constraints require NF4 Binoculars, and several detectors use project-specific retraining or readout choices documented in the appendices. Results should not be interpreted as universal evasion.

Ethics Statement

This work directly studies adversarial evasion of machine-generated-text detectors and is therefore dual use. The experiments target public research methods and local/reproducible detector implementations rather than deployed integrity services. We report negative as well as positive results, including RoBERTa's successful adaptation, because the defensive question—which attack mechanisms can be closed cheaply and which require new representations—is central to the study.

The trace-based framework can be used to improve robustness testing, attack-aware training, and detector design, but the same techniques could also be used to conceal machine-generated text. Our practical checks intentionally reject trivial encoding attacks and attacks that obtain evasion through obviously degraded writing; this does not remove the misuse risk of successful ordinary-language rewriting. Released artifacts preserve provenance, detector configurations, and experimental limitations so that results can be audited rather than treated as guarantees about real-world detection.

Code and Data Availability

An anonymized reproduction package contains the measurement, intervention, robustness, competitor, adaptation, and composition code used in this study together with saved manifests and evaluation artifacts. Derived data are released subject to the licenses and provenance requirements of the source datasets and model checkpoints. The repository distinguishes frozen SHADE outputs, optional SHADE+RAFT/competitor compositions, detector-specific adaptation splits, and technical dependency-trace experiments.

AI Assistance Disclosure

ChatGPT and Claude were used as research assistants for manuscript editing, code development and debugging, literature-search assistance, and discussion of experimental design and robustness analyses. The research questions, measurement constructions, primary hypotheses, experimental decisions, execution and

verification of experiments, and final interpretation of results were performed and approved by the authors. The authors take full responsibility for the content of the paper.

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A Reproducibility Details

Computational budget. Each individual experiment required fewer than six GPU-hours on one NVIDIA RTX 2000 Ada (8 GB). No experiment required distributed or multi-GPU execution.

Cífka released-code equivalence audit. We run the authors' unmodified released context length probing (CLP) implementation (Cífka and Liutkus, 2023), specifically GPT-2 (Radford et al., 2019) `run_probing()`, and our xent-only scaling implementation on the same two 257-token documents with the same FP32 model on CPU. Across 498 context-length comparisons ($k = 8, \ldots, 256$ for each document), mean, median, and maximum absolute error are all exactly 0, and Pearson correlation is 1.000. The larger experiments then use the xent-only implementation to avoid retaining full-vocabulary distributions.

Context utility. The independent scorer is Qwen3-0.6B-Base with a fixed 384-token scorer prefix and eight deterministic target positions. The reported context-utility profile is $C = (U(32), U(128), U(256))$ relative to the 8-token baseline. The original GHOSTBUSTER context extractor additionally evaluated intermediate windows $k = 16$ and $k = 64$; these do not alter the reported $U(32)$, $U(128)$, or $U(256)$ values because the maximum context window and target positions are unchanged. In the corrected GHOSTBUSTER observational analysis, documents are matched by original essay identifier before eligibility filtering, leaving 708 complete seven-way groups.

Generator-history intervention. To validate that context utility responds to the source generation process rather than only to static lexical correlates, we compare full-history generation with a condition that preserves the task prompt while restricting generated self-history. Qwen2.5-1.5B-Instruct uses 60 matched source pairs and SmolLM2 (Allal et al., 2025) uses 40. With $\Delta U(k) = U_{\text{cap}}(k) - U_{\text{full}}(k)$, Qwen2.5 gives -0.020, -0.405, and -0.407 at $k = 32, 128, 256$, while SmolLM2 gives -0.176, -0.492, and -0.578. In both families the long-range intervals exclude zero while the short-range effect is weak or unresolved. We use this result only as technical support that U-adjacent observables can encode process-sensitive contextual structure; U itself is not treated as a universal detector feature in the main attack evaluation.

RAID model provenance. We retain RAID's released generator labels and metadata. The six headline conditions correspond to GPT-2 XL, MPT-30B (MosaicML, 2023), Mistral 7B (Jiang et al., 2023), Llama 2 70B Chat (Touvron et al., 2023), ChatGPT, and GPT-4. The clean headline condition is greedy decoding ($T = 0$) with no repetition penalty and no adversarial attack; controlled alternatives use RAID's sampling condition ($T = 1$) and repetition-penalty setting $\Theta = 1.2$. For ChatGPT and GPT-4 we report the identifiers supplied by RAID rather than inferring an API snapshot not encoded in the released model label.

Data-pairing audit. During preparation of the reproduction package, we identified that

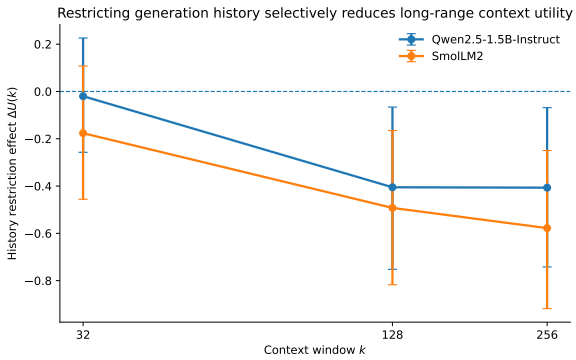


Figure 2: **Generator-side validation of context utility.** Restricting generated self-history has a weak or unresolved effect at $k = 32$ but clear negative effects at $k = 128$ and 256, replicated with Qwen2.5-1.5B-Instruct ($n = 60$) and SmolLM2 ($n = 40$). Error bars show 95% confidence intervals.

an earlier version of the GHOSTBUSTER observational analysis paired generator streams by enumeration index rather than original essay identifier. We discarded that analysis, reconstructed the corpus directly from the public source files, matched all seven streams by original document ID, reran context extraction, and report only the corrected 708-group source-matched result.

Ghostbuster eligibility audit. Of the 292 source groups excluded from the seven-way context analysis, 249 fail eligibility in only one of the seven streams. Human essays account for 227 of the 356 individual ineligible documents (22.7% of human essays), whereas each machine stream has an ineligibility rate between 0.1% and 3.0%. Thus the complete-case restriction is driven primarily by human document length. A pairwise-eligibility sensitivity analysis retains 773 human-eligible sources and yields pooled mean differences of 0.3186, 0.4398, and 0.4802 at $k = 32, 128, 256$, respectively, versus 0.3162, 0.4366, and 0.4772 under the strict seven-way analysis. The absolute shift is below 0.004 at every context length.

Frontloading. *Recurrence frontloading* $F(x)$ measures whether repeated content words are more concentrated between nearby sentences than distant ones, relative to shuffled sentence order. Let A_i be sentence i's content-word set, $J_{ij} = |A_i \cap A_j|/|A_i \cup A_j|$ its Jaccard overlap with sentence j, and P_b the sentence pairs in token-distance bin b. Define

$$O_b(x) = \frac{1}{|P_b|} \sum_{(i,j)\in P_b} J_{ij}, \quad R_b = O_b - S_b,$$

where S_b is the mean overlap after within-document sentence-order shuffling. With bins ordered by increasing distance,

$$F(x) = \frac{1}{3}\sum_{b=1}^{3} R_b(x) - \frac{1}{3}\sum_{b=7}^{9} R_b(x).$$

Positive F means greater shuffle-corrected recurrence at short distances; it is not a direct measure of semantic information density. The lexical filtering, bins and shuffle estimator are specified below.

Recurrence is computed on the decoded first 384 Qwen3-0.6B tokenizer tokens. We segment that prefix with NLTK (Bird et al., 2009) `sent_tokenize`. Each sentence is represented as a *set* of lowercase matches to `[a-zA-Z][a-zA-Z'-]*` after removing scikit-learn's `ENGLISH_STOP_WORDS` (Pedregosa et al., 2011) and words shorter than three characters; no lemmatization is applied. Sentence centers are approximate Qwen-token centers obtained by tokenizing each segmented sentence independently and cumulatively placing its token span. The recurrence extractor uses token-distance bins $(0, 32)$, $(33, 64)$, $(65, 96)$, $(97, 128)$, $(129, 160)$, $(161, 192)$, $(193, 224)$, $(225, 256)$, and $(257, 320)$. For each of 200 within-document shuffle draws (seed 42), the positional center slots and therefore the distance-bin pair geometry are held fixed while sentence lexical sets are randomly reassigned to those slots; this preserves the document's lexical inventory and positional geometry while destroying the observed sentence ordering. R_{near} averages the first three bins, R_{far} the last three, and $F = R_{\text{near}} - R_{\text{far}}$.

Same-family control. The cross-model base-versus-chat comparison can confound post-training with model family, scale, and capability. We therefore perform matched same-family comparisons on RAID under fixed news sources, greedy decoding, no repetition penalty, and no attack, recomputing frontloading with the same frozen 384-token recurrence extractor used in the headline analysis.

For Mistral, the base model has mean $F = 0.104$ while the chat variant is approximately

zero ($F = -0.0003$), yielding $\Delta F_{\text{chat}-\text{base}} = -0.105$ (95% CI $[-0.134, -0.076]$, $n = 115$). MPT shows an analogous pattern, falling from $F = 0.175$ in the base model to $F = -0.005$ in MPT-Chat ($\Delta F = -0.180$, 95% CI $[-0.253, -0.106]$, $n = 17$). Cohere is qualitatively different: both the base and chat variants are already near zero ($F = 0.0010$ and 0.0022, respectively), producing a precise null difference of $+0.0012$ (95% CI $[-0.0019, 0.0043]$, $n = 225$).

These controls therefore suggest that lower frontloading is not a universal consequence of chat post-training. Instead, some model families such as Mistral and MPT move from strongly frontloaded base outputs toward a near-zero regime, while Cohere appears to occupy that low-frontloading regime already at the base-model stage.

Decoding-intervention visualization. Figure 3 shows the paired effects and absolute movement toward the chat/aligned centroid reported in the main text.

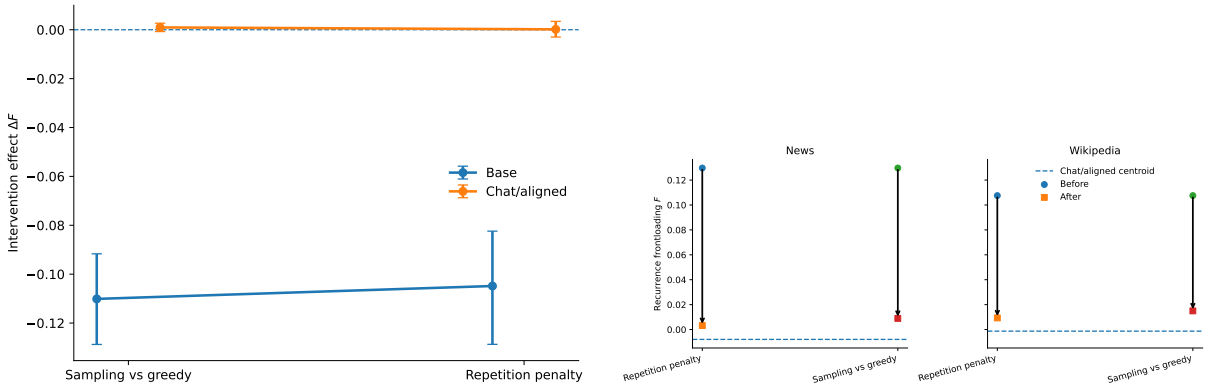
Dense context-probing audit and compression protocol. The direct Cífka-lineage experiments (Cífka and Liutkus, 2023) use the same loss quantity L_k but evaluate dense windows $k = 8, \ldots, N$. We separately audit the released context-probing code with GPT-2 and use a scalable xent-only implementation of the same windowing/diagonalization construction for the larger matched experiment. The source-attribution pool contains 146 complete GPT/Claude pairs after requiring all $L_8, \ldots, L_{256}$ values for both scorers. For each $N \in \{32, 64, 128, 256\}$, endpoint, geometric, and dense feature sets are evaluated on identical source-grouped splits. Regularization is selected using training data only.

The primary non-inferiority margin is $\delta = 0.01$ AUROC for $d_s = \text{AUROC}_s(M) - \text{AUROC}_s(D_N)$. At $N = 256$, Qwen geometric probing gives mean $d = +0.0158$ with 95% interval $[+0.0087, +0.0241]$; GPT-2 gives $d = -0.0071$ with $[-0.0153, +0.0019]$. Both use six measurements rather than 249. Qwen geometric probing passes the primary criterion at all four N, whereas GPT-2 passes at $N = 32, 64$ and fails at $128, 256$. We report these scorer-specific outcomes rather than pooling them into a universal sufficiency claim.

Effective-dimensionality protocol. The dimensionality analysis reuses the same 292 documents for each scorer. Raw covariance PCA is used only to describe the natural geometry of D_{256}; a standardized-geometry audit gives essentially the same raw effective dimensions. Shape geometry uses the deterministic endpoint-normalized features $S_k = L_k - L_{256}$ for $k = 8, \ldots, 255$. Attribution PCA is nested inside the training pipeline: each inner and outer training partition fits its own standardizer and PCA basis before logistic regression, preventing test-fold leakage. The tested ranks are $r \in \{1, 2, 3, 4, 6, 8, 16, 32, 64, 128\}$. We define the reported stable dimension as the smallest r such that the paired-bootstrap lower bound for $\text{AUROC}(\text{PC}_{1:r}) - \text{AUROC}(D_{256})$ is above -0.01 at that and every larger tested rank. This gives $r = 4$ for both Qwen3-0.6B and GPT-2.

WaveDetect split protocol. For each seed, the full 6,000-item original GPT/human manifest is permuted once and divided 80/20 before intersecting feature availability. The WaveDetect probability is first clipped to $[10^{-6}, 1 - 10^{-6}]$ and transformed to the logit $W = \log(p/(1-p))$. The meta-classifier receives this logit together with the requested C and/or F features, standardizes using training data only, and fits balanced logistic regression. Evaluation combines all available unseen-Claude rows with held-out original humans. Split-level deltas are always computed against the W baseline from the same seed.

Statistical discipline. Contrasts are paired by prompt/source wherever possible. Bootstrap intervals resample matched sources or use the saved confirmatory paired analyses. Repeated detector splits are described as split robustness rather than independent replications. We therefore report medians, interquartile ranges, ranges, and the number of positive split-level deltas rather than constructing a conventional confidence interval across the 20 overlapping splits. Exploratory short-text robustness and fine-grained attribution branches are not used as evidence for the headline claims.



(a) Paired intervention effects on F.

(b) Absolute movement toward the chat/aligned centroid.

Figure 3: **Decoding interventions move outputs along the frontloading axis.** Sampling and repetition penalties strongly reduce base-model F while chat/aligned systems remain approximately stable; both interventions move the base centroid near the chat/aligned region in news and Wikipedia.

B Scorer–Generator Lineage Diagnostics

B.1 Source-Level Context Signatures Generalize Across Scorers

The three-scorer factorization separates two levels of invariance. At the individual-document level, cross-scorer agreement is weak: at $U(256)$, Pearson correlations within a fixed source condition are generally between approximately -0.06 and 0.33. Thus $U_x(k)$ should not be interpreted as a scorer-invariant feature attached to each document.

Additive decomposition. For each context scale k and matched-essay bootstrap resample, let $Y_{s,g}(k)$ be the mean $U(k)$ for scorer s and source condition g. With $S = 3$ scorers and $G = 7$ source conditions, we use the balanced additive decomposition

$$\mu = \frac{1}{SG}\sum_{s,g} Y_{s,g}, \qquad A_s = \bar{Y}_{s\cdot} - \mu, \quad B_g = \bar{Y}_{\cdot g} - \mu, \tag{4}$$

$$I_{s,g} = Y_{s,g} - \mu - A_s - B_g. \tag{5}$$

Writing $SS_T = \sum_{s,g}(Y_{s,g} - \mu)^2$, the reported fractions are

$$\rho_{\text{scorer}} = \frac{G\sum_s A_s^2}{SS_T}, \qquad \rho_{\text{source}} = \frac{S\sum_g B_g^2}{SS_T}, \qquad \rho_{I} = \frac{\sum_{s,g} I_{s,g}^2}{SS_T}. \tag{6}$$

The balanced components are orthogonal, so the three fractions sum to one within each resample. We bootstrap matched essay identifiers while preserving all scorers and source conditions, and report the median and percentile interval of each fraction. This is a descriptive partition of variation among aggregate cell means, not a random-effects or causal variance decomposition.

At the source-population level, however, the long-range structure is substantially more stable. All 18 machine-source–by–context point-estimate contrasts (six machine conditions times $k \in \{32, 128, 256\}$) have the same machine-minus-human sign under all three scorers. The lone near-boundary case is Claude under GPT-2 at $U(256)$, where the mean contrast is $+0.0015$ and is not resolved from zero; Qwen and SmolLM2 give $+0.233$ and $+0.374$, respectively. The other long-range machine conditions remain positive across scorers, with especially consistent effects for the four GPT prompt/style variants.

Agreement in the *relative ordering* of machine source conditions also strengthens with range. At $U(128)$, machine-only Pearson correlations of the six-condition source signature are 0.971 (Qwen/GPT-2), 0.873 (Qwen/SmolLM2), and 0.807 (GPT-2/SmolLM2); at $U(256)$ they are 0.982, 0.776, and 0.828. Corresponding Spearman correlations are 1.000/0.886/0.886 at $U(128)$ and 1.000/0.943/0.943 at $U(256)$. By comparison, $U(32)$ is less stable, especially for GPT-2 versus SmolLM2. Matched-source bootstrap intervals remain wide because each signature contains only six machine conditions, but the long-range Pearson lower bounds are positive for all scorer pairs.

The additive decomposition reveals why long-range source signatures become more cross-

Metric	Qwen/GPT-2	Qwen/SmolLM2	GPT-2/SmolLM2
$U(32)$ Pearson	.786	.701	.712
$U(128)$ Pearson	.971	.873	.807
$U(256)$ Pearson	.982	.776	.826
$U(128)$ Spearman	1.000	.886	.886
$U(256)$ Spearman	1.000	.943	.943

Table 5: **Machine-source signature agreement across independent scorers.** Correlations compare the six machine-condition mean context-utility signatures on the same 128 matched GHOSTBUSTER essay identifiers. Long-range source ordering is substantially more stable than the short-range $U(32)$ signature.

scorer stable. The scorer main-effect fraction falls sharply with range, from 0.227 at $U(32)$ to 0.096 at $U(128)$ and 0.067 at $U(256)$, while the scorer $\times$ source interaction remains approximately constant at 0.12–0.14. Over the same range, the source-condition main effect grows from 0.628 to 0.777 and 0.802. Thus the increasing source share at long range is driven primarily by a reduction in scorer-specific calibration effects rather than by a disappearance of genuine scorer–source non-additivity. This also explains why $U(32)$ shows the least stable cross-scorer source signature: short-range utility retains a substantially larger scorer-specific component. At $U(128)$ and $U(256)$, the source-condition component is more than six times larger than the interaction component.

B.2 Model Lineage Appears in Context-Gain Magnitude, Not Normalized Shape

The six-family matrix asks whether known base/instruction lineage is visible in context-gain magnitude after removing scorer-specific scale, and whether any such alignment survives normalization of total gain amplitude. Averaged over families, the standardized home-family advantage is positive for raw long-context loss and for every tested context-utility scale:

$$D(L_{256}) = 0.265 \quad [0.163, 0.354], \tag{7}$$
$$D(U_{32}) = 0.250 \quad [0.156, 0.343], \tag{8}$$
$$D(U_{128}) = 0.178 \quad [0.090, 0.271], \tag{9}$$
$$D(U_{256}) = 0.194 \quad [0.102, 0.292]. \tag{10}$$

Intervals bootstrap the 46 matched prompt groups. L_{256} measures absolute predictive fit and is used only as a positive control. The diagonal remains after subtracting the short-context baseline; the gain-amplitude controls below test whether this residual alignment extends beyond overall context-benefit magnitude.

The exact family-label test strengthens this result (figure 4). Among all 720 possible one-to-one scorer–generator mappings, the true lineage mapping ranks first for L_{256} and $U(32)$, eighth for $U(128)$, and fourth for $U(256)$. The corresponding exact one-sided p-values are 0.00139, 0.00139, 0.0111, and 0.00556. Thus the known family genealogy is unusually aligned with the *magnitude* of additional-context benefit, not only with absolute predictive fit. As a tokenizer-level control, we apply the same exact $6! = 720$ family-mapping test to scorer-token efficiency (tokens per 1,000 characters). The true family correspondence is not unusual under this control (rank 225/720, exact one-sided $p = .3125$), indicating that the $U(128)$ and $U(256)$ lineage effects are not explained by a simple home-tokenizer efficiency advantage.

Gain-amplitude calibration control. To test whether the $U(k)$ diagonal instead reflects residual same-family calibration of context-gain magnitude, we normalize each prompt–scorer–generator response before rerunning the same exact 6! mapping test. The primary normalized late-context share is

$$H_{\text{end}} = \frac{U(256) - U(128)}{U(256)}, \tag{11}$$

and a denominator-robust companion uses

$$H_{\ell_2} = \frac{U(256) - U(128)}{\|(U_{16}, U_{32}, U_{64}, U_{128}, U_{256})\|_2}. \tag{12}$$

Both are invariant to any positive affine recalibration $L'_k = aL_k + b$: the additive offset cancels in $U(k)$ and the multiplicative factor cancels under normalization. As in the original mapping test, each scalar is then standardized within prompt and scorer across the six generators before the 720 assignments are enumerated.

Neither normalized control retains a resolved identity mapping. H_{end} ranks 114/720 (exact $p = .158$), with standardized identity advantage 0.073 and 95% prompt-bootstrap interval $[-0.043, 0.188]$. H_{ℓ_2} ranks 216/720 ($p = .300$), with advantage 0.031 and interval $[-0.083, 0.148]$. Because a small number of observations have near-zero $U(256)$, the ℓ_2 result is the less denominator-sensitive robustness

Control	Rank/720	Exact p	Identity advantage [95% CI]
Endpoint-normalized late share	114	.158	.073 [−.043, .188]
ℓ_2-normalized late share	216	.300	.031 [−.083, .148]

Table 6: **Gain-amplitude controls for the six-family lineage mapping.** After removing overall context-gain amplitude, the true scorer–generator family correspondence is no longer unusual among the 720 one-to-one mappings. Identity advantages are computed after the same within-prompt, within-scorer standardization as the original exact mapping analysis; intervals bootstrap the 46 matched prompt groups.

check; both lead to the same qualitative conclusion. The original $U(128)/U(256)$ mapping therefore supports an amplitude-dependent same-lineage context-gain association, but not a robust family-specific normalized response shape. Residual same-family calibration remains a viable explanation for the magnitude effect.

Figure 5 reports the full generator-identification matrices. The absence of a universal AUROC diagonal is scientifically useful: although raw context-gain magnitude favors the true family genealogy, the corresponding scorer is not generally the most discriminative instrument for identifying that generator, and the separate gain-amplitude controls above do not support a normalized-shape diagonal.

Generated corpus and eligibility. The lineage corpus contains 48 matched prompts and six generated continuations per prompt before filtering (288 generated documents). Mean generated length is approximately 503–512 generator tokens across families. Joint scorer-token eligibility retains 46 prompts, so every reported 6×6 comparison uses the same 276 generated continuations and all 1656 scorer–generation evaluations.

C Additional Context-Probing Analyses

Scorer–source factorization visualizations. Figure 6 shows the $U(256)$ aggregate source signature and the matched machine-minus-human contrasts used in section B.1. The figures emphasize the distinction between stable source-level structure and scorer-dependent magnitude.

This appendix expands the paper's second central question, the measurement geometry of the validated context-range trace. It contains the full dimensionality uncertainty analysis, marginal-gain decomposition, power-law diagnostics, and geometric-sampling derivation underlying the condensed main-text results.

Analysis definitions and conditional geometric-sampling result. For dense D_{256} we study the raw curve and the endpoint-normalized shape

$$S_k = L_k - L_{256}, \qquad k = 8, \ldots, 255, \tag{13}$$

using the participation-ratio dimension

$$d_{\mathrm{PR}} = \frac{\left(\sum_j \lambda_j\right)^2}{\sum_j \lambda_j^2}. \tag{14}$$

Uncertainty in these descriptive dimensions is estimated by matched-`pair_id` bootstrap resampling. For task-effective dimension, PCA is fit strictly within each training fold after feature standardization; PCA is representation compression only and still requires all dense measurements.

Define the adjacent marginal gain

$$m_x(d) = L_{d-1}(x) - L_d(x), \qquad d = 9, \ldots, N. \tag{15}$$

By telescoping,

$$L_k(x) - L_N(x) = \sum_{d=k+1}^{N} m_x(d). \tag{16}$$

We fit the mean marginal response in 16 logarithmic distance bins as $\bar{m}(d) = ad^{-\alpha}$ and bootstrap matched source pairs. Under the idealized continuous form with $\beta = \alpha - 1 > 0$,

$$S(k) = \frac{a}{\beta}\left(k^{-\beta} - N^{-\beta}\right). \tag{17}$$

For piecewise-linear interpolation in $x = \log k$ on a geometric grid with maximum log spacing h, the standard interpolation remainder gives

$$\frac{\|S - \widetilde{S}\|_\infty}{|S(k_0)|} \le \frac{\beta^2 h^2}{8\,[1 - (k_0/N)^\beta]}. \tag{18}$$

For fixed normalized reconstruction tolerance this implies $O(\log N)$ sampled context lengths. This bound applies only to the idealized integrated power-law response; it is not an empirical upper bound on noisy observed CLP curves and does not establish attribution non-inferiority.

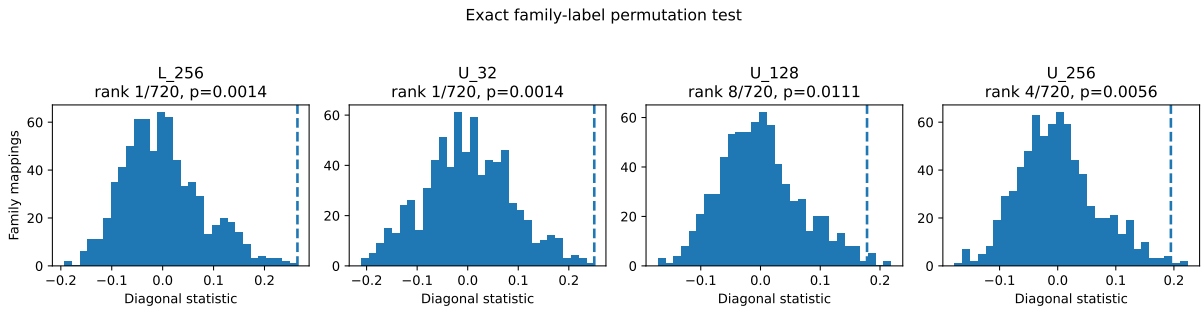
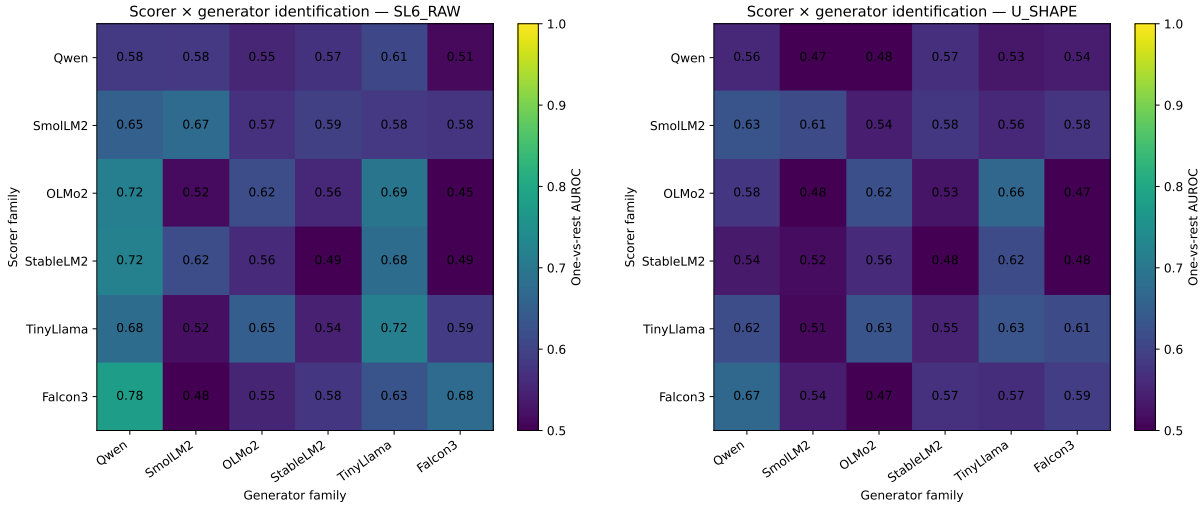


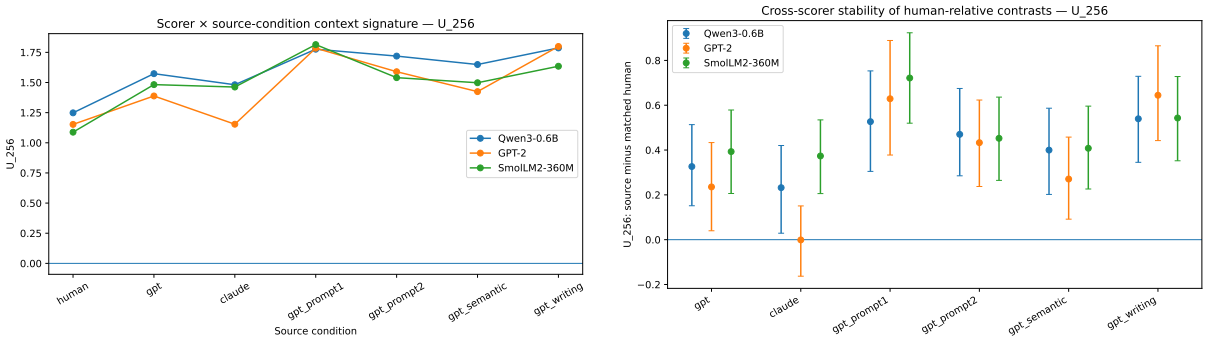
Figure 4: **Exact scorer–generator family-mapping test before gain-amplitude normalization.** Dashed lines show the statistic for the true base/instruction family correspondence; histograms enumerate all $6! = 720$ one-to-one mappings. Raw long-context fit is a positive control. The identity correspondence is unusually strong for context-gain magnitude at $U(128)$ (rank 8/720, $p = .0111$) and $U(256)$ (rank 4/720, $p = .0056$); amplitude-normalized controls are reported in table 6.



(a) Raw six-point loss profile.

(b) Context-gain shape $(U_{16}, \ldots, U_{256})$.

Figure 5: **Generator-family identification across scorer families.** Each row uses one frozen scorer family; each column is one-vs-rest AUROC for a generator family under grouped prompt splits. Unlike the direct contextual-affinity statistic, the classification matrices do not show a statistically resolved global identity diagonal.



(a) Mean $U(256)$ source-condition signature.

(b) Matched source-minus-human contrasts with 95% bootstrap intervals.

Figure 6: **Scorer × source-condition factorization at long range.** The broad source-level pattern is preserved across Qwen3-0.6B, GPT-2, and SmolLM2-360M, while contrast magnitudes remain scorer-dependent.

Family	Instruction/chat generator	Base-family scorer
Qwen	`Qwen/Qwen2.5-0.5B-Instruct`	`Qwen/Qwen2.5-0.5B`
SmolLM2	`HuggingFaceTB/SmolLM2-360M-Instruct`	`HuggingFaceTB/SmolLM2-360M`
OLMo2	`allenai/OLMo-2-0425-1B-Instruct`	`allenai/OLMo-2-0425-1B`
StableLM2	`stabilityai/stablelm-2-zephyr-1_6b`	`stabilityai/stablelm-2-1_6b`
TinyLlama	`TinyLlama/TinyLlama-1.1B-Chat-v1.0`	`TinyLlama/TinyLlama-1.1B-intermediate-step-1431k-3T`
Falcon3	`tiiuae/Falcon3-1B-Instruct`	`tiiuae/Falcon3-1B-Base`

Table 7: **Exact checkpoints used in the six-family lineage experiment.** Each generated continuation is evaluated by every base-family scorer; "home family" denotes the corresponding lineage pairing used by the exact 6! mapping test. Model-family references: Qwen2.5-0.5B-Instruct (Qwen Team, 2024a), SmolLM2 (Allal et al., 2025), OLMo2 (Team OLMo et al., 2024), StableLM2 (Bellagente et al., 2024), TinyLlama (Zhang et al., 2024), and Falcon3 (Falcon-LLM Team, 2024).

C.1 Measurement Geometry of the Context-Range Trace

A first compression test on 2,860 matched GPT/Claude source pairs shows that the four-point profile $(L_8, L_{32}, L_{128}, L_{256})$ is already dominated by its long-context endpoint: L_{256} reaches mean AUROC 0.7087 versus 0.7100 for the full four-point curve, retaining 99.35% of its above-chance performance. Adding L_8 raises this to 0.7090. This motivates a stronger test against genuinely dense context-length probing rather than against only four sampled points.

Using dense Cífka-style profiles (Cífka and Liutkus, 2023) makes the compression question direct (figure 7). At $N = 256$, Qwen3-0.6B gives mean AUROC 0.8482 with E_{256}, 0.8667 with six-point G_{256}, and 0.8509 with 249-point D_{256}. GPT-2 gives 0.7454, 0.7442, and 0.7512, respectively. Sparse values exceeding dense values should not be interpreted as information created by compression; with finite samples and regularized classifiers they are consistent with improved statistical efficiency from a lower-dimensional representation. The important observation is that moving from six geometrically spaced measurements to all 249 integer context lengths does not open a clearly stronger attribution regime.

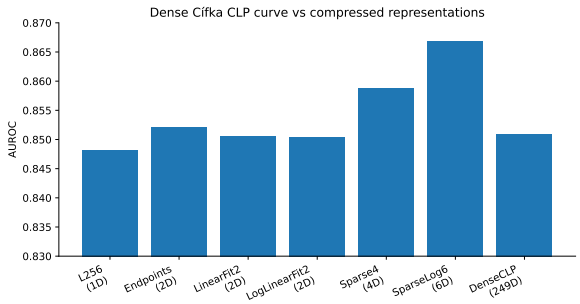
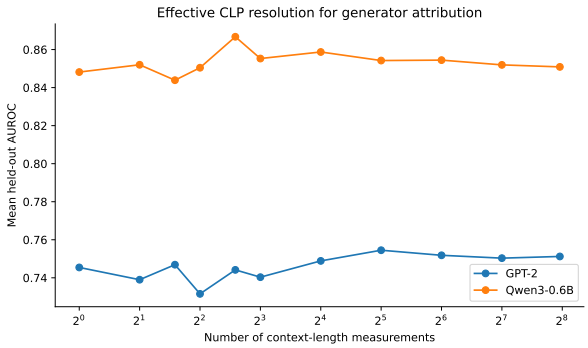


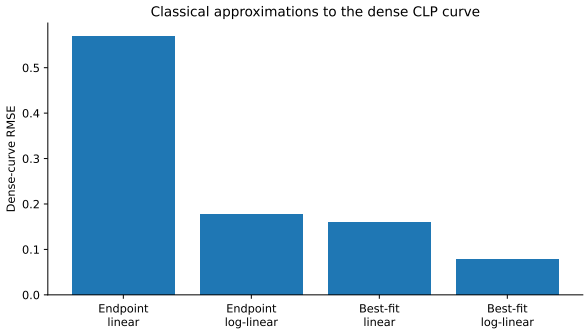
Figure 7: **Direct compression of dense Cífka-style context profiles** (Cífka and Liutkus, 2023). Source-attribution AUROC at $N = 256$ is compared across endpoint, low-dimensional fitted or sampled summaries, six-point geometric probing, and the full 249-feature dense profile. Sparse representations can numerically exceed the dense classifier under finite-sample regularization; the comparison tests information retention, not whether compression creates information.

Resolution sweep and curve reconstruction. The headline non-inferiority analysis compares endpoint, geometric, and dense representations at four maximum contexts. Two complementary diagnostics provide a finer view. First, we vary the number of retained context lengths from a single endpoint through the full 249-feature profile. Attribution performance saturates early and does not improve monotonically as additional context points are added (figure 8a). Second, we ask a different question: how well can low-dimensional summaries reconstruct the *numerical dense curve*? Log-context interpolation and fitting substantially improve reconstruction relative to ordinary endpoint-linear interpolation (figure 8). The two panels should not be conflated: task-information retention can be high even when pointwise curve reconstruction is imperfect.

Effective dimensionality of DenseCLP. Figure 9 separates geometric variance from task-effective dimension. For raw D_{256}, $d_{\mathrm{PR}} = 1.078$ for Qwen3-0.6B (95% matched-`pair_id` bootstrap interval $[1.064, 1.095]$) and 1.131 for GPT-2 ($[1.104, 1.163]$). Endpoint-normalized shape remains low-dimensional, with $d_{\mathrm{PR}} = 1.850$ for Qwen ($[1.696, 2.031]$) and 1.726 for GPT-2 ($[1.594, 1.875]$). The leading raw PC correlates strongly with L_{256} ($r = 0.984$ for Qwen and 0.973 for GPT-2), showing that the dominant



(a) Source-attribution AUROC as context-profile resolution increases.



(b) Dense-curve reconstruction error for low-dimensional approximations.

Figure 8: **Resolution saturation and geometric structure of dense context profiles.** Increasing the number of sampled context lengths beyond a small sparse set yields little or inconsistent attribution benefit, while reconstruction of the dense loss curve is substantially better in log-context features. These diagnose discriminative and geometric compressibility, respectively.

raw direction is largely an overall predictive-loss axis. In the train-only attribution pipeline, raw PCA is stably non-inferior from $r = 4$ onward for both scorers. Shape-only PCA performs substantially worse, indicating that useful attribution directions combine absolute loss level with contextual response rather than simply following the highest-variance endpoint-normalized modes.

Adjacent marginal gains are qualitatively much higher-dimensional, with descriptive point estimates $d_{\mathrm{PR}} = 50.4$ for Qwen and 59.5 for GPT-2. We do not attach ordinary percentile-bootstrap intervals to these high-dimensional PR estimates because cluster resampling with replacement reduces the number of distinct source pairs and produces a systematic downward shift in the nonlinear PR statistic; the dimensional separation from the integrated shape is nevertheless very large.

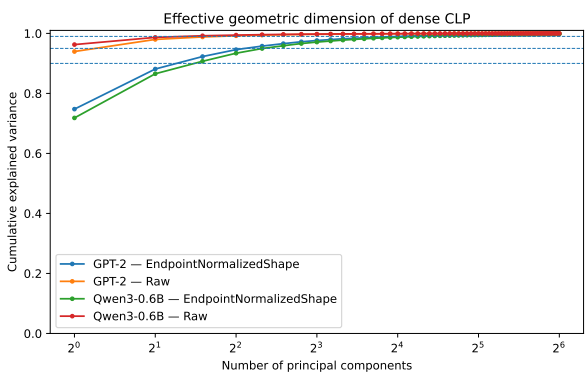
From high-dimensional marginal gains to low-dimensional CLP. Figure 10 visualizes the decomposition in equation (16). The adjacent marginal-gain matrix is high-dimensional for both scorers, whereas cumulative integration produces the much lower-dimensional endpoint-normalized shape. Independently permuting documents within each distance column before integration raises the resulting shape dimension, showing that deterministic smoothing is the dominant but not exclusive explanation.

Scorer	α	Bound	Ideal	Observed
Qwen3-0.6B	1.425	.0141	.0122	.0167
GPT-2	1.351	.0105	.0093	.0139

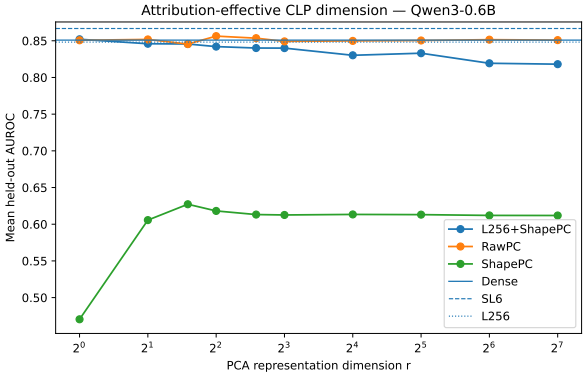
Table 8: **Six-point geometric-sampling theory versus observed mean CLP.** "Bound" is the normalized supremum-error bound for the *idealized integrated power-law model* in equation (18); it is not an empirical ceiling for the observed curve. "Ideal" is the exact interpolation error of the fitted power-law response, while "Observed" includes model mismatch and residual document structure and can therefore exceed the theoretical bound. Errors are normalized by $|\bar{S}(8)|$.

Power-law marginal envelope. Across $d = 9, \ldots, 256$, mean adjacent gain is close to linear in log–log features. Matched-pair bootstrap gives $\alpha = 1.351$ for GPT-2 (95% interval $[1.318, 1.385]$) and 1.425 for Qwen3-0.6B ($[1.396, 1.456]$), with mean $R^2 = 0.991$ and 0.997, respectively. These ordinary CLP marginal gains extend prior context-probing observations (Cífka and Liutkus, 2023). In contrast, Barenholtz (2026) measures an order-specific persistence signal relative to shuffled context and reports near-$1/d$ decay. The observables differ: SHADE uses ordinary CLP marginal gains mainly to study measurement geometry and compressibility, not to claim discovery of a new scaling law.

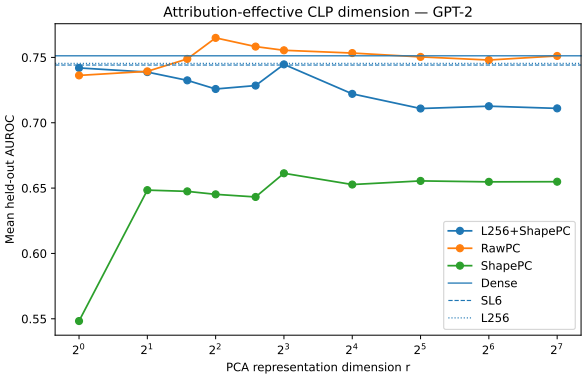
Geometric-sampling theory and reconstruction. Figure 12 compares the conditional power-law curvature bound, exact error of the fitted ideal response, and observed



(a) Cumulative explained variance for raw and endpoint-normalized CLP.

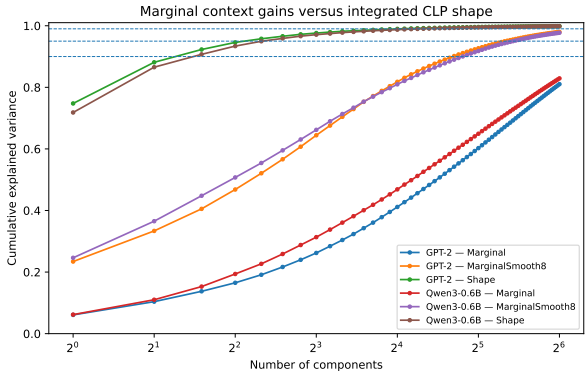


(b) Held-out attribution performance versus PCA rank for Qwen3-0.6B.

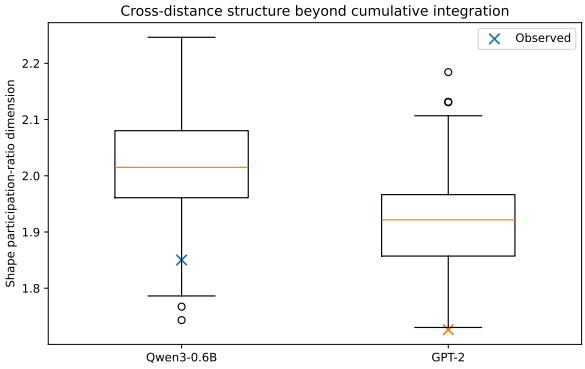


(c) Held-out attribution performance versus PCA rank for GPT-2.

Figure 9: **Geometric and attribution-effective dimension of dense context-length probing.** The 249-feature raw curve is highly low-rank, and a small number of train-only PCs recovers dense-profile attribution performance. PCA is a diagnostic of representation redundancy, not acquisition efficiency: all 249 L_k values are still required before projection.



(a) Explained variance before and after cumulative integration.



(b) Observed shape dimension versus the column-permutation null.

Figure 10: **Cumulative probing compresses high-dimensional local context gains.** Adjacent context increments contain substantial local variation, but their cumulative transform is nearly low-rank. The permutation null preserves distance-wise marginal distributions while removing cross-distance covariance, separating operator-induced smoothing from additional response structure.

mean-shape interpolation error over measurement resolutions. The three curves track one another closely at low and moderate resolution; the observed curve departs at high resolution as model mismatch and residual structure dominate the now-tiny ideal interpolation error.

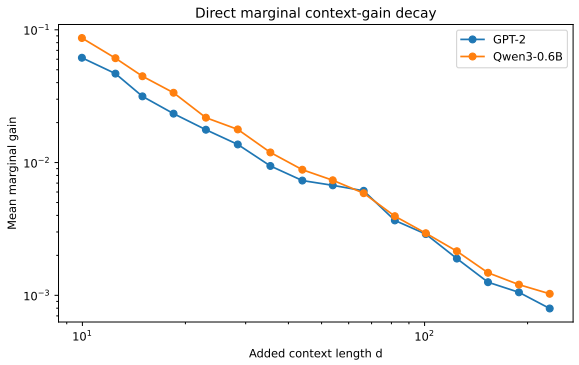


Figure 11: **Mean marginal predictive gain decays approximately as a power law.** The binned adjacent gain $\bar{m}(d) = \mathbb{E}[L_{d-1} - L_d]$ is nearly linear in log–log features over $d = 9$–256 for both frozen scorers.

Figure 13 shows the complementary document-level result: interpolation in $\log k$ is substantially better than ordinary interpolation in k at low resolution, while both converge as sampling becomes dense.

Structured context topology beyond a single long-context loss. The strong compression of ordinary context-length probing does not imply that the completed-text signal is exhausted by L_{256}. In an exploratory source-grouped analysis, a structured context-topology block T based on alternative patterns of context availability adds mean AUROC +0.0109 beyond a strong baseline already containing L_{256} (median +0.0152; positive on 13 of 20 robustness splits). By contrast, adding L_{256} alone to the same strong baseline gives only +0.0010 mean AUROC. Figure 14 visualizes the comparison. We treat this as supporting evidence that redundancy is specific to densely resampling the same clean context-length axis, not evidence that all dependency organization collapses to a single perplexity-like scalar.

D Detector Stress-Test Visualizations

Numerical information-overlap results. Across the 20-split unseen-Claude protocol, $W+C$ gains 0.00098 mean AUROC and $W+F$ gains 0.00417; $W+C+F$ gains 0.00499 AUROC and 0.0718 TPR at 1% FPR, positive on all 20 splits for both metrics (table 9). Mean F1 and AUPRC gains are 0.0200 and 0.00193, also positive on every split. Split-level dispersion is shown in figure 15. For $W+C+F$, median AUROC gain is 0.00490 (IQR $[0.00470, 0.00554]$) and median TPR gain 0.0727 (IQR $[0.0478, 0.0978]$); because splits overlap, these are dispersion diagnostics rather than independent-split confidence intervals.

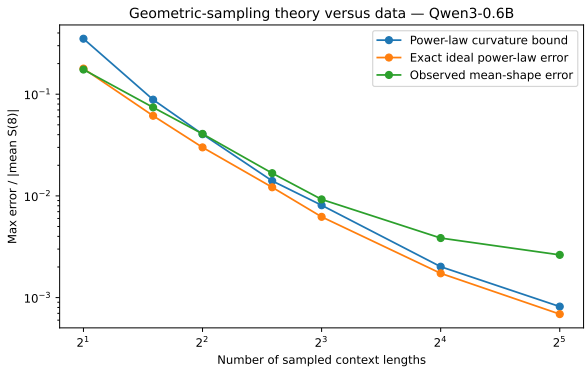
Feature set	Mean ΔAUROC	Mean ΔTPR@1%FPR
W	0.00000	0.0000
$W+C$	0.00098	0.0207
$W+F$	0.00417	0.0345
$W+C+F$	0.00499	0.0718
$W+R$	0.00471	0.0430
$W+C+R$	0.00560	0.0735

Table 9: **WaveDetect complementarity across 20 protocol-matched splits.** Values are mean improvements relative to the frozen WaveDetect score W. Here C denotes the context-utility block, F the scalar frontloading feature, and $R = (R_{\text{near}}, R_{\text{far}})$ the richer two-dimensional recurrence summary. The 20 splits overlap and are therefore treated as split-robustness checks rather than independent replications.

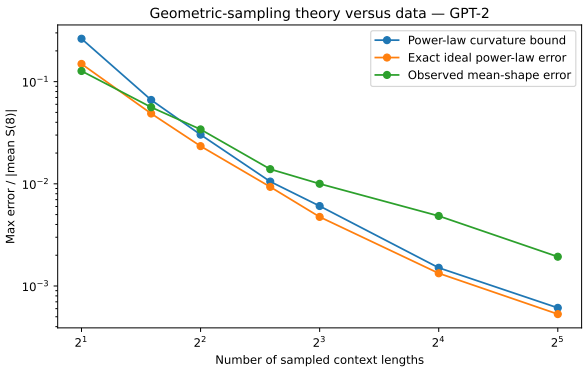
The richer recurrence block $(R_{\text{near}}, R_{\text{far}})$ reaches +0.00560 AUROC and +0.0735 TPR when paired with C. Since $F = R_{\text{near}} - R_{\text{far}}$, the scalar retains about 89% of the richer block's AUROC gain and 98% of its low-FPR gain while remaining directly interpretable.

A larger RAID stress test gives a more conservative picture for the post-compression context features. Averaged over all 11 attacks and 153 common sources, WaveDetect has source-averaged accuracy 0.9492 and $W+$SL6 reaches 0.9534, a mean gain of +0.00416. The paired source-level 95% interval $[-0.0128, +0.0223]$ includes zero, and no individual attack resolves a significant $W+$SL6 gain; the per-attack intervals are shown in figure 16. Likewise, $W+L_{256}$ does not show a resolved gain on any attack. We therefore do not claim that sparse context probing improves a frontier detector on RAID. Instead, the positive but unresolved SL6 result is consistent with small residual multi-scale information in a regime where the detector is already strongly saturated.

Unseen-Claude complementarity. Figure 15 shows split-level dispersion for the original Ghostbuster transfer experiment summarized in table 9. The same 20 overlapping protocol-matched splits are shown for AUROC and TPR at 1% FPR; they are robustness checks rather than independent replications.



(a) Qwen3-0.6B.



(b) GPT-2.

Figure 12: **Conditional geometric-sampling bound versus observed CLP.** The generic curvature bound for the fitted integrated power-law response, its exact interpolation error, and empirical mean-shape error follow similar resolution scaling.

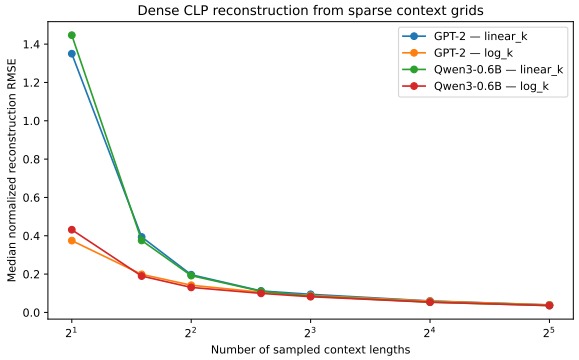


Figure 13: **Document-level dense-curve reconstruction from sparse grids.** Log-context interpolation has a large advantage at low measurement resolution, consistent with the integrated power-law geometry. Individual-document error remains materially larger than mean-curve error because local marginal responses contain high-dimensional residual variation.

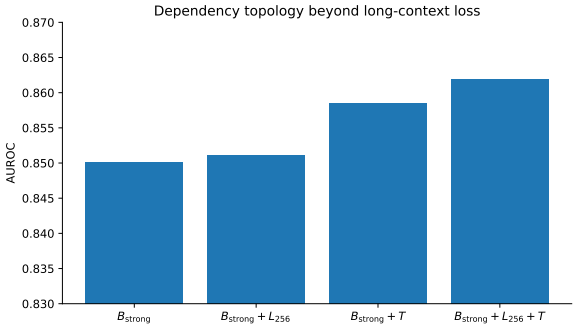


Figure 14: **Structured context topology retains information beyond L_{256}.** A topology block based on structured patterns of available context provides residual source-attribution value when added to a strong baseline already containing the long-context endpoint. This analysis is exploratory and is used to delimit the interpretation of clean-curve compression.

Scaled RAID stress test. On the 153-source cross-fitted RAID experiment, the attack-averaged $W + \text{SL6}$ gain is $+0.00416$ in source-level accuracy, but its paired 95% interval $[-0.0128, +0.0223]$ crosses zero. The per-attack confidence intervals in figure 16 likewise do not resolve a consistent improvement. This figure is therefore presented as a ceiling-limited stress test of residual information, not as evidence that sparse context probing improves a frontier detector.

E Additional Recurrence-Block Comparison

For detector complementarity, the richer recurrence summary $(R_{\text{near}}, R_{\text{far}})$ gives mean gains of $+0.00560$ AUROC and $+0.0735$ TPR at 1% FPR when combined with C. Since $F = R_{\text{near}} - R_{\text{far}}$ exactly, including all three would be linearly redundant. The corresponding F-only combined model gives $+0.00499$ and $+0.0718$. We therefore retain F in the main formulation because it captures the dominant interpretable recurrence contrast while preserving most of the downstream gain.

F SHADE Construction and Stagewise Audits

From measuring traces to intervening on them. The preceding experiments ask which traces of generation survive in completed text. A complementary question is whether these

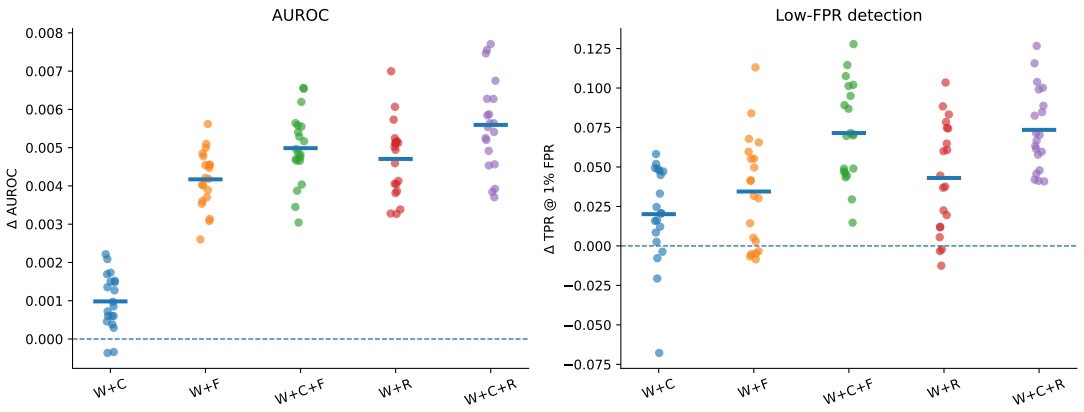


Figure 15: **Residual information beyond WaveDetect on unseen Claude.** Points are protocol-matched splits; horizontal segments are split means. $W + C + F$ is positive on all 20 splits for both AUROC and TPR at 1% FPR. $W + R$ and $W + C + R$ provide richer-recurrence robustness references.

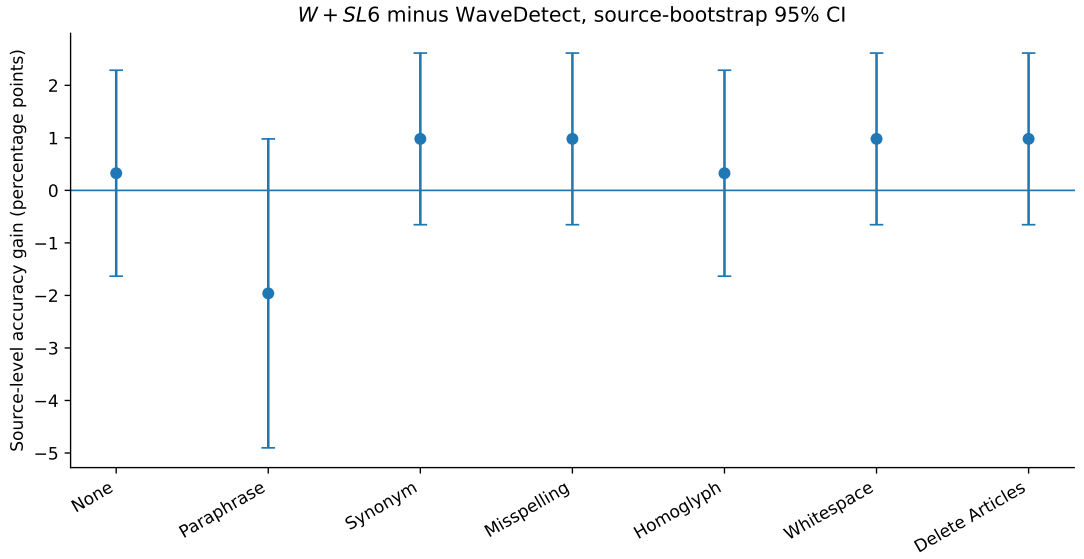


Figure 16: **Incremental $W + \mathrm{SL6}$ gain on the scaled RAID stress test.** Source-level cross-fitting is evaluated across clean text and 11 attacks. Point estimates are generally small and the confirmatory intervals do not support a detector-performance claim over the already saturated WaveDetect baseline.

traces can be changed while preserving the text's communicative function. We investigate this through staged, human-referenced interventions: first lexical and punctuation statistics, then richer syntactic and dependency observables. This appendix records the construction details and stagewise audits for the SHADE pipeline summarized in the main text. The evidence concerns feature-based detectors and bounded twenty-example adaptation experiments; it does not establish evasion of all detectors. The threat model is motivated by extensive evidence that paraphrasing, editing, and style humanization can degrade machine-text detectors while text quality and adaptive defense remain separate concerns (Krishna et al., 2023; Wang et al., 2024b; Macko et al., 2024; Gu et al., 2026).

F.1 Three Practical Robustness Checks

We propose a stricter operational notion of attack success, called *substantive evasion*. A score reduction alone is insufficient: the attack should survive canonicalization, avoid deliberate linguistic and source-content degradation, and withstand a specified amount of defender adaptation. These criteria define our evaluation scope rather than declaring other threat models invalid. In particular, representation-level attacks remain legitimate robustness tests. Our

contribution is the conjunction of the following requirements, not a claim that normalization, quality evaluation, or adversarial training is individually new.

Canonical Integrity: evasion after canonicalization. An attack should change ordinary language rather than exploit invisible or confusable encodings. For the English keyboard-normalized setting, we propose a canonical alphabet containing printable ASCII letters, digits and punctuation, together with explicitly permitted whitespace. A declared, versioned normalizer N maps accepted typographic variants to this representation and handles disallowed formatting characters. Attack evaluation is performed on $N(A(x, p))$, where x is the source text, p is the task prompt when available, and A is the attack. The same preprocessing applies to human controls and detector calibration data.

The essential requirement is persistence of evasion after this transformation, not necessarily byte-for-byte identity before and after normalization. For example, an ordinary dash may canonicalize without undoing a lexical rewrite. Conversely, deleting an inserted zero-width character may undo a purely representational perturbation. RAID includes homoglyph and zero-width-space attacks (Dugan et al., 2024); our proposed criterion explicitly evaluates such attacks after the declared preprocessing. Large authorship-obfuscation sweeps further show that attacks can remain effective while some perturbation families visibly damage readability, reinforcing the need to separate canonicalization from linguistic non-degradation (Macko et al., 2024). Unicode normalization and confusable handling are distinct mechanisms, so a normalizer must specify both rather than assuming that NFKC alone reverses every attack (The Unicode Consortium, 2025). Restricting the character inventory also does not prevent every encoding artifact, such as deceptive ASCII substitutions; canonicalization checks must be accompanied by a separate non-degradation assessment.

For this study, Canonical Integrity means that evasion does not rely on invisible characters, homoglyph substitutions or encoding tricks. Our operators change ordinary words, sentence boundaries and punctuation, and scoring normalizes newlines. We regard this as satisfying the intended practical requirement; the stricter canonical-alphabet protocol above is an optional extension, not an additional completion gate. This does not assert invariance to every possible preprocessing pipeline.

Linguistic Integrity / Non-degradation: excluding corruption-based evasion. This is a modest admissibility criterion relative to the original generated text. Its purpose is to exclude attacks that obtain evasion by deliberately introducing misspellings, grammatical breakage, incoherent or repetitive filler, or major deletion or distortion of source content. The reference text may itself contain errors: retaining its material content is not a claim of truthfulness, human-equivalent writing quality, or formally verified semantic equivalence. The criterion does not require an attack to improve the source's writing or establish independent task quality.

Local likelihood/grammar guards and author inspection support this criterion. Local fluency does not ensure global coherence: likelihood and lexical statistics can miss contradictions, causal inconsistencies and topical disconnection (Liu et al., 2026a). During development, meaning-changing lexical substitutions were identified and excluded from the adopted pipeline; such safeguards assist selection without certifying preservation for every output. No blinded multi-rater study or numerical quality score is claimed. This qualitative non-degradation check has a different epistemic role from Canonical Integrity, which is mechanically verifiable for a declared normalization procedure, and Adaptive Camouflage, which is tested through quantitative post-exposure detector performance. The three should not be read as equally measured empirical endpoints.

Adaptive Camouflage: evasion that survives exposure. A successful attack against a frozen detector may simply create a new, easily learned fingerprint. We therefore propose measuring robustness after the defender observes a small labeled sample of attacked outputs. Let d_m denote a detector updated using m attack examples, an unchanged base training set and a declared training budget. The attack is fixed before this adaptation experiment; the remaining attacked test texts and their source

prompts are disjoint from the adaptation and tuning sets. Report detection-recovery curves over m, with multiple adaptation samples and training seeds, and recalibrate thresholds on independent human data after each update.

This criterion operationalizes information-theoretic coherence as a finite-data, model-relative test. It is motivated by distributional proximity, but does not prove it. For attacked and human distributions P_A and P_H, conditioned on the same prompt/genre sampling scheme,

$$\sup_B |P_A(B) - P_H(B)| = \mathrm{TV}(P_A, P_H). \quad (19)$$

Small total variation would bound every decision region's detection advantage; failure of a particular finite-sample retraining procedure does not imply small total variation. We therefore reserve distributional indistinguishability as a hypothesis, not a conclusion from a low frozen-detector score. This link between best-achievable detection and total-variation distance also appears in theoretical analyses of adversarial machine-text detection (Sadasivan et al., 2025).

Adaptive evaluation has relevant precedent: OUTFOX co-adapts attacker and detector through adversarial examples, Zhou et al. study iterative adversarial learning, and DAMAGE uses data-centric augmentation for humanized text (Koike et al., 2024; Zhou et al., 2024; Masrour et al., 2025). Paraphrase-resilience studies likewise show that high clean performance need not imply robustness to rewriting (Shportko and Verbitsky, 2025). A paraphrasing service, including QuillBot, cannot be ruled out by name; it must be tested under the same exposure and quality protocol. Sections F.9, F.16, and F.19 report twenty-example adaptation experiments for the FZ118, FZ186, and FZ301 stages, respectively.

Joint success and coverage. For a detector threshold τ_α calibrated to human FPR α, a quality-qualified success indicator is

$$S_{\alpha,m}(x,p) = \mathbf{1}\{N(y) \in \mathcal{C}\}\, Q(x, N(y), p, g) \cdot \mathbf{1}\{d_m(N(y)) < \tau_\alpha\}, \qquad y = A(x,p), \quad (20)$$

where $\mathcal{C}$ is the declared canonical text space. An incomplete or invalid output counts as failure in the attempted-cohort success rate. Conditional detector performance on supported, completed texts should be reported separately. The present ROC analysis is such a conditional analysis, without the complete normalization and quality qualification.

F.2 A Staged Intervention Strategy

Surface control followed by structural control. We organize the program by measurement families rather than assume a universal ordering of detector strength. The initial layer contains inspectable lexical, length, readability and punctuation statistics. The next implemented stage adds POS statistics (Section F.11); planned extensions add entity/discourse features, probability-based summaries such as Ghostbuster, and sequence-level methods such as WaveDetect, alongside our context-utility and recurrence observables (Fröhling and Zubiaga, 2021; Darwinkel et al., 2024; Verma et al., 2024; Liu et al., 2026c). Each detector remains a separate evaluation target; success on a weaker or related feature family is not evidence of transfer to the others.

The feasibility hypothesis is that many useful signals are redundant and can be controlled through a smaller set of linguistically meaningful features. Our earlier context-probing results support this for specific measured profiles: raw participation-ratio dimensions are approximately 1.08 and 1.13, endpoint-normalized shape dimensions are 1.85 and 1.73, and four train-only PCs recover the tested attribution performance. However, adjacent-gain dimensions are approximately 50.4 and 59.5. These observations motivate low-dimensional structural control; they do not establish that all structural features are low-dimensional, independently editable, or sufficient for universal detector evasion.

Preserve useful features; intervene on residual evidence. The implemented sequence is Unique intervention, sentence-mean treatment, and discovery-gated punctuation intervention. Unique intervention uses Qwen2.5-1.5B-Instruct regeneration and bidirectional prefix feedback on the share of distinct alphabetic words. Sentence treatment targets mean words and characters per sentence using same-genre human references. Within-document vari-

ance is not an objective in this experiment. Neither stage is required to achieve every target before entering the supported-case analysis.

After these treatments, a discovery set ranks the mean signed contribution gap for each frozen-detector feature against its matched human pool. If $c_j(x)$ is the tree-path contribution of feature j, define

$$\widehat{\Delta}_j = \frac{1}{n_D} \sum_{i \in D} \left[c_j(x_i^{US}) - \frac{1}{|H_i|} \sum_{h \in H_i} c_j(h) \right], \tag{21}$$

where US denotes Unique plus sentence-mean treatment. Only positive features among the top 30 are selected. This gate is frozen before evaluation-case punctuation editing; comma is included regardless of its rank. Periods remain protected even if selected. Relative and per-sentence punctuation rates are separate features: only selected denominators drive the objective, although a single mark-count edit affects both.

The discovery gate selected colon and hyphen rates in both denominators, together with comma. Semicolons, en dashes and em dashes were protected. This implements a central design distinction: ability to enforce a statistic does not imply that changing it is beneficial. Earlier exploratory enforcement moved semicolons away from a human-ward contribution; gating prevents that unnecessary intervention in this run.

Vertex search and target assignment. Eligible nontrivial punctuation is processed before commas. Candidate conversions pass through words or commas; direct exchanges between different nontrivial marks are prohibited. Other nontrivial counts are locked during each pass. Local GPT-2 likelihood ranks candidates, while acceptance requires improvement in the assigned objective, preservation of sentence count and periods, and bounded regression in sentence-mean control. Candidate exhaustion refers only to the implemented vertex inventory, not every grammatical rewrite.

Unbumped targets use matched-human Q20–Q60 bands. The pilot also retains a probabilistic genre-mean adjustment when a local band lies entirely outside a narrow interval around the genre human mean on the training-AI side. The artificial interval uses 0.5 times the bootstrap standard deviation of the estimated human mean, not individual-document dispersion. Selected adjustments undergo stochastic integer-count rounding with fixed seeds. These are control heuristics, not confidence regions or guarantees of distribution matching.

Stage	Mean score	95% CI	AUC
Original	.5732	[.5397, .6113]	.8472
Unique + means	.5267	[.4909, .5630]	.7533
Final gated	.4817	[.4517, .5146]	.6485

Table 10: Frozen-detector scores and ROC AUC on 30 supported evaluation cases. Score intervals are paired case-bootstrap summaries; AUC entries are point estimates against 122 genre-weighted held-out humans.

F.3 Frozen FZ118 Pilot Results

Cohort and access model. We evaluate a restricted reproduction of the feature-based detector of Fröhling and Zubiaga (2021), denoted FZ118. The local ExtraTrees configuration (Geurts et al., 2006) uses 114 of the 118 extracted features after excluding four parenthesis-rate features. The attacker has access to this frozen detector's feature contributions for discovery; this is not a detector-agnostic black-box attack. The original C02r experiment selected 95 non-poetry RAID test texts, excluding the earlier pilot and exploratory cohort. The fixed split contained 54 discovery and 41 evaluation cases. There were 94 completed generations and one runaway failure. Requiring at least 15 same-genre human training references within a factor of 1.25 of treated length excluded 20 further cases, leaving 44 discovery and 30 evaluation cases. The evaluated genres are abstracts, books, news, recipes, Reddit and Wikipedia; the supported evaluation cohort contains no reviews.

The ROC comparison uses 122 individual held-out human documents, weighted to match the evaluation AI genre mixture. These differ from the training reference pools used for treatment and feature-gap calculation. The same human score distribution and the same 30 AI sources are used across the three stages. ROC operating points are empirical test-set summaries, not deployment thresholds estimated from an independent calibration set.

Human FPR (%)		AI TPR (%)		
Budget	Achieved	Original	U + means	Final
1	0.95	13.33	10.00	3.33
5	4.48	23.33	13.33	3.33
10	9.45	50.00	30.00	10.00
20	19.87	70.00	56.67	26.67

Table 11: Low-FPR empirical operating points. Thresholds use the most permissive human-score cutoff within each FPR budget, with tied scores treated together. All stages share each cutoff.

Overall score movement. Mean AI score decreases from .5732 to .5267 after Unique plus sentence-mean treatment, and to .4817 after punctuation editing (table 10). The paired total change is $-.0915$ (95% CI $[-.1284, -.0563]$); the punctuation-stage change is $-.0450$ ($[-.0742, -.0203]$). Scores decrease for 26 of 30 cases overall and 22 of 30 during punctuation editing. The first contrast includes regeneration as well as Unique and sentence control; this run lacks an ordinary-regeneration arm and does not isolate the causal contribution of Unique feedback alone. A score near .5 is not by itself a chance-level discrimination result.

Strongest attenuation at low FPR. At an achieved FPR of 4.48%, detections fall from 7/30 to 1/30, an 85.7% relative reduction. At 9.45% FPR, TPR falls from 50% to 30% and then 10%, an 80% overall reduction (table 11). The final curve is near the diagonal at several low-FPR operating points, while the overall AUC of .6485 shows residual ranking information. We describe this as substantial suppression of low-FPR detection by the frozen FZ118 model, not complete neutralization or demonstrated commercial-detector evasion. A substantially below-diagonal curve would also require checking reverse discrimination.

Enforcement, preservation and residual signals. Assigned targets are met in 28/30 colon, 22/30 hyphen and 30/30 comma cases: 80/90 eligible case–mark objectives. Protected marks are not included in this success count. Period and sentence counts are preserved for all 30 cases; semicolon and dash per-sentence rates remain unchanged after punctuation editing. Character-normalized rates can shift slightly because character counts change.

Colon and comma features leave the displayed final top 30 contribution gaps. Remaining leading features include the share of words in the Google top-10,000 list (gap .00604), POS unigram-overlap bins 9 (.00598) and 7 (.00591), mean words per sentence (.00558), and sentence count (.00538). Relative-hyphen contribution decreases from .00625 to .00413, whereas its per-sentence counterpart changes less. These tree-path contributions describe the frozen model; they are not independent causal effects.

Target attainment is distinct from human matching: final hyphen and comma rates fall below the matched-human means, with paired intervals excluding zero in both denominators. This motivates reviewing target assignment rather than merely increasing enforcement. Length-related residuals may reflect genre, sampling or detector shortcuts, but their presence does not by itself establish overfitting. POS organization and common-word share are deferred to later interventions.

F.4 Generator-Relative Admissibility: A Diagnostic Example

Lexical identity is an overly restrictive proxy for communicative utility when a task permits variation. We therefore distinguish strict source-preserving rewriting from task-preserving adaptation. The latter may admit a change in an incidental detail or an ingredient choice, provided that such variation is allowed by the task specification. This permission must be declared independently of detector outcomes; it cannot be inferred solely from the generator's preference. We introduce *generator-relative admissibility* as an operational candidate filter within this setting, rather than as a replacement for independent utility assessment.

Let h denote the common token prefix, including the rewrite prompt and source passage, and let z_0 and z be the ordinary and proposed token paths from a shared search anchor through their respective word endings. For the frozen generator q, define

$$\ell_q(z \mid h) = \sum_{t=1}^{|z|} \log q(z_t \mid h, z_{<t}),$$
$$R_q(z, z_0 \mid h) = \exp[\ell_q(z \mid h) - \ell_q(z_0 \mid h)]\,. \tag{22}$$

A proposal passes the additional filter when $R_q \geq \rho$; the diagnostic uses $\rho = 0.10$. A rejected proposal triggers ordinary fallback with-

out further candidate search. These are joint token-path probabilities, potentially including punctuation and varying token counts, not length-normalized scores or probabilities renormalized over a set of semantic alternatives.

In a post hoc replay of the failed recipe case, `gate_0004`, the decision at word 82 proposed *corn* instead of the ordinary continuation *potato*, before *starch*. The recorded path probabilities were 0.373857 for *potato* and 0.155847 for *corn*, yielding $R_q = 0.416862$. The proposal therefore passed the 0.10 threshold. This illustrates an admissible swap under the generator-relative rule: the generator assigns appreciable probability to the alternative at the same controlled prefix. It does not establish that the ingredients are interchangeable in every recipe. Under a task allowing ingredient variation, this is a candidate adaptation; under the original instruction to preserve the source's facts, it remains a content change requiring explicit review. Probability measures plausibility under q, not factual correctness, safety, or semantic equivalence.

The guarded replay reached EOS at 224 words with 168 unique words and 11 accepted substitutions, compared with 225 words and 175 unique words for ordinary regeneration. The unguarded controlled replay was manually stopped after 332 words of repetitive output. Ordinary continuation from the saved prefix through word 48 also reached EOS at 225 words. These single-case diagnostics motivate the filter and expose the distinction between avoiding degeneration and preserving information. They neither establish corpus-level fidelity nor modify the frozen pilot results above; the guard and adaptation criterion require separate prospective evaluation.

F.5 Three-Stage Algorithm: Controlled Generation, S/W/C, and Punctuation

We formalize the intervention as $x^G = G(x, p)$, $x^S = S(x^G; H)$, and $x^P = P(x^S; H, \mathcal{G})$, where p is the rewriting prompt, H is a training-human reference pool and $\mathcal{G}$ is the discovery-selected feature gate. We propose the specific prefix-feedback controller and gated punctuation search below; generic constrained decoding, sentence editing and punctuation conversion are not themselves claimed as new. S/W/C denotes sentence/word/character statistics. In the present implementation, this second stage edits sentence boundaries to control words and characters per sentence; it does not independently optimize word length. All three stages are heuristic searches with explicit failure or partial-attainment outcomes. Neither likelihood admissibility nor metric preservation guarantees linguistic or source-content non-degradation.

Reference policy and access. The generator is Qwen2.5-1.5B-Instruct (Qwen et al., 2024; Qwen Team, 2024c). Prefix references use same-genre human text at the same absolute alphabetic-word position and require at least 15 surviving reference prefixes. For downstream editing, the original human pool contains up to 30 same-genre training documents selected by proximity in log word count, within a factor of 1.25 of treated length, with at least 15 references. The sensitivity study keeps the parent pools fixed and requires every pool member to satisfy the length condition in every arm. Targets are not estimated from the held-out human ROC sample. Discovery uses the frozen detector's feature contributions, but individual candidate edits are ranked by language-model scores and target progress, not by repeated detector-score minimization.

F.5.1 Controlled Generation

Let $w_{1:n}$ be committed alphabetic words, V_n the number of distinct lowercase forms, and $k_n = V_n/n$. At each supported position, human prefix quantiles $Q_{.2}(n), Q_{.4}(n), Q_{.6}(n)$ define a bidirectional controller. Below $Q_{.2}$ it requests a novel word; above $Q_{.6}$ it requests a previously observed word. Hysteresis keeps an active mode until $Q_{.4}$ is reached. Control is inactive during the initial 15% of source-word length and when reference support is insufficient. This avoids imposing short-prefix or extrapolated targets.

The ordinary greedy continuation first supplies a complete word. If its novelty already agrees with the requested mode, it is retained. Otherwise a best-first search explores token paths from an anchor preserving the committed text. Paths are ordered by cumulative negative log probability. Word-forming tokens must

exceed an absolute probability floor

$$\theta(q) = .005 + \frac{.095}{1 + \exp(4(q-1))}, \qquad (23)$$

where q is distance from $Q_{.4}$ in the requested direction, normalized by the corresponding quantile width with floor $1/n$. Larger deviations admit more alternatives. A greedy boundary-confirmation token may fall below this floor only when it closes an already spelled word without changing its letters. The search returns the first novelty-compatible completed path; it does not optimize a global text objective.

The additional guard compares the proposed and ordinary paths from the same anchor through their word-ending tokens using R_q from Section F.4. Only proposals with $R_q \geq \rho$ are accepted. The development setting is $\rho = .05$, compared with .10 and an unguarded arm. These are joint path probabilities, not absolute word probabilities; tokens spanning the word ending can include punctuation. Lookahead beyond the ending token is excluded from the ratio. Rejection immediately selects ordinary fallback rather than restarting search for another ratio-compatible word.

Each word search is limited to 500 completed candidates and interrupted after an expansion reaches 500 expanded nodes or 5,000 queued branches. Because the queue is checked after expansion, 5,000 is a trigger rather than a strict peak-memory bound. Operational-budget fallback is distinguished from exhaustive failure to find an alternative. Generation stops unsuccessfully on three consecutive identical word blocks of length at least 12, or the unfinished-output length guard at twice source length. These safeguards bound specific failure modes; they neither detect every near-repetition nor guarantee EOS. The committed-word ledger separates irreversible output from tentative token lookahead, and the ratio guard limits departures from the ordinary continuation without imposing a blanket noun lock.

Algorithm 2 Guarded prefix-feedback generation

Require: Prompt, generator q, prefix references, ρ, search budgets

Initialize committed words, seen forms and mode
while EOS has not been reached **do**
  $z_0 \leftarrow$ ordinary continuation to next closed word
  Update mode from committed k_n and references
  $z \leftarrow z_0$
  if mode active and z_0 violates requested novelty **then**
    $z_* \leftarrow$ budgeted best-first word search at θ
    if z_* exists and $R_q(z_*, z_0) \geq \rho$ **then**
      $z \leftarrow z_*$
    else
      Record ordinary fallback and reason
    end if
  end if
  Commit selected word; update counts and checkpoint
  if repetition or length safeguard fires **then**
    return incomplete output and failure status
  end if
end while
return completed text and decision trace

F.5.2 S/W/C Operation

Sentence splitting has established precedent in Split and Rephrase (Narayan et al., 2017); iterative candidate editing with a fluency/meaning objective is developed by Kumar et al. (2020); explicit control of simplification attributes is studied by Martin et al. (2020). We use these as methodological foundations, not as claims to reproduce their trained models or complete algorithms. Our implementation is a restricted, bidirectional boundary-editing procedure with a human-reference band objective, rather than a general simplifier that always shortens text.

For alphabetic word count W, sentence count S, and trimmed sentence strings s, define

$$f_W(x) = W/S, \qquad f_C(x) = \frac{1}{S}\sum_s |\,\mathrm{strip}(s)|. \qquad (24)$$

The latter is not the detector's total-character-count divided by S. For each statistic $m \in \{W, C\}$, the fixed reference pool gives bounds

$[l_m, u_m] = [Q_{.2}, Q_{.6}]$. The objective is

$$D(x) = \sum_m d_m(x),$$
$$d_m(x) = \frac{\max\{l_m - f_m(x), 0, f_m(x) - u_m\}}{\max\{u_m - l_m, 1\}}. \quad (25)$$

Only the means are controlled in C02r/C02s; within-document variances are not targets. The ordered lowercase alphabetic-word sequence and each word's length are invariant, preserving unique-word share and alphabetic characters-per-word statistics. Capitalization at newly introduced boundaries may change.

Proposals split at commas or semicolons joining independently parsed clauses, merge adjacent independent sentences by replacing a period with a semicolon, or combine nonoverlapping split and merge operations. The repair adds subjectless base-form verbal imperatives without subordinate markers and training-compatible imperative merges; its new merges do not cross newlines. Original candidates are retained. The imperative test is a parser heuristic and can misclassify fragments; manual inspection remains necessary. Candidate sentence count must change by $+1$, -1, or 0 for these operations respectively, and remain at least two.

A frozen GPT-2 scorer evaluates the edit plus up to 32 following whitespace-delimited words, conditioned on up to 48 preceding words. Let $\bar{\ell}$ be the mean token log probability of this scored continuation under each alternative, including the fixed right context. Define

$$M(e; x) = \exp\{\min(0, \bar{\ell}(x \oplus e) - \bar{\ell}(x))\}. \quad (26)$$

This length-normalized score is distinct from the generator's joint-path ratio R_q. Unscorable windows, including context overflow, are rejected. A candidate must strictly decrease D and satisfy

$$M \geq b(q), \quad b(q) = .05 + \frac{.90}{1 + \exp(4(q-1))}, \quad (27)$$

with $q = \max_m d_m(x)$. The exploratory repair uses $b_{\text{repair}}(q) = \max\{.05, b(q)/2\}$. Among eligible edits, larger M takes priority, followed by objective reduction and a deterministic tie-break. The summed objective allows one dimension to worsen if total distance improves; it is not monotone for each feature.

Algorithm 3 Reference-band S/W/C editing

Require: Text x, fixed bands, proposal inventory, bar b

Save word signature; initialize visited texts
while $D(x) > 0$ **do**
  Enumerate split, merge and boundary-shift candidates
  Remove visited texts and signature/count violations
  Keep strict D improvements with $M \geq b$
  if no eligible candidate remains **then**
    return x with partial-attainment status
  end if
  Apply candidate maximizing $(M, \text{loss reduction})$
  Record edit and mark resulting text visited
end while
return x with both bands attained

Boundary shifts cannot change W/S when W and S are fixed, but can affect the character statistic. Moreover, an attainable WPS band must contain W/s for an integer $s \geq 2$. This necessary condition does not establish that grammatical proposals reach that count or that both bands are jointly reachable. Candidate exhaustion is therefore reported separately from successful control.

F.5.3 Punctuation Operation

Our punctuation procedure combines a discovery-selected gate, training-human targets and constrained local vertex search. A vertex is a concrete span replacement, not an arbitrary substitution of punctuation identities. The baseline inventory removes or introduces eligible commas, converts colons through connective wording, and edits hyphenated compounds supported by human training text. Nontrivial marks are processed before commas; direct exchanges between nontrivial marks are excluded. Semicolon and dash counts remain protected in the frozen pilot gate. The original pass also preserves each of the counts of periods, question marks and exclamation marks.

For count $c_m(x)$ and exposure $E_d(x) \in \{C(x), S(x)\}$, let $r_{md}(x) = c_m(x)/E_d(x)$. If an exact count t_m is assigned, $L_m(x) = |c_m(x) -$

$t_m|$. Otherwise,

$$L_m(x) = \sum_{d \in \mathcal{G}_m} L_{md}(x),$$

$$L_{md}(x) = \frac{\max\{l_{md} - r_{md}(x), 0, r_{md}(x) - u_{md}\}}{\max\{u_{md} - l_{md}, 1/E_d(x_0)\}}, \tag{28}$$

where x_0 is the input to the punctuation pass. The local bounds are training-human Q20–Q60 values. Exact targets use the genre adjustment described above: when a local band lies beyond the artificial genre-mean interval in the training-AI direction, a distance-dependent probability $p_{\min} + (1 - p_{\min})/(1 + \delta)$ triggers the adjustment. Here $p_{\min} = .10$ and δ is the separation between the local band and genre interval divided by $\max\{\sigma_g, 10^{-6}|\mu_g|, 10^{-12}\}$, where σ_g is the bootstrap standard deviation of the human mean. The maximum trigger probability across selected denominators is used. Stochastic rounding of genre mean times exposure produces t_m, with a fixed case/mark seed. Exact-target attainment and local-band compatibility are different outcomes; exact targets can override already satisfied local bands.

The original punctuation pass ranks by M but imposes no numerical M acceptance floor. Its preservation rules permit incidental comma changes, lock other nontrivial marks, preserve sentence count, and bound regression from the input sentence-mean distances with raw-distance tolerances of 0.5 words per sentence and 3.0 characters per sentence. Hyphen edits can change alphabetic tokenization, so this stage does not inherit the S/W/C word-signature invariant. After sentence repair we recompute exposure-dependent targets with the same random keys and rerun punctuation only when sentence-stage text changed.

Explicit extensions. The exclamation ablation follows the original punctuation pass. It swaps standalone sentence-final **!** and **.**, excludes mixed terminal punctuation and selected abbreviation/initial cases, and preserves each sentence's alphabetic-word membership as well as W, S, C. It targets local training-human bands without the genre adjustment, and requires $M \geq b_{\text{repair}}(L_!)$. It intentionally relaxes the earlier period/exclamation count locks. A subsequent hyphen-only diagnostic broadens compounds across training genres and recognizes quantity-plus-unit recipe markers, while protecting numeric strings and other punctuation counts. It uses the original punctuation ranking without a likelihood floor. A marker removed in this way is replaced by a space: ingredient-list readability is not guaranteed by numerical preservation and requires review.

Algorithm 4 Discovery-gated punctuation vertex search

Require: x_0, fixed gate, reference bands and assigned targets

$x \leftarrow x_0$; visited $\leftarrow \{x\}$; done $\leftarrow \emptyset$
for active marks m, with comma last **do**
  while $L_m(x) > 0$ **do**
    Enumerate span replacements for m
    Reject visited candidates and preservation violations
    Reject sentence-mean regression beyond tolerance
    Reject loss of an attained target in done
    Keep strict L_m improvements scorable by GPT-2
    if no eligible candidate remains **then**
      Record candidate exhaustion; **break**
    end if
    Apply largest-M candidate; tie-break by loss reduction
    Update visited texts and edit trace
  end while
  if $L_m(x) = 0$ **then**
    Add m to done
  end if
end for
return edited text, per-mark losses and audit

F.6 Guard Sensitivity and Subsequent Discovery Repairs

The following results are separate from the 30-case C02r evaluation above. C02s regenerates all 54 discovery sources under four arms with common safeguards. Ordinary, .10 and .05 arms complete 54/54 cases; the unguarded controller completes 53/54. Mean accepted substitutions are 0, 3.91, 5.83 and 10.52, respectively. Common-prefix band attainment is .3842, .4538, .5037 and .6465. Thus the guard trades some control strength for generator-

Condition	Score	AUC	$TPR_{\leq 5\%}$
Original AI	—	.8800	40.74
Ordinary regeneration	—	.8479	40.74
Unguarded final	—	.6316	18.52
.10 final	.4936	.6459	14.81
+ S/W/C repair + punct.	.4791	.6210	7.41
+ exclamation operation	.4689	.5775	7.41

Table 12: Same 27 discovery sources. Lower scores, AUC and TPR indicate reduced detection in the fixed AI-positive orientation. TPR is percent at achieved human FPR 4.92%. Dashes denote scores not tabulated here. AUC and TPR have no uncertainty intervals in this analysis.

relative admissibility; the observed completion difference is one case, not a guarantee of stability.

Downstream comparisons retain 27 sources completed and reference-supported in every arm, including the unguarded arm. The other 27 sources are excluded from this conditional analysis, not counted as successful attacks. The ROC reference consists of 106 held-out humans, weighted so that each genre's total weight matches its frequency among these 27 sources. These human documents were not used to set edit targets; they are reused for descriptive ROC estimation and are not an independent deployment calibration set. Discovery sources and observed outcomes informed the repairs.

Diagnosing incomplete S/W/C control. Before repair, 16/27 .05 texts meet WPS targets and 12/27 meet both means. Punctuation pushes none of the 16 out of the WPS band. All 11 WPS misses admit an integer sentence count; four recipe cases generate no original candidates, and 29 improving split/merge candidates across other endpoints fail the sentence-score threshold. Expanded proposals alone repair one case; lowering the bar alone repairs six; their combination also repairs six while reducing mean normalized WPS distance on the 11 stalled cases from .6884 for bar-only to .3312. Combining repaired endpoints with untouched cases yields 22/27 WPS and 18/27 joint attainment before punctuation is rerun. This is an endpoint-repair ablation, not a fresh end-to-end run of the revised sentence algorithm.

Detection after repair. The S/W/C repair plus rerun punctuation decreases mean score by .01456 (paired 95% interval $[-.03426, .00077]$). Adding exclamation editing yields a total decrease of .02474 ($[-.04454, -.00765]$), with decreases on 11/27 sources. Four single-character !→. changes, each in a different document, reduce AUC further from .6210 to .5775. At achieved FPR 9.63%, TPR changes from 33.33% to 25.93% and then 22.22%; at 19.18% it changes from 40.74% to 37.04% and then 33.33%. The requested 1% budget achieves zero empirical FPR in this human sample and must not be described as measurement at exactly 1% FPR.

Residual contributions and hyphen coverage. After exclamation editing, leading signed gaps relative to same-genre held-out humans include long-word share (.00427), hyphen rate per sentence (.00362), relative hyphen rate (.00305), period rate per sentence (.00305), and WPS (.00278). These use a different reference baseline from the original matched-training-human ranking. Tree-path contributions sum with the fixed forest bias to the predicted score; they are not SHAP values or independent causal effects. Changes to other features can alter a feature's attribution without changing its raw value.

The later hyphen extension changes only one recipe: seven marker removals reduce its hyphen count from 11 to 4, while target attainment remains 18/27 overall. Mean score falls from .468876 to .467741; the paired interval for the change is $[-.003406, 0]$ to rounding. No ROC or contribution update for this last extension is included in Figures 17–18. This localized result should not be conflated with the preceding cohort-wide comparison.

Reproducibility and interpretation. Figures are regenerated from repository CSVs and exported as both PDF and PNG with a retained plotting script and source data. Score intervals resample the 27 cases jointly across conditions with 3,000 bootstrap replicates; they condition on fixed references and the selected cohort. Target attainment, EOS completion and detector attenuation are reported separately from quality-qualified success. Human review fields have not been completed. These adaptive discovery results motivate prospective validation. The subsequent retraining experiment is reported below; semantic fidelity and transfer to other detectors remain separate questions.

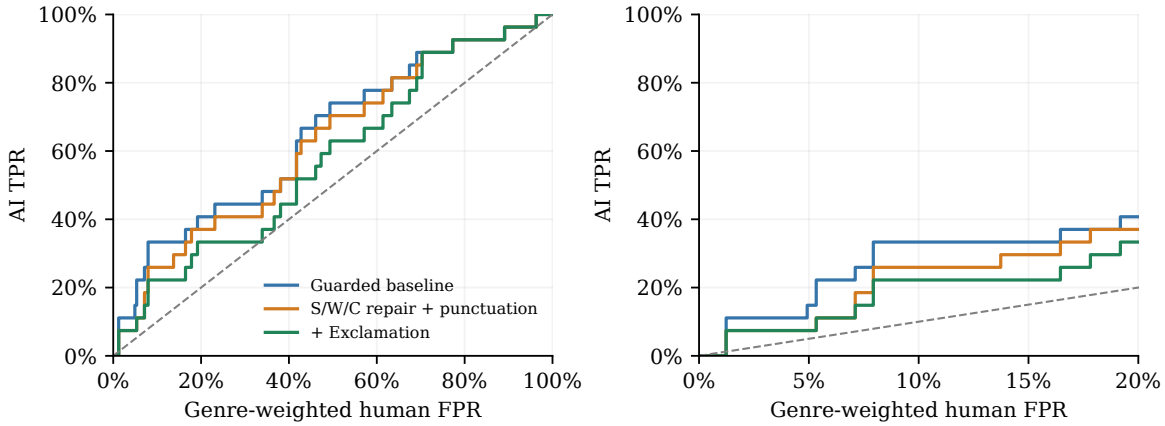


Figure 17: Empirical ROC staircases from saved individual threshold points, before and after discovery repairs. Both panels use the same 27 AI sources and 106 genre-weighted humans. The right panel expands the low-FPR region. The diagonal is a chance reference, not an equivalence test; AUC .5775 does not establish neutralization.

F.7 Long-Word Share: Constrained Lexical Substitution

The fourth feature category is lexical length, following lexical diversity, sentence structure and punctuation. We treat it as a separate operation on the repaired text, avoiding simultaneous control of two lexical objectives during generation. Candidate generation, contextual selection and ranking follow the established lexical-simplification framework (Paetzold and Specia, 2015); context-sensitive substitution has earlier precedent in Biran et al. (2011). We use WordNet synsets (Miller, 1995) and a simplified, Lesk-inspired gloss-overlap heuristic (Lesk, 1986). We do not reproduce those systems or claim lexical substitution or sense disambiguation as new. Our implementation adds a bidirectional human-band objective, exact lexical-diversity preservation, compensating pairs and two distinct likelihood gates.

Let W, V, L count alphabetic tokens, distinct lowercase forms and tokens of at least seven characters. The objective is distance of L/W from the fixed training-human Q20–Q60 band, divided by its width (floored at 10^{-12}). For a single occurrence replacement $a \to b$, the predicted type-count change is

$$\Delta V = \mathbb{1}[b \notin \mathcal{V}(x)] - \mathbb{1}[c_x(a) = 1]. \qquad (29)$$

Candidates fall into four groups: previously seen/new forms crossed with short/long forms. Single edits require $\Delta V = 0$; pairs combine $+1$ and -1 proposals at distinct positions. Since paired edits can interact, we recount the combined text rather than assuming these changes add exactly. Both shortening and lengthening are permitted when they approach the assigned band.

Context and preservation checks. We consider nouns, verbs, adjectives and adverbs, excluding named entities and stopwords. A sense is retained only when it uniquely maximizes a positive overlap between sentence-context lemmas and its definition/example lemmas. Tied and zero-overlap senses are rejected. Candidates are ASCII alphabetic synset members: bare lemmas are used only for uninflected sources; otherwise compatible inflected forms must already occur in the document. Reparsing must preserve the edited tokens' coarse and fine POS tags. The combined output must preserve W, V, the ordered sentence word-count vector and nonalphabetic material. Relative to the operation input, mean sentence-character drift is at most 3 and $|C_0/C - 1| \leq .02$, bounding relative punctuation-rate drift when counts are unchanged. These are operational checks, not a semantic guarantee.

Likelihood gates and deterministic search. A local GPT-2 gate compares joint probabilities of the replacement and original word paths given up to 48 preceding whitespace-delimited words. Tokens crossing the word boundary are included. Each member of a pair is checked again with the other substitution present. Sepa-

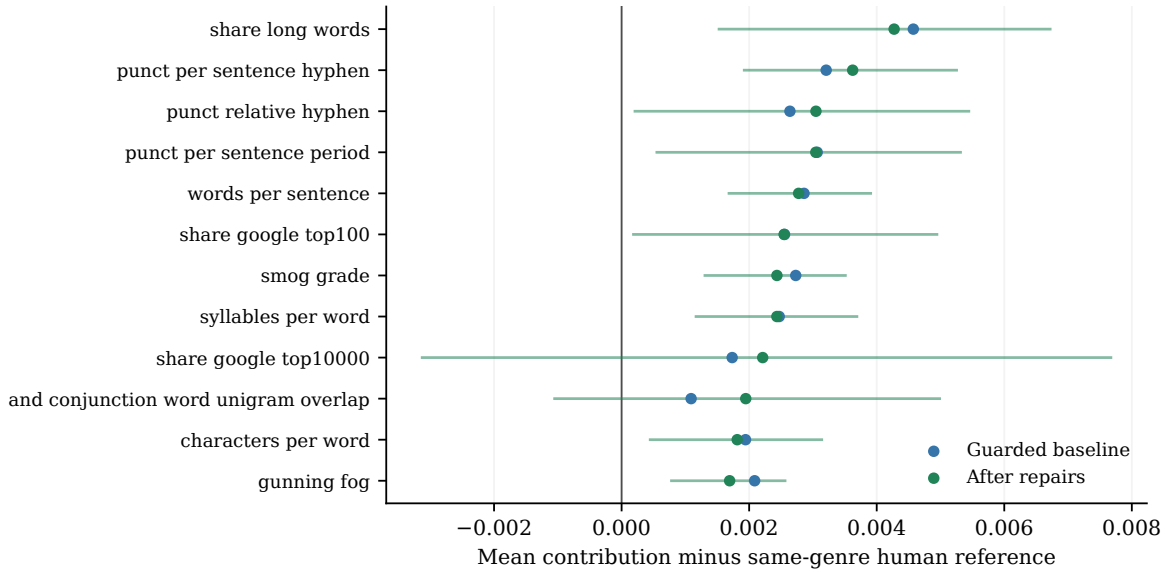


Figure 18: Largest 12 remaining mean signed contribution gaps after the exclamation operation, compared with the preceding .05 baseline. Intervals are pointwise 95% case-bootstrap intervals conditional on fixed same-genre human references. Feature selection and repeated development are not covered by these intervals.

rately, the combined edit must satisfy the mean-token contextual score $M \geq .95$ defined in Section F.5.2. Local ratio floors $\lambda \in \{.20, .10, .05\}$ are independent of this contextual floor. Search uses position-round-robin screening of at most 128 proposals per step, pools of at most 24 candidates per type-count sign, at most 24 combined candidates for contextual evaluation, and 12 accepted-step iterations. Ranking prioritizes the minimum local ratio, then target gain and fewer edits, with stable enumeration resolving ties. Determinism is conditional on frozen resources and numerical runtime.

Development results and rejected alternatives. An earlier broad proposal inventory accepted 13 operations (eight pairs, five singles), increasing band attainment from 5/27 to 8/27 and reducing rounded AUC from .571 to .563. Inspection exposed meaning-changing substitutions, including *buttermilk*→*butter*; these results are not adopted as quality-qualified successes. Stricter sense and individual-likelihood screening accepted only two single substitutions at local floor .95, with rounded AUC .569. This revision addresses the failure of a high aggregate contextual likelihood to certify each individual substitution.

The subsequent floor sweep starts each arm independently from the same 27 hyphen-repaired texts. Floors .20 and .10 produce the same five accepted single edits, reach 6/27 bands and yield mean normalized distance .842467. Floor .05 adds one substitution, also reaches 6/27 and yields .839052. Rounded AUCs are .568, .568 and .569, respectively, versus .571 before this operation. For .20/.10, mean score change is −.000601 (paired 95% interval $[-.001417, -.000058]$); for .05 it is −.000496 ($[-.001332, .000108]$). TPR remains 7.41% at achieved FPR 4.92% and 22.22% at 9.63%; at 19.18%, it falls from 33.33% to 29.63%. All arms preserve unique-word share. The marginal benefit motivates ending lexical-length development and retaining the pre-long-word pipeline for the subsequent adaptation experiment.

F.8 Full-Cohort Completion and Resistant Cases

We subsequently complete downstream processing for the remaining 27 discovery sources, starting from their saved completed .05 controlled generations. The existing 27 outputs are preserved. Parent sentence-reference pools are retained where available; otherwise up to 30 same-genre training-human texts are selected by nearest absolute log word-length ratio, with at least 15 required. These reference-policy

Algorithm 5 Type-count-preserving long-word correction

Require: Repaired text x_0, fixed band, local floor λ

$x \leftarrow x_0$; initialize visited texts
for at most 12 iterations **do**
  if $L(x)/W(x)$ is in band **then**
    break
  end if
  Generate sense-compatible candidates in four groups
  Screen individual POS and local likelihood
  Form zero-ΔV singles and compensating pairs
  Recount invariants; retain strict target improvements
  Recheck pair-local ratios and require $M \geq .95$
  if no eligible candidate remains **then**
    break
  end if
  Apply best ranked candidate; record trace and visited text
end for
return x, attainment status and review record

extensions are recorded explicitly and do not reinstate the earlier all-arm eligibility condition. Completion means that each operation ran to its stopping rule, not that all feature targets were attained.

Control strength differs across cohorts. Final controlled-generation uniqueness is above/below/in-band/unsupported for 10/1/10/6 of the new sources versus 5/5/14/3 of the original sources. Mean supported-prefix band attainment is .3480 versus .6536, despite 8.15 versus 3.52 replacements per text. A diagnostic pilot regenerates the five new sources with largest supported upper-band excess, activates downward control at the human median, quarters downward token thresholds and doubles search budgets while retaining the .05 path floor. None reaches the band; four retain the same final uniqueness statistics. Case 47 retains 116 distinct forms in 154 words and only one replacement, despite full reference coverage. This establishes resistance to the tested controller, not impossibility of meaning-preserving rewriting. All resistant cases remain in the final cohort; stronger-pilot outputs are not substituted.

Model	AUC	TPR	Test FPR
Frozen / base replay	.6202	20.59	6.94
Rewritten augmentation	.6587	23.53	6.44
Ordinary augmentation	.6281	17.65	6.38

Table 13: Adaptation-held-out discovery sources ($n = 34$). TPR and realized test FPR are percentages at a nominal 5% validation FPR budget; achieved validation FPRs are 4.87, 4.35 and 4.53%, respectively. All models use their own calibrated threshold.

F.9 Adaptive Camouflage at Twenty-Example Exposure

We freeze a source-level split of all 54 completed texts into 20 adaptation and 34 test examples using deterministic genre allocation and a fixed seed. All variants of an adaptation source remain outside the test set. The 34 sources are held out from detector adaptation, but have appeared in discovery diagnostics; they are not a fresh confirmatory sample. Long-word edits and the stronger five-case pilot are excluded from these inputs.

We compare the frozen FZ118 implementation, exact base-training replay, replay augmented with the 20 rewritten AI examples, and a matched control augmented with ordinary generations of those 20 sources. Additional examples have unit weight. The saved ExtraTrees configuration is retained (600 trees, minimum leaf size 3, square-root feature subsampling and fixed seed); the implementation uses 114 features after the four parenthesis features are excluded from the 118-feature extractor. Base replay reproduces frozen predictions. Human validation data calibrate each model separately using the rule score $> \tau$; 128 held-out test humans estimate realized FPR, weighted to the 34-source genre mix. We use 2,000 within-genre bootstrap resamples, paired across models. Intervals condition on this adaptation set, trained models, human calibration thresholds and observed genre composition.

Adaptation recovers AUC by .038539 over the frozen model (paired 95% interval [.012641,.065742]) and .030588 over ordinary augmentation ([.004621,.057285]). Thus expo-

sure learns some residual signal; recovery is not zero. Nevertheless, at the nominal 5% operating point it detects only 8/34 rewritten examples versus 7/34 for the frozen model. The adapted model misses 76.47% while rejecting 6.44% of weighted test humans. At nominal 10% and 20%, adapted TPR is 29.41% and 44.12%, with realized test FPR 10.71% and 21.35%. These are substantial operational limitations at the tested exposure budget, not a universal commercial-viability determination.

F.10 Phase-One Conclusion and Validation Status

We conclude Phase One's algorithm-development stage with substantial FZ118 attenuation and limited detection sensitivity after the tested defender update. Four feature categories have explicit, reproducible operators: lexical diversity through controlled generation, sentence statistics through S/W/C edits, punctuation through local vertex search, and lexical length through constrained substitution. The long-word extension is documented as a marginal-benefit optional operation, not credited to the adaptation results. Stage-specific invariants preserve selected measurements; they do not imply that every output reaches every target or that feature contributions are independent causal effects.

The three criteria are our proposed operational distinction between substantive evasion and changes cheaply reversed by preprocessing or modest labeled exposure; they are not presented as an established community standard. Under this framing, Phase One meets the intended requirements: ordinary-language operations avoid encoding-based evasion, local guards and author inspection support the qualitative non-degradation criterion relative to the generated source, and the twenty-example adaptation test leaves detection sensitivity limited despite measurable recovery. We therefore conclude Phase One successfully at the tested scope. This conclusion allows resistant cases and incomplete feature-band attainment: the objective is a distribution-level shift, not universal per-document acceptance. It does not establish resistance to larger adaptation budgets or all detectors, and it does not replace deployment-specific utility analysis.

F.11 POS Extension: Frozen Detector and Evaluation Protocol

We next retain the completed 54-source discovery cohort and extend the detector to a project-defined FZ186 configuration. This adds 68 features to the 118-feature extractor: for each of 17 Universal POS tags, we compute its share of non-space tokens and the mean, population standard deviation, and maximum of its counts across sentences. Punctuation is included in the token denominator; sentences with non-space tokens contribute even when the target tag count is zero. This is an explicit project extension of the feature-based approach of Fröhling and Zubiaga (2021), not a verified reconstruction of the original authors' complete POS inventory. The operational detector uses all 186 features, including parentheses. A 182-feature matched-exclusion configuration is a diagnostic ablation, not the detector used below.

The ExtraTrees detector is trained on original RAID training data and frozen throughout the POS interventions. Steps 5–8 follow the completed surface pipeline; numbering continues the feature-family program, without implying that the optional Phase-One long-word operation contributed to the FZ186 baseline. Steps 5–6 are implemented in C03c, steps 7–7.5 in C03d, and step 8 and the final FZ186 robustness analysis in C03e.

Let $c_j(x)$ denote an exact forest-path contribution to the AI-class probability, so that $p_{\mathrm{AI}}(x) = b + \sum_j c_j(x)$ for a fixed forest bias b. For genre g, let r_{gj} be the mean contribution on the fixed test-human reference of that genre. We report the mean gap $G_j = n^{-1} \sum_i [c_j(x_i) - r_{g_i j}]$ and block sums $G_B = \sum_{j \in B} G_j$. References remain fixed after editing. Consequently, a paired gap change equals the paired contribution change; it is not a raw POS-frequency difference. Human-reference gaps are descriptive, while edit selection has access to the frozen detector. Probability reduction and ROC attenuation are secondary to targeted contribution changes.

Operating thresholds are fixed using genre-weighted validation-human scores. For requested FPR α, we select the lowest candidate threshold whose weighted calibration FPR is at most α, with detection defined by strict score

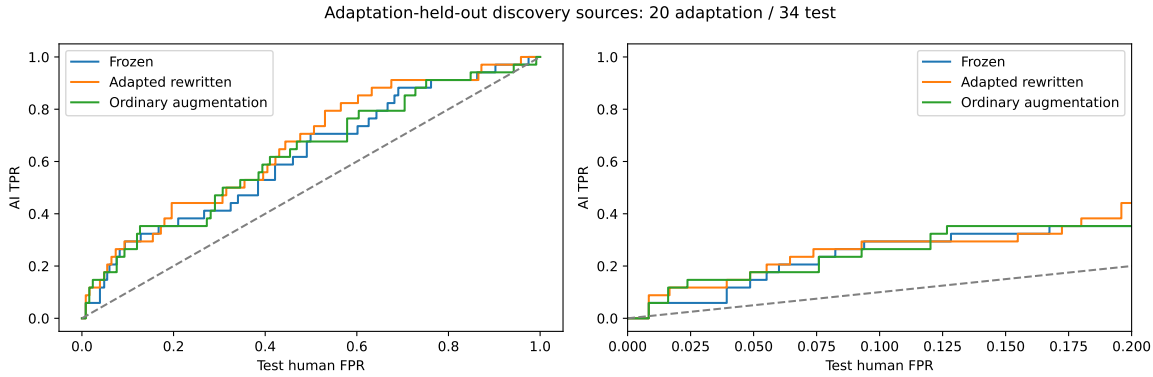


Figure 19: Twenty-example adaptation: empirical ROC curves on 34 adaptation-held-out discovery sources and 128 genre-weighted test humans. The right panel expands the low-FPR region. The notebook exports this figure directly as `C02w_adaptive_ROC.pdf`; Table 13 reports validation-calibrated operating points.

exceedance. Test-human FPR is measured separately and is unchanged between frozen intervention stages. The same 54 AI documents, including unchanged documents, enter each stage comparison. These are exploratory, detector-aware results on development sources rather than held-out attack generalization. A separate twenty-example adaptation experiment is reported after the frozen-FZ186 endpoint.

F.12 Step 5: Targeted X-Tag Cleanup

An audit decomposes X-tagged tokens by their surface roles rather than treating X as a homogeneous lexical category. In the fixed AI texts, the principal concentrations are recipe list labels and angle prefixes: 52 list-item-tagged tokens occur in seven documents, and 13 angle markers occur in two. Two editorial `[edit]` residues occur in one book text. The recipe greater-than markers occur in one of nine AI recipe documents and none of 21 test-human recipes; this is a sample observation, not a universal authorship rule. Foreign words, URLs, brackets, and other fragments are inspected separately, but the implemented operation does not inject foreign expressions or delete all such tokens indiscriminately.

Three deterministic operators address identified structures. First, remove bracketed editorial `[edit]` labels while retaining ordinary uses of the word. Second, when the original recipe has at least three line-leading greater-than prefixes and their count matches the current text, convert those prefixes to hyphens. Third, identify consecutive numbered instruction runs of at least three items in the original source, allowing excerpts to begin after item one. Token-sequence alignment and nearby lexical anchors locate the corresponding current labels, including labels whose punctuation or line breaks changed during earlier processing. Replace confirmed labels with `Step n:`, preserving their number and order; skip ambiguous alignments. This source alignment distinguishes instruction labels from ingredient quantities and other numerals.

The edit-residue, angle-prefix, and step-label operators affect respectively one document/two spans, one document/12 spans, and seven documents/37 spans. Their cumulative arm changes nine of 54 documents. The summed X contribution gap falls from .026349 to .003803, with paired change −.022546 and reported document-bootstrap 95% interval [−.032593, −.011174]. The gain is accompanied by punctuation displacement: the new step labels increase colon contributions.

Step 5.5: punctuation restoration. We therefore rerun the existing colon controller followed by the comma controller on the step-5 output, retaining frozen references and earlier sentence-control safeguards. The colon pass additionally permits the colon in confirmed recipe `Step n:` labels to become a space. This cleanup reduces the relative-colon gap from .005067 to −.001274 and the per-sentence-colon gap from .003249 to −.001097. The X block remains at .003809, approximately 85.5% below its pre-step-5 gap. Thus the cleanup retains the targeted effect while addressing a concrete induced feature increase. AUC falls from .800940

before step 5 to .731318 after step 5.5; mean AI score falls from .540866 to .510462.

F.13 Step 6: Deterministic ADV Substitution

The subsequent audit finds lower ADV frequency and per-sentence counts in treated AI than in the genre-weighted human reference. For example, token shares are .026415 versus .034692, and mean sentence maxima are 1.925926 versus 2.800129. Positive ADV contributions therefore do not imply that adverbs should be removed. We generate localized paraphrase candidates that increase the measured ADV count, then evaluate their actual contribution effects.

Candidate retrieval and mechanical screening. We use the English PPDB 2.0 L/all release, which supplies automatically extracted paraphrases and entailment-relation annotations (Pavlick et al., 2015). We retain observed, contiguous alphabetic spans of one to eight words and replacement spans of one to eight words whose stored relation is `Equivalence`. Identity pairs are excluded; duplicate pairs retain the higher PPDB score. Source spans do not cross sentence or punctuation boundaries and exclude named-entity spans and NUM/PROPN tokens. Only stored directions are used. Each occurrence replacement is evaluated independently against the step-5.5 document. Full-document retagging must increase the net ADV count and preserve the ordered NUM/PROPN token sequence, explicit dependency-tagged negation, punctuation sequence, and sentence count. These checks are operational constraints, not semantic or grammatical guarantees.

The scan processes 18,206,988 PPDB rows and yields 10,557 observed phrase pairs and 15,015 contextual candidates. Of these, 14,530 do not increase ADV and 14 fail protected-structure checks, leaving 471 candidates across 49 documents. A preliminary phrase-to-single-word search had zero ADV coverage; broadening the phrase inventory resolves that coverage failure. The final selection uses likelihood gates rather than a manually curated list of approved substitutions.

Broad and local contextual likelihood gates. For an original span a and replacement a', take up to 48 preceding whitespace-delimited words L and 48 following words R, crossing sentence boundaries when available. A frozen GPT-2 scorer, with the same pinned revision as the punctuation experiment, computes mean causal token log-likelihood on the edited span and following context, conditioned on L. Define

$$\Delta\ell_q = \bar{\ell}_q(a', R \mid L) - \bar{\ell}_q(a, R \mid L). \tag{30}$$

The broad gate requires $\Delta\ell_q \geq -1.0$ nats/token. A second gate uses the same preceding context but scores only the span plus the first eight following model tokens, also requiring a difference of at least -1.0. Each side is tokenized separately; boundary-straddling tokens are included, and available following context is used at document ends. The local gate limits dilution of an edit-local penalty by a long unchanged suffix. These permissive thresholds are development choices, not calibrated grammaticality probabilities. Scoring uses fixed resources, stable enumeration, batching, and cached results; determinism is conditional on the numerical runtime.

The broad gate retains 429 candidates across 48 documents. Of these, 406 reduce the ADV block contribution; 221 across 38 documents also avoid an increase in the combined non-ADV contribution. The local gate retains 204 of the 429 candidates across 43 documents. For final selection, require both likelihood gates, a strictly negative ADV contribution change, and a nonpositive combined non-ADV change (with 10^{-12} numerical tolerance). Rank candidates by the largest ADV decrease, then higher local likelihood, then stable candidate identifier. Apply at most one candidate per document; documents without eligible candidates remain unchanged. Individual non-ADV features may increase even when their combined change is nonpositive.

Accepted one-swap results. The selected arm changes 30 of 54 documents. Across all 54, its mean ADV contribution change is $-.005449$ (document-bootstrap 95% interval $[-.008246, -.003053]$); the total contribution change is $-.009495$ ($[-.013889, -.005641]$). These intervals resample documents after selection and do not account for policy search or estimate held-out performance. The aggre-

Stage	AUC	Mean AI score
Before step 5	.800940	.540866
After step 5.5	.731318	.510462
After step 6	.712270	.500967

Table 14: Sequential frozen-FZ186 evaluation on the same 54 development sources and fixed genre-weighted test humans. A mean score near .5 does not imply indistinguishability: after step 6 AUC remains .712270.

Requested FPR	Test FPR	Step 5.5 detected/54	Step 6 detected/54
.01	.015988	10	9
.05	.038906	16	16
.10	.050044	16	16
.20	.118355	22	21
.30	.233352	30	29
.50	.405241	38	38
1.00	1.000000	54	54

Table 15: Fixed validation-calibrated operating points before and after ADV treatment. The final ADV contribution reduction produces modest threshold-crossing changes, including no change at the requested 5% and 10% budgets.

gate acceptance conditions themselves favor negative changes, so the intervals should not be read as an independent test of the selection rule. The accepted texts were retained by the author as the next experimental baseline; this is not a blinded or quantitative fidelity assessment. PPDB labels and likelihood gates do not establish meaning preservation, and the permissive gate can admit grammatical or tense-changing substitutions.

Remaining evidence after ADV. After step 6, NUM token share (.004195) and ADP sentence maximum (.004079) lead the remaining mean human-adjusted contributions, followed by NUM sentence mean (.003262) and ADP sentence standard deviation (.003216). ADV sentence maximum falls from .004484 to .002536, but the ADV family is not eliminated. These residuals motivate C03d rather than implying that all POS evidence has been removed.

F.14 Steps 7–7.5: ADP and NUM

Step 7 targets the ADP family identified after ADV. For each genre, the controller freezes the ceiling of the human median sentence maximum and protects sentences already at or above that threshold. It searches for an ADP-boosting edit in a below-threshold sentence, then attempts compensating ADP reductions in other below-threshold sentences to restore the incoming token share. Both broad and local contextual-likelihood floors remain at -1.0 nats/token. Final acceptance requires improvement in the ADP contribution block, non-increasing combined non-ADP contribution, a non-decreasing sentence maximum, and increased sentence-level ADP standard deviation. Compensation is best-effort rather than exact. The accepted run changes 15/54 documents; ten accepted documents receive compensation edits, while 25 cases have no reachable boost under the recorded bounds and 14 have no final accepted candidate.

Step 7.5 then addresses NUM. An initial bidirectional cardinal-integer pilot contributes three accepted conversions. The final deterministic pass converts years recognized by the implemented rules, including linked range endpoints, to long forms such as *two thousand nineteen*. Fractions are audited but not treated: in the tested examples, spelling them out does not reduce NUM count and lowers NUM token share. The recognizer leaves an unclassified `2003` in `gate_0051` unchanged, so the operation should not be described as exhaustive year conversion. Unlike the ADP search, the final year pass is not selected by likelihood or detector improvement.

Across the fixed 54 documents, AUROC falls from .712270 after ADV to .696681 after ADP and .651342 after NUM; mean AI score falls from .500967 to .493817 and .478065, respectively. NUM leaves the displayed top-30 residual list, while long-word share and unique-word share again become leading positive features.

F.15 Step 8: Lexical Micro-Substitution

Post-NUM audits compare each document with the same-genre human reference distribution. Mean long-word share is .287766 versus a genre-matched human mean of .251386, with 45/54 cases above the same-genre median. Mean unique-word share is .595792 versus .576374, with 36/54 above the same-genre median. Step 8 therefore treats lexical length as the primary feature and lexical diversity as a conditional secondary feature.

Candidate proposals are one-word WordNet

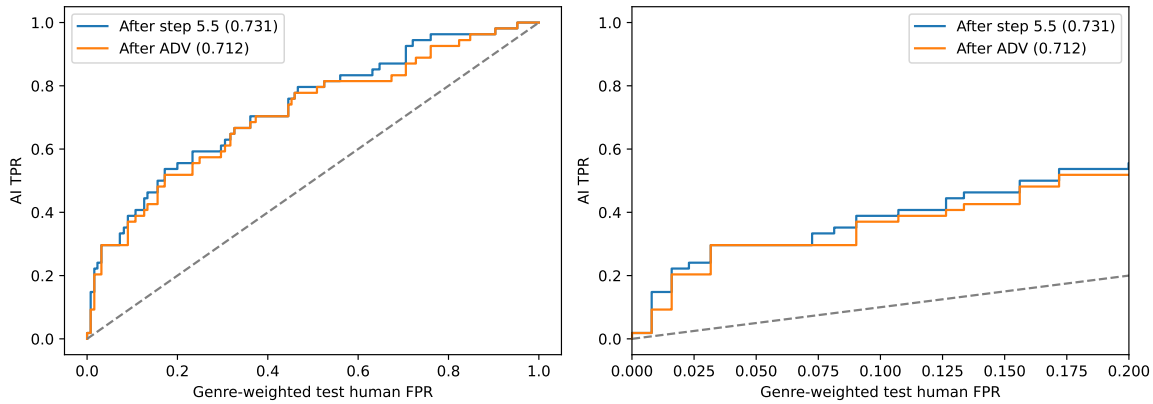


Figure 20: Frozen-FZ186 ROC before and after the accepted ADV pass, with a low-FPR enlargement. Both stages use the same test-human weights; validation-calibrated operating points are reported separately in Table 15.

synonyms for nouns, verbs, adjectives and adverbs, excluding stopwords, named entities and protected proper nouns. A replacement must reparse as one alphabetic token with the same coarse and fine POS tag. Frozen GPT-2 word-level and continuation likelihood ratios provide permissive local gates. Routing is case-specific. If both shares are already at or below the genre median, the text is untouched. If long-word share is high, the search proposes long-to-short replacements; when uniqueness is also high, replacements that merge a source type into a type already present in the document receive priority. When uniqueness is already satisfied, a long-word edit must preserve the distinct-type count exactly, including by replacing all occurrences of a repeated source type when needed. Unique-only cases seek a type merge that preserves the long-word count.

The bounded search edits 27/54 documents using 52 accepted operations and 57 token replacements. Afterward, 9/54 cases are at or below their same-genre long-word median, 19/54 are at or below their unique-word median, and 5/54 satisfy both. Forty-eight cases terminate with no likelihood/POS-valid candidate, five were already below both medians, and one has no lexical proposal. Thus the operation is a conservative partial correction rather than full lexical matching. Its detector effect is correspondingly small: mean AI score changes from .478065 after NUM to .476988 and AUROC from .651342 to .650639.

Endpoint and residual evidence. The full staged intervention reduces frozen-FZ186 AUROC by .225169 relative to the original AI texts and by .150301 relative to the post-FZ118 baseline. The calibration-threshold staircase shows that this is not only an averaging effect: the current texts are substantially less detectable than both earlier stages across much of the low-realized-FPR region (figure 22). The final contribution ranking, however, no longer isolates one dominant AI-like feature family. Positive residuals are distributed across long-word share, ADJ sentence-count dispersion, unique-word share, PROPN sentence mean, CCONJ token share, and smaller readability, punctuation, overlap and POS features. Step 8 itself moves contributions only slightly. We therefore stop FZ186-specific feature engineering at this point and treat the diffuse residual structure as the input to the next, richer detector stage.

Stage	AUC	Mean AI score
Original AI text	.875808	.604191
Post-FZ118 (steps 1–4)	.800940	.540866
Step 6 — ADV	.712270	.500967
Step 7 — ADP	.696681	.493817
Step 7.5 — NUM	.651342	.478065
Step 8 — LEX	.650639	.476988

Table 16: Frozen-FZ186 evaluation across the complete intervention program on the fixed 54 development sources. Original and post-FZ118 rows show the same FZ186 detector applied before the POS-specific stages. The final mean score below .5 is not an equivalence result: AUROC remains above chance.

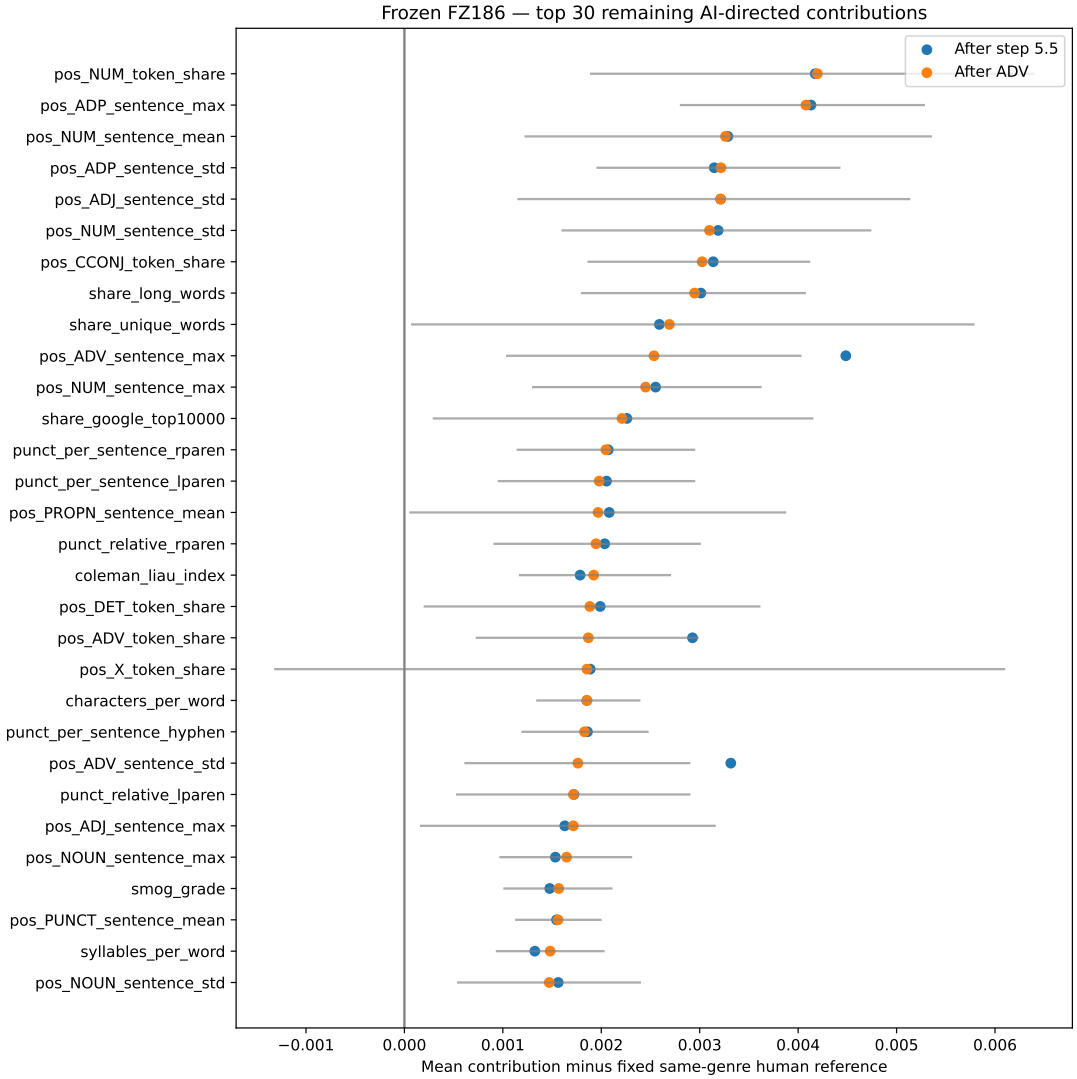


Figure 21: The 30 largest remaining mean human-adjusted feature contributions after ADV treatment, compared with step 5.5. Intervals use genre-stratified document bootstrap resampling with fixed human references and are descriptive after candidate selection.

F.16 FZ186 Adaptive Camouflage at Twenty-Example Exposure

We next test whether the final FZ186 endpoint merely creates an easily learned replacement fingerprint. Using the same deterministic genre-allocation rule and fixed seed as the earlier adaptive experiment, we split the 54 final Step-8 outputs into 20 adaptation and 34 test examples. The adapted detector clones the frozen FZ186 ExtraTrees configuration, retains the original RAID training set, appends the 20 current outputs with AI label 1 and refits. The 34 remaining outputs are not used for fitting. Each detector separately rescored the same frozen human calibration and test pools, and the detailed threshold staircase uses strict score exceedance with detector-specific calibration thresholds.

Detector	AUC	Mean test-AI score	Mean human score
Frozen FZ186	.647886	.478855	.40299
Adapted FZ186 (+20)	.684254	.497615	.41099

Table 17: FZ186 Adaptive Camouflage on the 34 adaptation-held-out final texts. The adapted model sees 20 other final outputs in addition to the unchanged original RAID training set. Human calibration and test documents are rescored separately by each detector.

Twenty-example exposure recovers AUC by .036368. This is measurable learning of residual structure, not zero recovery, but it remains far below the .875808 separation of the frozen FZ186 detector on original AI text. The

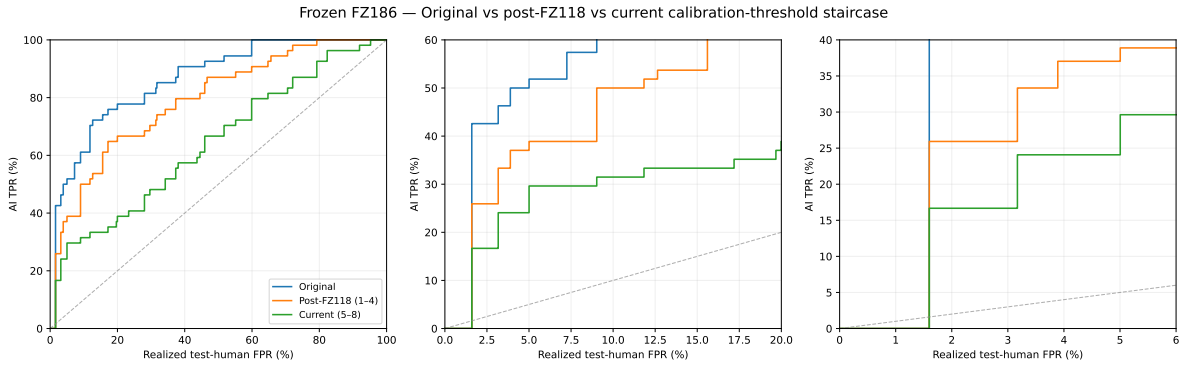


Figure 22: **Frozen-FZ186 attenuation across the full staged intervention.** Calibration-threshold TPR staircases compare the original AI texts, the post-FZ118 baseline after steps 1–4, and the final Step-8 outputs on the same 54 sources. Thresholds sweep the distinct calibration-human scores with strict score exceedance; the horizontal axis is realized genre-weighted test-human FPR. AUROC is computed separately from the full weighted ranking.

calibration-threshold staircase shows localized recovery in the low-FPR region: at some realized false-positive rates the adapted detector detects additional held-out texts, while at other low-FPR thresholds the frozen detector remains as good or better (figure 23). The adapted curve therefore does not uniformly dominate the frozen curve. This experiment supports only a bounded Adaptive Camouflage statement: modest labeled exposure partially restores FZ186 discrimination but does not recover the original separation under this fixed 20/34 split. The 34 sources have appeared in earlier discovery diagnostics, the split is not repeated, and larger adaptation budgets or different detector families remain untested.

F.17 FZ186 Conclusion

The project-defined FZ186 stage ends after Step 8 and the adaptation check. Relative to the original AI texts, the same frozen detector moves from AUC .875808 and mean AI score .604191 to AUC .650639 and mean score .476988. Relative to the already treated post-FZ118 baseline, the POS-era stages reduce AUC from .800940 to .650639. The final residual contributions are diffuse rather than dominated by a single obvious feature family, and additional lexical micro-edits yield diminishing returns. Together with the partial, not complete, twenty-example retraining recovery, this provides the stopping criterion for FZ186 and motivates moving to the richer FZ301 feature family. It does not establish detector-agnostic indistinguishability or resistance to broader retraining.

Stage	AUC	Mean AI score	Median A
Original AI text	.875380	.599424	.
Full FZ118 — post Step 4	.801146	.544962	.
Full FZ186 — post Step 8	.699558	.500605	.
Step 9 — ORG explicitization	.697009	.498378	.
Step 10 — CARDINAL insertion	.671326	.476821	.

Table 18: Frozen-FZ301 evaluation on the same 54 development sources. The first three rows isolate how much of the 301-feature detector remains after the previously completed intervention stages; Steps 9 and 10 are evaluated without retraining the detector.

F.18 FZ301 Extension: Steps 9–10

The next project-defined detector retains the fixed 54-source cohort and expands the frozen feature vector to 301 features, including the richer named-entity statistics used in the C04 manifest. Training remains on the regular RAID training set; the intervention texts are never used to fit the frozen detector. On the same detector, AUROC is .875380 for the original AI texts, .801146 after the full FZ118 steps 1–4, and .699558 after the complete FZ186 Step-8 endpoint. Mean AI score moves from .599424 to .544962 and .500605, respectively. Thus FZ301 is stronger than FZ186 on the final Step-8 texts, but does not restore the original-text separation.

Why ORG is targeted. The post-Step-8 contribution audit shows that low organization-mention statistics remain among the strongest positive AI-directed residuals. In particular,

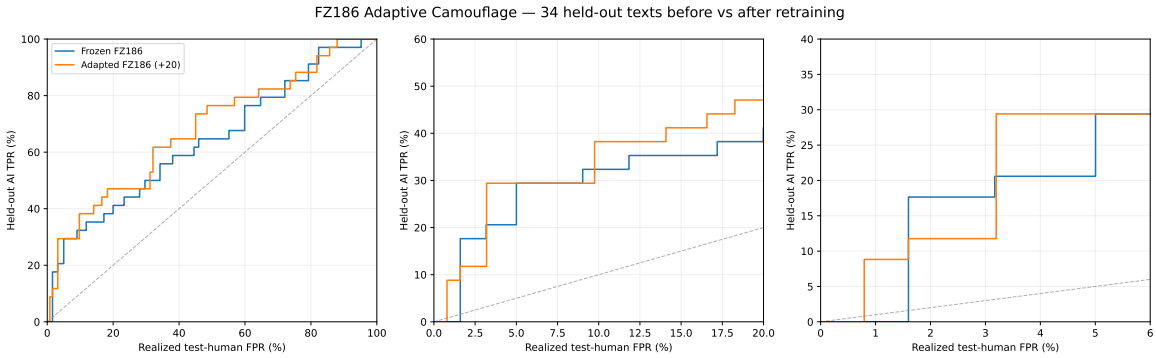


Figure 23: **FZ186 Adaptive Camouflage after twenty-example exposure.** Calibration-threshold TPR staircases compare the frozen FZ186 detector with a detector refit after adding 20 final Step-8 outputs to the unchanged RAID training set. Evaluation uses the remaining 34 final outputs and detector-specific rescoring of the same human calibration and test pools. Recovery is localized rather than uniform across the low-FPR region.

low values of `ner_ORG_sentence_mean`, `ner_ORG_sentence_std`, and `ner_ORG_sentence_max` are associated with positive tree-path contribution under the frozen forest. A first local-reference controller confirms the intended direction but has very low coverage. We therefore use *ORG Referential Explicitization*: a constrained sentence rewrite that repeats an organization already present in the source rather than inventing a new entity. Entity-controlled generation provides precedent for controlling entity appearance (Aggarwal et al., 2023; Dong et al., 2021), and paraphrase-based NER work motivates explicit preservation checks for named-entity spans (Sharma et al., 2023).

The accepted generator is Qwen2.5-1.5B-Instruct (Qwen et al., 2024; Qwen Team, 2024c) with deterministic beam search. Only below-target documents containing an existing ORG are promptable; zero-ORG documents remain unchanged. A candidate must remain exactly one sentence, contain exactly one additional surface occurrence of the selected organization, increase spaCy ORG count, preserve every original named-entity surface and number, introduce no new entity identity, preserve original verb lemmas and negation, retain at least 90% of source content lemmas, and remain within a bounded length ratio. Surviving candidates are ranked by the same pinned causal-LM likelihood scorer used in earlier interventions. Frozen FZ301 score is not part of candidate ranking.

Qwen generates 391 candidates over the 12 below-target documents that contain an ORG. Thirty-eight pass all guards, covering 8/12 promptable cases. Fourteen additional below-target documents contain no existing ORG and are intentionally left unchanged rather than receiving fabricated entities. This sharply improves candidate coverage over an earlier FLAN-T5 pilot, whose outputs overwhelmingly failed to add the requested explicit entity mention.

Detector and contribution movement. Step 9 reduces AUROC from .699558 to .697009 and mean AI score from .500605 to .498378, changes of −.002549 and −.002227, respectively. The calibration-threshold staircase shows small improvements at several low-realized-FPR operating points but no uniform curve domination (figure 24). The more informative effect is feature-specific: ORG sentence mean, standard deviation and maximum all move left in the human-adjusted contribution comparison, while most unrelated features remain nearly stationary (figure 25). After the intervention, `ner_CARDINAL_sentence_max` becomes the leading displayed residual, with ORG mean and standard deviation still present but reduced.

Step 10: CARDINAL controlled insertion. The Step-9 audit finds all four CARDINAL statistics below their same-genre human references, with mean AI-minus-human differences of −.033391 for entity share, −.058549 for sentence mean, −.161746 for sentence standard deviation, and −.650023 for sentence maximum.

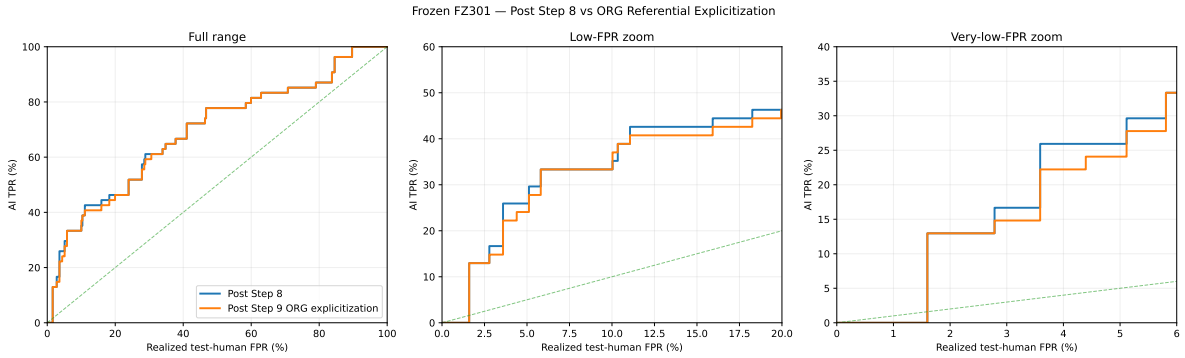


Figure 24: **Frozen-FZ301 response to ORG Referential Explicitization.** Calibration-threshold TPR staircases compare the post-Step-8 texts with the accepted Step-9 outputs on the same 54 sources. The intervention produces small low-FPR improvements rather than a uniform shift. AUROC is computed separately from the full weighted ranking.

The sentence-maximum feature is the leading displayed post-Step-9 contribution. We therefore reuse the controlled-insertion method of Kumar et al. (2022), restricted here to counts entailed by explicit source enumerations; the insertion algorithm itself is not a contribution of this work.

The accepted Step-10 run changes 29/54 documents using 51 accepted edits and raises detected CARDINAL mentions from 168 to 219. Twenty-five cases reach the assigned count target, four exhaust candidates, two have no valid enumeration candidate, and 23 are already at or above target. Frozen-FZ301 AUROC falls from .697009 to .671326, mean AI score from .498378 to .476821, and median AI score from .478403 to .459692. The calibration-threshold staircase is lower over much of the low- and very-low-FPR region (figure 26). In the contribution comparison, all four `ner_CARDINAL_*` features leave the displayed top 30; the remaining positive residuals are distributed across ORG, lexical-diversity/common-word, PERSON, and POS features (figure 27). This supports a simple CARDINAL-count deficiency under the frozen detector while leaving substantial non-CARDINAL residual structure.

Post-Step-10 ORG diagnostic (excluded from the pipeline). Residual ORG features reappear among the leading post-Step-10 contributions, so we ran an exploratory Step 10.9 to test whether they represented a remaining controllable writing tendency. The diagnostic used same-genre human Q20–Q80 bands and tested cross-sentence fusion, coreference-chain rewriting, source-grounded entity restoration, ORG de-explicitization, and sentence splitting. These operations follow established sentence-fusion and coreference-transformation settings (Yuan et al., 2021; Wilkens and Todirascu, 2020), while the source-grounded restoration arm follows prior entity-guided generation work (Dong et al., 2021; Aggarwal et al., 2023). The NER measurements themselves are produced by spaCy (Honnibal et al., 2020). Step 10.9 is diagnostic only: it is not included in the accepted intervention pipeline, and the FZ301 endpoint remains Step 10.

The diagnostic shows why. The residual ORG direction is not globally monotone: books and reviews appear to require substantially more ORG mentions, whereas recipes and Wikipedia-like texts often require fewer or less concentrated mentions. In the book subset, five of seven Step-10 AI documents contain zero spaCy ORG spans, and their corresponding original source texts also contain zero ORG spans. Nevertheless, the fixed human-book reference has nonzero medians for ORG entity share (.175), sentence mean (.437), sentence standard deviation (.477), and sentence maximum (1). Inspection reveals that this gap is driven in part by inconsistent entity typing rather than a coherent semantic distinction. Human book passages include spans such as *Princess Guenevere*, *Nila*, and *Crack House* labeled as `ORG`, while comparable AI-book spans such as *Stevens* are labeled `PERSON` and *Darlington Hall* as `FAC`. Other human examples likewise assign `ORG` to character-like spans such as *Batty*. These labels are not a plausible stylis-

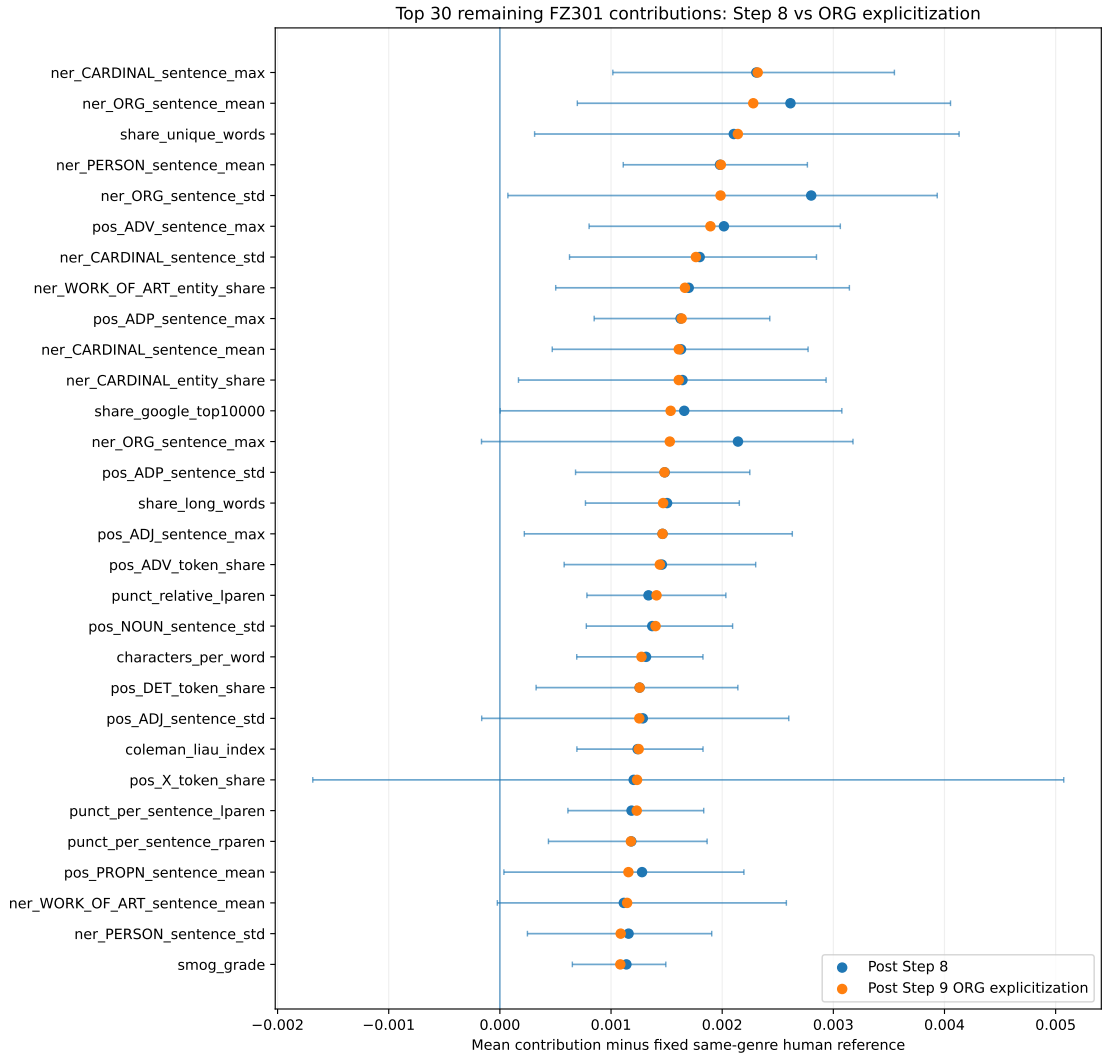


Figure 25: **FZ301 human-adjusted contribution movement after Step 9.** Points compare post-Step-8 and post-Step-9 mean tree-path contribution gaps against fixed same-genre human references; gray intervals are Step-9 genre-stratified document-bootstrap intervals. ORG mean, standard deviation and maximum move in the intended direction, whereas most other leading residuals change little.

tic target for rewriting.

A genre-conditioned controller therefore finds only sparse, localized opportunities: 9 exploratory edits across 8/54 documents, concentrated in recipes and Wikipedia-like texts, while the strongest apparent deficits in books and reviews cannot be repaired without fabricating entities absent from the source content. We interpret this residual as detector overfitting to a mixture of cohort composition and extractor-specific NER errors rather than a stable AI-writing trace. Here *overfitting* refers to reliance on dataset/extractor artifacts that do not survive semantic inspection, not merely to classifier train–test variance. Continuing to optimize these features would risk changing subject matter or deliberately reproducing annotation mistakes, so we stop FZ301 after Step 10 rather than intervene on Step 10.9.

F.19 FZ301 Endpoint and Twenty-Example Adaptive Camouflage

The accepted FZ301 pipeline therefore ends at Step 10. The canonical wrap-up compares the same frozen detector on Original AI, post-FZ118 Step 4, post-FZ186 Step 8, and post-FZ301 Step 10. AUROC decreases monotonically from .875380 to .801146, .699558, and .671326, while mean AI score decreases from .599424 to .544962, .500605, and .476821. The calibration-threshold staircase shows the

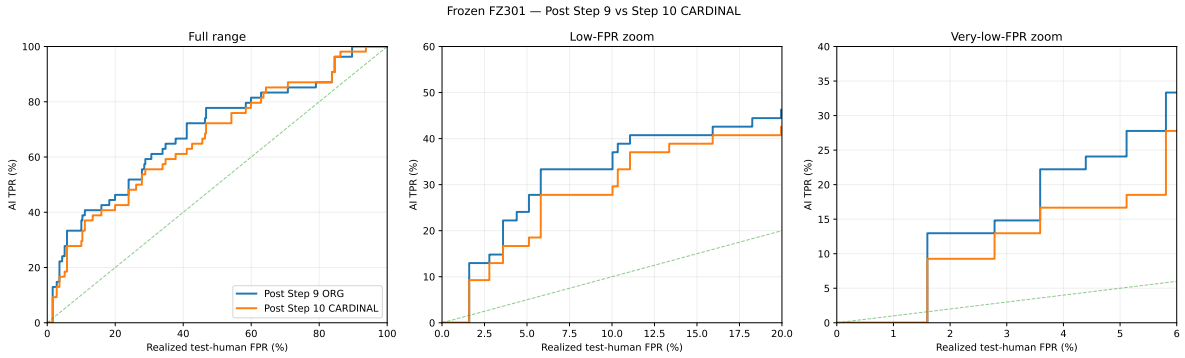


Figure 26: **Frozen-FZ301 response to Step-10 CARDINAL controlled insertion.** Calibration-threshold TPR staircases compare the accepted Step-9 and Step-10 texts on the same 54 sources. The Step-10 curve is lower across much of the low-realized-FPR region; AUROC is computed separately from the full weighted ranking.

Detector	AUC	Mean test-AI score	Median test-AI score
Frozen FZ301	.677416	.492403	.484809
Adapted FZ301 (+20)	.707792	.514390	.500258

Table 19: FZ301 Adaptive Camouflage on the 34 adaptation-held-out Step-10 texts. The adapted detector sees 20 other Step-10 outputs in addition to the unchanged regular RAID training set. Human calibration and test pools are rescored separately by each detector.

same progression in the low- and very-low-FPR regimes (figure 28).

To test whether this endpoint merely creates an easily relearned fingerprint, we run a separate Adaptive Camouflage experiment. A fixed seed selects 20 of the 54 Step-10 texts uniformly without replacement as adaptation examples; the remaining 34 are used only for evaluation. A fresh clone of the FZ301 ExtraTrees configuration is fit on the unchanged regular RAID training set plus the 20 exposed Step-10 AI examples with unit weight. The frozen and retrained detectors each obtain their own thresholds from the same fixed validation-human pool, and realized FPR is measured on the same fixed test-human pool. This follows the same bounded-exposure logic used for the earlier adaptive experiments and is consistent with prior work showing that adversarial/data augmentation can recover part of detector discrimination (Koike et al., 2024; Masrour et al., 2025; Zhou et al., 2024).

Twenty-example exposure recovers AUROC by .030377. This is measurable relearning of residual structure, but the adapted detector remains far below the .875380 original-text separation of frozen FZ301. The low-FPR staircase shows localized rather than uniform recovery: the adapted detector gains at some stringent thresholds, including an earlier nonzero detection step at very low realized FPR, while the curves overlap or cross at others (figure 29). We therefore interpret the result as partial adaptive recovery, not adaptation resistance or distributional indistinguishability. The experiment uses one random 20/34 split, one exposure budget, one classifier family, and no repeated adaptation seeds.

FZ301 stopping criterion. Together, the canonical four-stage attenuation, disappearance of CARDINAL from the displayed residual top 30, the Step-10.9 ORG diagnostic, and only partial twenty-example adaptive recovery provide the stopping point for the feature-based FZ301 stage. Residual pressure after Step 10 increasingly reflects already-treated surface features, heterogeneous content composition, and extractor-specific NER behavior rather than a new clean intervention target. We therefore move to stronger probability- and structure-based detector families rather than continue fitting the 301-feature detector.

F.20 Frozen Ghostbuster Transfer Baseline on NFZ

We next evaluate whether the completed FZ intervention sequence transfers to the substantially different probability-based feature construction of Ghostbuster (Verma et al., 2024). We train one frozen Ghostbuster-

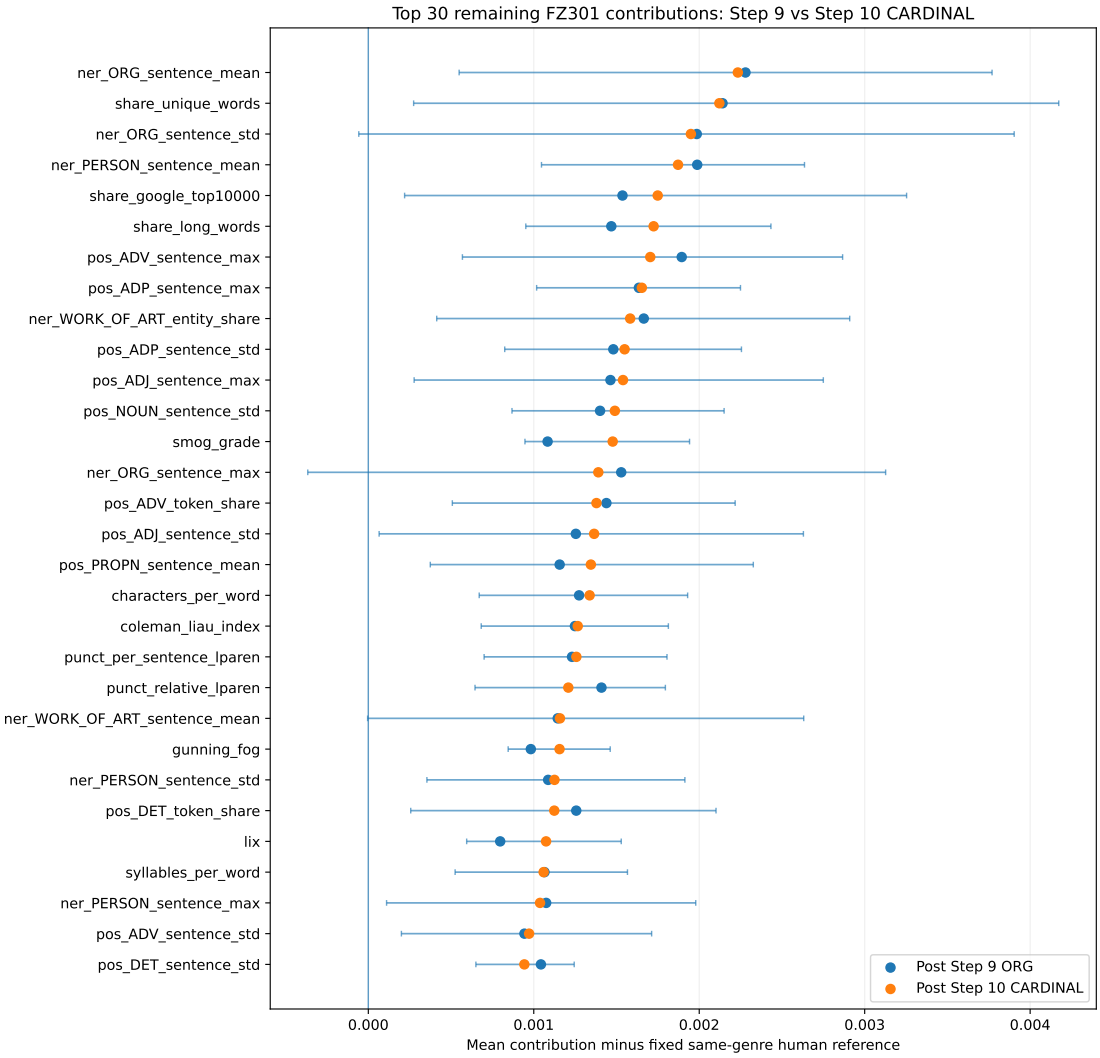


Figure 27: **FZ301 human-adjusted contribution movement after Step 10.** Points compare post-Step-9 and post-Step-10 mean tree-path contribution gaps against fixed same-genre human references; gray intervals are Step-10 genre-stratified document-bootstrap intervals. All four CARDINAL features leave the displayed top 30 after the intervention.

method reproduction on the regular RAID training split only and then score the same fixed 54 sources at Original AI, post-FZ118 Step 4, post-FZ186 Step 8, and the accepted post-FZ301 Step-10 endpoint, which we denote *NFZ*. None of the 54 intervention texts enters feature selection, normalization, or classifier fitting.

The reproduction follows the released architecture as closely as practical: four aligned token-probability streams, the released Brown-corpus unigram and Kneser–Ney trigram components, the released vector/scalar operators, depth-3 symbolic search over 2,534 candidate expressions, the seven manually designed features, train-set normalization, forward feature selection, and five-fold calibrated logistic regression. The forward search selects seven symbolic expressions, giving 14 final features after the seven manual features. Because the original GPT-3 `ada` and `davinci` completion endpoints are retired and cannot reproducibly score new NFZ texts, we substitute frozen GPT-Neo-125M and GPT-Neo-1.3B models (Black et al., 2021) that share the GPT-2/R50K tokenization used by the released pipeline. We therefore treat this as a GHOSTBUSTER-method reproduction rather than the authors' original checkpoint. As a sanity check, the frozen reproduction reaches RAID test AUROC .908633 before the fixed-54 transfer evaluation.

The transfer is substantial despite the ab-

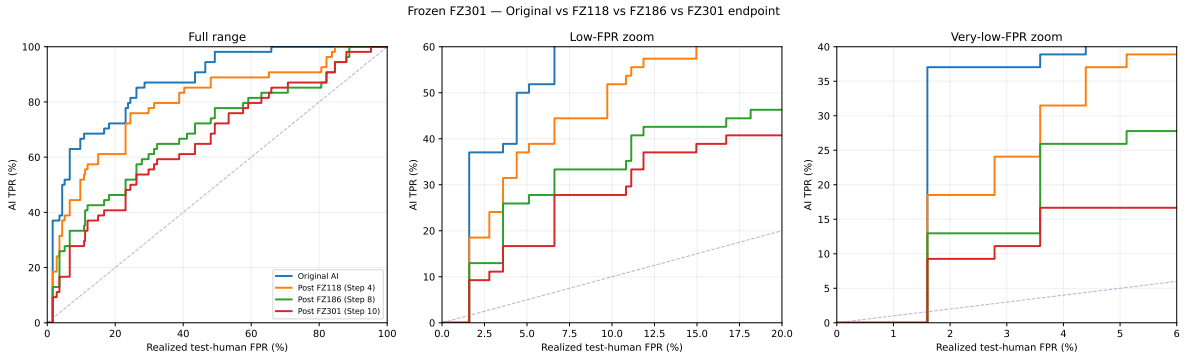


Figure 28: **Frozen-FZ301 attenuation across the accepted intervention sequence.** Calibration-threshold TPR staircases compare Original AI, post-FZ118 Step 4, post-FZ186 Step 8, and the accepted post-FZ301 Step-10 endpoint on the same 54 sources. The largest attenuation appears in the low-FPR region, and Step 10 further lowers the post-Step-8 curve. AUROC values are computed separately from the full weighted rankings.

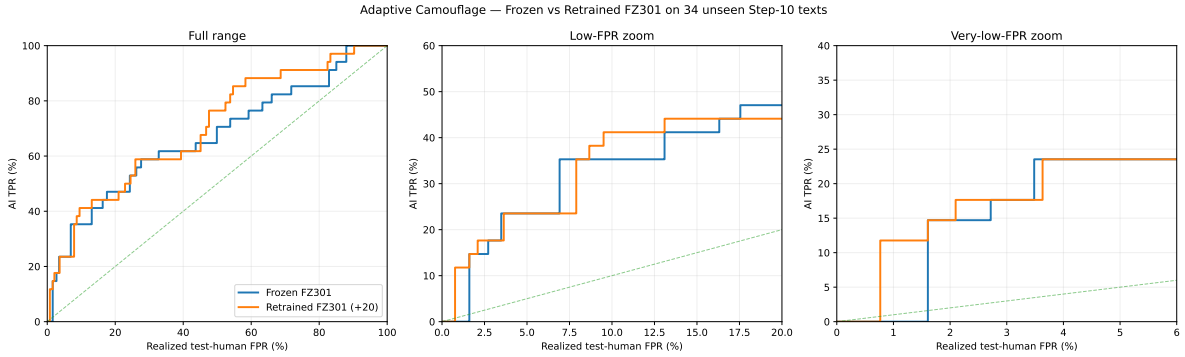


Figure 29: **FZ301 Adaptive Camouflage after twenty-example exposure.** Calibration-threshold TPR staircases compare frozen FZ301 with a detector refit after adding 20 accepted Step-10 outputs to the unchanged RAID training set. Evaluation uses the remaining 34 Step-10 outputs and detector-specific calibration on the same human pools. Retraining recovers some low-FPR sensitivity, but the curves do not show uniform domination.

Stage	AUC	Mean AI score	Median AI score
Original AI	.917333	.738358	.933199
Post FZ118 — Step 4	.820165	.568135	.610294
Post FZ186 — Step 8	.744837	.465446	.398728
NFZ — Step 10	.717215	.432177	.346497

Table 20: Frozen GHOSTBUSTER-method reproduction on the same 54 intervention sources. No GHOSTBUSTER-aware intervention is used in any row. The score columns are calibrated probabilities within this detector and should not be compared numerically with the independently calibrated FZ scores.

sence of any GHOSTBUSTER-aware optimization. AUROC falls by .200118 from .917333 on the original texts to .717215 on NFZ, and the mean calibrated AI score falls from .738358 to .432177. The intermediate stages are monotone as well: .917333 → .820165 → .744837 → .717215. Thus the surface/POS/entity interventions developed against FZ detectors also remove a large amount of information exploited by the probability/ngram feature geometry of this frozen GHOSTBUSTER reproduction. At the same time, the result does not collapse the two detector families into one: GHOSTBUSTER remains stronger than frozen FZ301 on NFZ (.717215 versus .671326 AUROC) and retains visibly stronger discrimination at stringent low-FPR operating points (figure 30). We therefore interpret C05 as evidence of strong cross-detector feature overlap, not as evidence that GHOSTBUSTER is defeated or equivalent to a feature-only detector.

This baseline is the starting point for the C05 series. Subsequent notebooks keep the C05 detector frozen and initially inspect its individual manual and selected symbolic features on NFZ

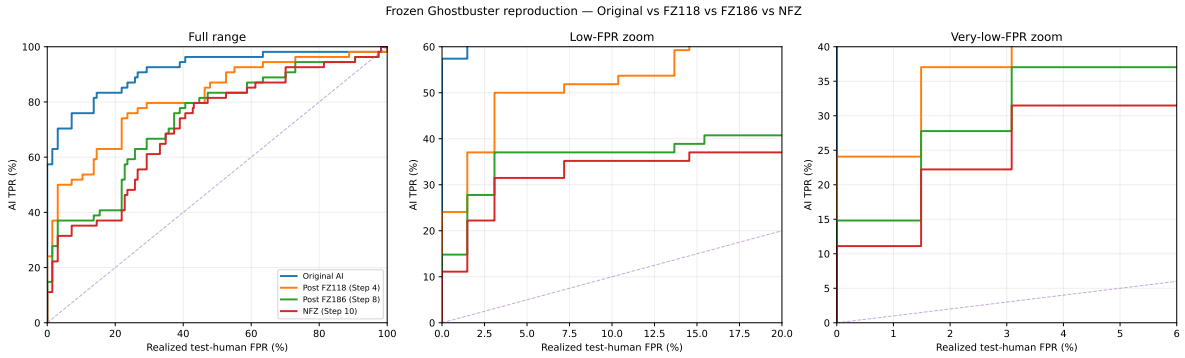


Figure 30: **Transfer of FZ neutralization to a frozen GHOSTBUSTER reproduction.** Calibration-threshold TPR staircases compare Original AI, post-FZ118 Step 4, post-FZ186 Step 8, and NFZ on the same 54 sources. The detector was trained and frozen before these texts were scored. FZ-era interventions substantially attenuate GHOSTBUSTER despite never using its score or features, while the very-low-FPR panel shows that residual high-confidence discrimination remains.

before attempting any GHOSTBUSTER-specific intervention.

F.21 U-Series: Surprisal-Aware Generation with Online Coherence Recovery

The U-series moves from feature-specific rewriting to a generation-time question: can a controller increase the frequency of human-like high-surprisal events while preserving the established lexical intervention and preventing visible degeneration? We use the fixed twenty-case U01 cohort and `meta-llama/Llama-3.2-3B-Instruct` as the generator (Grattafiori et al., 2024; Meta, 2024). Prompts include the original task and the previous AI answer (*WITH_PREV*). The existing Unique controller remains primary: it steers the running share of distinct alphabetic words toward matched human reference bands. When Unique is inactive or all viable candidates are Unique-equivalent, decoding falls through to unrestricted sampling with temperature 1.0 and `top_p`= 1.0. This hybridization is motivated by the broad sensitivity of output statistics to decoding strategy (Wiher et al., 2022); here the objective is not generic diversity but a scorer-robust change in realized token surprisal.

U01 decoding verification. For each generated document we evaluate post-warmup token surprisal $S_t = -\log q(x_t \mid x_{<t})$ under three frozen observers: Qwen3-0.6B-Base, GPT-2, and SmolLM2-360M. The first 20% of the matched OG scorer-token length is excluded and conditions are compared on common token support. Human references exhibit a consistently heavier surprisal-survival tail than OG RAID AI text and regenerated arms. In the immediate U01d decoding check, temperature-1 sampling raises paired mean surprisal relative to greedy Llama by .1093, .1186, and .1180 nats under Qwen3, GPT-2, and SmolLM2, with positive-case fractions .65, .70, and .75. The twenty-case bootstrap intervals cross zero, so this is a directional pilot result rather than a population-level significance claim. Temperature .7 is inconsistent and temperature 1 with `top_p`= .9 moves in the wrong direction overall.

U02 hybrid controller and lexical control. The first U02 hybrid run improves the running Unique trajectory while increasing surprisal relative to Greedy+Unique. On common supported word positions, mean distance to the human Unique band falls from .02645 to .01196, normalized distance falls from .4777 to .2390, and mean in-band occupancy rises from .5247 to .6604. The initial raw Hybrid run also exposes an important failure mode: three outputs develop visibly incoherent suffixes. Because repairing those suffixes increases rather than decreases the independent-scorer surprisal signal, the Hybrid gain is not explained by word-salad tails alone; nevertheless, the mixed repaired panel is not used as the final robustness result.

We therefore add an online coherence controller and regenerate all twenty cases from scratch. The controller monitors rolling generator predictive entropy over sixteen committed model tokens. If the rolling mean exceeds 5

Condition	GPT-2	Qwen3	SmolLM2
Human	3.2056	2.6433	2.5490
OG	2.5485	1.7916	1.8010
Greedy+Unique	2.5027	1.6562	1.6782
Raw Hybrid	2.6338	1.9642	1.9340
Full system	2.5991	1.7769	1.8296

Table 21: Mean post-warmup token surprisal (nats) on the fixed twenty-case U cohort under three independent scorers. The Full system is regenerated from scratch with the online coherence controller; Raw Hybrid is the earlier uncontrolled hybrid generation and is retained only as a diagnostic.

nats for eight consecutive tokens, generation rewinds to the latest complete-word checkpoint at least 24 model tokens before the detected danger run. Unique is temporarily disabled during recovery and temperature-1 sampling regenerates through the alarm word position before normal Hybrid control resumes. The entire lexical/controller state is restored at rewind; the sampling RNG continues forward so a rejected branch is not deterministically replayed. A hard budget of 24 coherence rewinds guarantees finite completion, after which a deterministic greedy/no-Unique fallback is available. When the empirical human Unique reference expires, we do not extrapolate a lower band: only the last supported upper ceiling is retained, so post-reference control can decrease excessive uniqueness but never force it upward.

On the fresh run, all twenty cases terminate at EOS. Eighteen never trigger the coherence alarm. One case triggers once and recovers after one rewind. The remaining case repeatedly re-enters the high-entropy regime, reaching the 24-rewind bound and then completing through the deterministic greedy fallback. Across the cohort there are 26 alarms and 25 executed rewinds; only one alarm occurs while Unique is active, whereas 23 occur during no-Unique recovery, indicating that the observed coherence collapse is not reducible to the lexical intervention. The post-reference ceiling controller activates in four cases and no final output exceeds the retained ceiling. The two alarmed final outputs were manually inspected and found coherent; this is author inspection, not blinded human evaluation.

The fresh Full system preserves the intended directional movement relative to Greedy+Unique under all three observers. Paired case-level Full-minus-Greedy mean differences are +.1054 nats for GPT-2 (95% bootstrap CI $[-.0570, .2567]$), +.1225 for Qwen3 ($[-.0718, .3137]$), and +.1426 for SmolLM2 ($[-.0545, .3327]$), with positive-case fractions .60, .65, and .65. The pooled survival rate is also higher than Greedy+Unique at each prespecified threshold $\tau \in \{4, 6, 8, 10\}$ under every scorer. Human text remains substantially above the Full system: paired Human-minus-Full differences are .6423, .8865, and .7431 nats under GPT-2, Qwen3, and SmolLM2, respectively. Thus U02 establishes cross-scorer directional movement toward the human surprisal profile while retaining bounded coherence recovery, but it does not close the human–machine gap.

Scope. U01–U02 are development experiments on only twenty matched cases. The Full-minus-Greedy confidence intervals include zero, the entropy threshold and rewind policy were diagnosed on this development cohort, one case required the finite greedy fallback, and qualitative coherence inspection is not a blinded human study. These results therefore support the controller as a robust engineering baseline for the next intervention stage, not a population-level claim that sampling or the U02 state machine universally humanizes model output.

F.22 U03: Direct Prompting and an Independent Information Pool

U03 tests whether changing the information requested by the prompt can address the remaining surprisal deficit. We compare the saved U02 Full system, a direct concrete-detail instruction (DP), and an independently generated information pool on the same twenty development cases. The cohort contains four abstracts, four books, and three cases each from news, recipes, Reddit, and Wikipedia. Original RAID AI and matched human texts provide evaluation references.

Interventions and information access. Both new arms retain the Llama generator, WITH_PREV context (the original task and original RAID AI answer), primary Unique controller, and secondary $T = 1.0$, $p = 1.0$ sampling described above. DP appends the following fixed instruction:

> Keep the response grounded in specific people, places, organizations, objects, events, and other concrete referents that are relevant to the subject. As the response develops, introduce specific referents when they naturally advance the content, and return to previously introduced referents when useful. Prefer advancing the subject matter with concrete details over repeatedly restating broad themes, relationships, or implications. Keep those details coherent with the original task and the supplied previous answer; do not introduce unrelated specifics merely for variety.

The information-pool arm instead uses a separate greedy call to the same generator, conditioned on the original task only, to propose up to twelve relevant candidate assertions within a 768-token budget. Its genre-general instruction requests appropriate factual, explanatory, procedural, or explicitly hypothetical details, rather than assuming every task is a book summary. The entire pool is then supplied alongside the original AI answer for a fresh rewrite. This generate-then-condition design is related to generated-knowledge prompting (Liu et al., 2022); here the pool is an unverified model output, not retrieved evidence. The experiment neither schedules gradual release nor adds a model-based fact-checking stage.

No subject-specific human reference text is supplied to either generation stage. The existing aggregate human-derived Unique bands remain available to the controller; matched human texts are used only in the evaluation reported here. Thus additional pool content comes from the generator's parametric knowledge and can contain unsupported claims.

Run contract. The new arms use one seed replicate per case, a 640-token rewrite budget, and a ten-rewind cap followed by greedy/no-Unique fallback. All twenty saved U02 baselines are reused, including `gate_0016`; their historical 24-rewind cap is retained in provenance. This is an existing-system comparison, not a rerun with identical rewind budgets. The baseline has twenty EOS completions, 25 total rewinds, and one greedy fallback. DP has sixteen EOS completions, four token-capped outputs, sixteen rewinds, and no fallback. The pool has eighteen EOS completions, two token-capped outputs, one rewind, and no fallback. All twenty cases per arm, including token-capped outputs, enter the analysis.

Condition	T_4 (%)	T_6 (%)	T_8 (%)
Human	38.603	22.597	12.400
Original RAID AI	24.701	13.279	6.570
U02 baseline	25.131	13.163	6.582
Direct prompt	32.275	19.086	10.460
Information pool	22.263	10.871	5.202

Table 22: U03 whole-text word-surprisal tail shares under Qwen3-0.6B, averaged equally over twenty cases per condition. Human references are evaluation-only.

Whole-text word-surprisal diagnostic. Unlike the post-warmup token analysis in U01–U02, U03 scores complete texts with `Qwen/Qwen3-0.6B` (Yang et al., 2025; Qwen Team, 2025b), not the Base checkpoint. Teacher-forced scoring excludes the generation prompt and adds no special tokens; the first model token has no scored predecessor. Words are identified by a fixed regex allowing internal hyphens and apostrophes. A word's surprisal is the sum of finite token surprisals whose character spans overlap that word; a boundary-spanning token can contribute to more than one word. Words with no scored token are omitted. For each case and arm, we compute

$$T_\tau = \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \mathbf{1}[S(w) > \tau], \qquad \tau \in \{4, 6, 8\}. \tag{31}$$

Here $\mathcal{W}$ denotes scored words. We report equal-case means, not pooled-word rates. These values are therefore not directly interchangeable with the earlier token-level results.

Results and retained configuration. DP raises mean tail shares over the U02 baseline by 7.143, 5.923, and 3.879 percentage points at thresholds 4, 6, and 8, respectively. The corresponding case counts with an increase are 17, 16, and 16 out of twenty. Relative to the baseline-to-human mean gaps, DP closes approximately 53%, 63%, and 67%, while remaining below the human mean at every threshold (table 22 and figure 31).

The upfront pool moves in the opposite direction: its mean changes from baseline are −2.869, −2.292, and −1.380 percentage points, with increases in only 7, 6, and 7 cases. DP exceeds the pool in nineteen of twenty cases at each threshold. The pool's lower rewind count does not establish better coherence or factual accuracy. We retain it as a negative result for

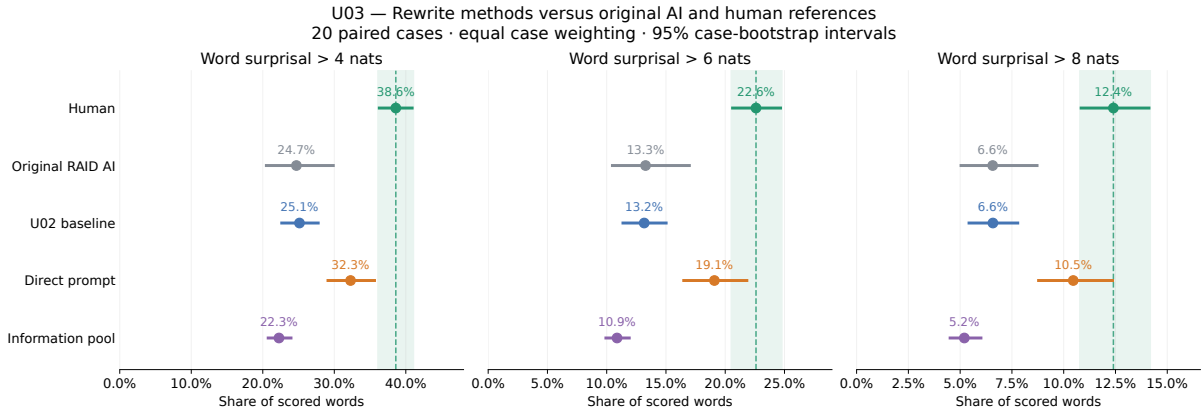


Figure 31: U03 five-condition comparison. Points are equal-case means; intervals use 10,000 case-bootstrap resamples with the same sampled case indices across conditions (seed 20260919). Shading marks the human mean's interval. These intervals describe case uncertainty, not variation across generation seeds.

this upfront implementation; controlled release remains untested.

The configuration carried forward is therefore *Llama + WITH_PREV + Unique primary + SAMPLE_T10 secondary + coherence rewind + DP*, with the ten-rewind default and finite greedy fallback. This decision is based on development-set performance. Higher word-surprisal tails do not by themselves demonstrate new factual information, factual correctness, fidelity, detector evasion, or equivalence to human writing. The single scoring checkpoint, single seed replicate, unequal historical rewind caps, differing output lengths, and retained truncations limit the inference. Held-out evaluation and quality assessment are still needed; U03 does not establish a general advantage across scorers or genres.

F.23 U04: Mid-History Suppression on Top of Direct Prompting

U04 asks whether history reweighting improves dependency organization beyond the accepted U03 direct-prompt (DP) system. Its motivation is frontloading: sustaining informative development rather than concentrating it at the beginning. The primary development comparison is therefore M75 versus U03 DP, not versus original RAID AI. Context utility is a proxy for the benefit of longer context; it is not itself a direct measure of semantic novelty or the temporal location of information.

Intervention and sweep. We retain Llama, WITH_PREV original RAID AI, Unique as primary controller, secondary temperature-1 sampling, DP, and the ten-rewind recovery policy. Generated-history keys are divided by model-token lag from the next predicted token into front (1–63), mid (64–128), and end (129+) bands. These are lag ranges, not sections of the output. At every layer and head, a key in the selected band receives additive attention-logit bias $\log \lambda$ before softmax; prompt keys receive zero additional bias. Thus attention weights are multiplied by λ and renormalized. No tokens are removed or positions renumbered, and normalized prompt attention can change despite its unit multiplier. The same weighting remains active during greedy/no-Unique fallback.

The initial sweep runs six settings on the same ten cases: each of the three bands at $\lambda = .75$ or 1.25, with other multipliers equal to one. This produces sixty saved outputs (52 EOS and eight token-capped). Front-.75 and mid-.75 are then extended to the other ten U03 cases, with the same per-case seed policy and 640-token cap. We retain mid-.75, denoted M75, based on the frequency of positive paired context-utility changes and author inspection of all twenty M75 outputs, which found them coherent and of good quality. This is qualitative author review, not blinded or quantitative fidelity evaluation.

Frozen context-utility protocol. Unlike U03's whole-text word-surprisal diagnostic, U04 uses the frozen `Qwen/Qwen3-0.6B-Base` observer (Yang et al., 2025; Qwen Team, 2025a). For scorer-token positions indexed from zero,

Batch	n	ΔU	95% CI	$+/-/=$
Initial ten	6	1.2258	[.8700,1.6677]	6/0/0
Additional ten	8	-.0129	[-.7006,.4604]	5/2/1
Combined	14	.5180	[-.0389,.9758]	11/2/1

Table 23: Primary U04 development comparison: M75 minus U03 DP on cases with common five-condition U128 support. Counts report positive, negative, and unchanged paired differences. The additional batch does not reproduce the initial mean gain, although positive cases remain more frequent than negative cases.

we take eight targets

$$\mathcal{T} = \{128, 146, 164, 182, 200, 218, 236, 255\}$$

in the first 256 tokens and compute

$$U(128) = \frac{1}{8} \sum_{t \in \mathcal{T}} [-\log q(x_t \mid x_{t-8:t}) + \log q(x_t \mid x_{t-128:t})], \tag{32}$$

where slices exclude t and context positions are reset as in U01b. Scoring adds no special tokens and requires at least 256 scorer tokens. There are eight target positions, each evaluated under two context lengths. The sparse probe can yield variable case estimates. U256 was also inspected but needs 512 tokens; we report U128 for broader support. Means weight cases equally. Percentile intervals use 10,000 paired case-bootstrap resamples (seed 20260920), describing case uncertainty rather than generation-seed variability. Token-capped outputs remain eligible if sufficiently long.

Main result: incremental change over U03 DP. Fourteen of the twenty cases have U128 support across the five displayed conditions (human, original AI, DP, front-.75, and M75). On these same cases, mean $U(128)$ rises from 1.2915 for DP to 1.8095 for M75: a paired increase of .5180 nats. Eleven cases increase, two decrease, and one is unchanged. The paired 95% interval is $[-.0389, .9758]$. The human and original-AI means on this subset are 1.5186 and 1.2806, respectively.

The selection emphasizes directional consistency (11/14 positive) rather than only the mean of the additional batch. Nevertheless, the batch heterogeneity and combined interval crossing zero remain part of the result. Front-.75 produces a smaller combined mean gain of .2573 (CI $[-.3487, .8190]$; nine increases and five decreases). We adopt M75 as an engineering default, not as a demonstrated population-wide optimum. This comparison uses the reported fourteen-case common-support sample; it does not imply that only fourteen DP–M75 pairs are individually scoreable.

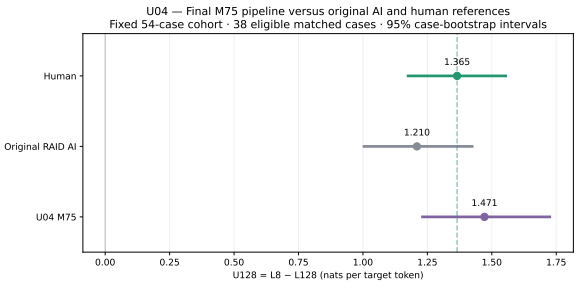


Figure 32: Supporting U04 full-pipeline comparison on 38 common eligible cases from the fixed 54. Points are equal-case U128 means with paired case-bootstrap intervals. The other sixteen cases fail the length requirement in at least one condition; this is not a 54-case U128 estimate.

Extension to the fixed 54 sources. After selection, M75 generates the remaining 34 fixed-cohort sources, producing 25 EOS completions and nine token-capped outputs. All 54 M75 texts are saved; completed outputs are checkpointed individually for resume. The remaining 34 have no corresponding U03 DP arm, so this extension evaluates the complete selected pipeline against original AI and human references rather than isolating the incremental effect of mid-history suppression.

Thirty-eight cases meet the three-group length requirement. Human, original-AI, and M75 mean U128 values are 1.3652, 1.2096, and 1.4714. M75 minus original AI is +.2619 (CI $[.0022, .5385]$), with 21 increases and 17 decreases. Across all forty eligible M75–original-AI pairs, the mean difference is +.1929 (CI $[-.0792, .4705]$), median .0901, with 21 increases and 19 decreases. Within those forty pairs, positive changes average .8310 nats and negative changes have mean magnitude .5125: gains are approximately 1.62 times as large as losses. This magnitude asymmetry is measured from case deltas, not inferred from confidence-interval endpoints.

Historical NFZ comparison. We additionally score the historical post-Step-10 NFZ out-

puts under the same observer. At this U04 checkpoint, M75 had not yet undergone Steps 2–10; C06 below evaluates that completed sequence. On 29 cases eligible across all four groups, human, original AI, NFZ, and M75 means are 1.3574, 1.1162, 1.3024, and 1.5497. NFZ is closer to the human mean, while M75 produces the larger upward shift. M75 minus NFZ is +.2473 (CI [−.1271, .6294]); across all 31 eligible pairs it is +.1444 (CI [−.2391, .5257]). On the 29-case subset, M75 gains over original AI average .9267 versus loss magnitude .2652, but the stronger asymmetry reflects a different eligibility subset. NFZ is a supplementary pipeline comparison, not an ablation of M75 or evidence that the two pipelines' gains will add.

U-series conclusion and C-series handoff. The U-series closes with the accepted configuration *Llama + WITH_PREV + Unique primary + SAMPLE_T10 secondary + rewind + DP + mid-history* $\lambda = .75$, retaining unit front/end multipliers, a ten-rewind cap, and history-weighted greedy/no-Unique fallback. U03 supplies the direct-prompt improvement in word-surprisal tails; U04 adds a directionally favorable development comparison against DP in longer-context utility. Higher U128 alone does not establish reduced frontloading, factual improvement, human equivalence, or detector evasion. Sparse target sampling, one generation replicate, adaptive development selection, varying eligibility, and retained truncations bound the inference. The 54-source result includes the selection cases and is not an independent held-out confirmation. Subject-specific human prose remains evaluation-only; aggregate Unique reference bands remain permitted. We now return to the C-series with the 54 saved M75 outputs as the new input texts for subsequent Steps 2–10 and detector evaluation, preserving historical NFZ and frozen detector artifacts for comparison.

F.24 C06: Combined Generation and Editing Against Frozen Ghostbuster

C06 completes the intervention sequence by applying the accepted Steps 2–10 to all 54 saved U04 M75 outputs. We call these input texts *NEW*, and their post-Step-4, post-Step-8, and post-Step-10 versions *NS4*, *NS8*, and *NS10*. Historical post-Step-8 and post-Step-10 outputs remain *FZ8* and *NFZ*. NEW retains Llama-3.2-3B-Instruct, WITH_PREV original RAID AI, primary Unique control, secondary temperature-1 sampling, bounded coherence rewind, direct prompting, and mid-history $\lambda = .75$. NS10 is the final frozen configuration of SHADE. No subject-specific human reference prose enters generation or editing; aggregate human feature targets remain available.

Stage	AUROC	Mean score	Median score
Original RAID AI	.917333	.738358	.833199
NEW	.755985	.453842	.435328
NS4	.535734	.269410	.209114
NS8	.470598	.220559	.136851
NS10	.458931	.216516	.136851
Historical FZ8	.744837	.465446	.398728
Historical NFZ	.717215	.432177	.346497

Table 24: C06: the same frozen C05 Ghostbuster reproduction evaluates all 54 sources at each stage. Scores are AI-class probabilities. Historical endpoints use the earlier generation pipeline.

Pipeline and frozen evaluation. Steps 2–4 apply the accepted sentence-statistic and punctuation controls. Steps 5–8 apply X-tag cleanup, the intervening punctuation repair, ADV substitution, ADP boost/compensation, NUM/year conversion, and lexical micro-substitution. Steps 9–10 apply ORG referential explicitization and entailed enumeration-count CARDINAL insertion. NS8 and NS10 are saved separately, including unchanged cases. Existing likelihood and linguistic guards are retained. All 54 sources enter each comparison, including token-capped generation inputs; no new generation is selected using Ghostbuster scores.

We reuse the frozen C05 Ghostbuster-method reproduction (Verma et al., 2024), its 14 feature features, normalization, classifier, and GPT-Neo surrogate streams. Validation-human thresholds, strict score-greater-than-threshold decisions, and genre-weighted test-human FPR follow the established protocol. The comparisons therefore concern one fixed detector and source cohort, not independently fitted stage-specific detectors.

Combined-pipeline result. Frozen-detector AUROC decreases from .917333 on original AI to .458931 on NS10, a .458402 reduction. NS8 improves over historical FZ8

by .274239 AUROC, and NS10 improves over historical NFZ by .258284. Surface editing remains consequential after the U-series: NEW-to-NS4 reduces AUROC by .220251, with a further .065136 decrease through NS8. These observations support complementarity between the selected generation configuration and the subsequent editing pipeline. They do not isolate M75's causal contribution, since NEW also includes the earlier U-series choices. The U04 context-utility evidence addresses dependency organization more directly than detector scores do.

NS10 improves aggregate AUROC over NS8 by .011667, but their threshold curves cross: Steps 9–10 do not improve every operating point. NS10's staircase lies below the diagonal over much of the full range, with exceptions at low FPR. Below-chance AUROC describes this detector's ranking on the evaluated sample; it is not proof of distributional identity with human writing.

Residual feature audit. We retain the C05a score-space diagnostic: for feature j, replace only its value by the fixed same-genre test-human mean and compute

$$\Delta_j(x) = p_{\mathrm{AI}}(x) - p_{\mathrm{AI}}(x_{j \leftarrow \bar{x}_{H,g,j}}).$$

Positive values identify residual AI-directed features. These one-feature counterfactuals are neither additive score decompositions nor effects of realizable text edits. On NS10, the largest mean decrease is .09661 (95% case-bootstrap interval [.06072, .13502]), lowering the score in 47 of 54 cases. It comes from the selected expression `unigram-logprobs v-> trigram-logprobs v-< ada-logprobs s-l2`. Under the released left-to-right Boolean operators and positive surrogate probabilities, this is effectively the square root of the count of tokens whose trigram probability is at least their unigram probability. It is sensitive to both very local probability structure and text length, rather than exclusively to long-range dependencies. We record this residual instead of adding a further intervention; isolating local predictability from length is deferred.

Detector	AUROC	Mean score	Median score
Frozen	.459875	.227968	.154963
Adapted (+20)	.540521	.303047	.222271

Table 25: C06c on the same 34 adaptation-held-out SHADE sources. The adapted detector trains on the original RAID training set plus twenty SHADE positives.

F.25 C06c: Twenty-Example Adaptive Camouflage on SHADE

We select twenty of the 54 SHADE texts uniformly without replacement after sorting case identifiers (NumPy seed 20260921), save that split, and reserve the other 34 for evaluation. Following the earlier adaptive protocol, the defender is refitted on the unchanged regular RAID training set plus the twenty exposed SHADE examples labeled AI, with unit example weights. The 14 selected features and original normalization remain fixed. Logistic classifiers and their five-fold sigmoid calibration are refitted; symbolic feature search is not rerun. None of the 34 held-out SHADE examples enters fitting or calibration. "Held out" here means held out from defender adaptation: the cohort is not an independent holdout from intervention development.

Each detector receives its own thresholds from the same validation-human pool. Realized FPR is measured on the same fixed test-human pool, with genre weights matching the 34 held-out sources. Thus the two detectors share examples and evaluation policy, but not numerical score thresholds.

Adaptation increases AUROC by .080647 as computed from unrounded scores, from .459875 to .540521. This is measurable recovery, but the resulting point estimate remains near chance and the full-range staircase broadly follows the diagonal. Recovery is visible at stringent operating points: at the threshold selected for at most 5% validation-human FPR, the adapted model detects four of 34 texts (11.765% TPR) with 3.509% realized test-human FPR. The corresponding frozen-model operating point detects three of 34 (8.824%) at 7.474% realized FPR. These particular rows have different realized FPRs and are not a matched-FPR effect estimate; the full staircase provides that comparison.

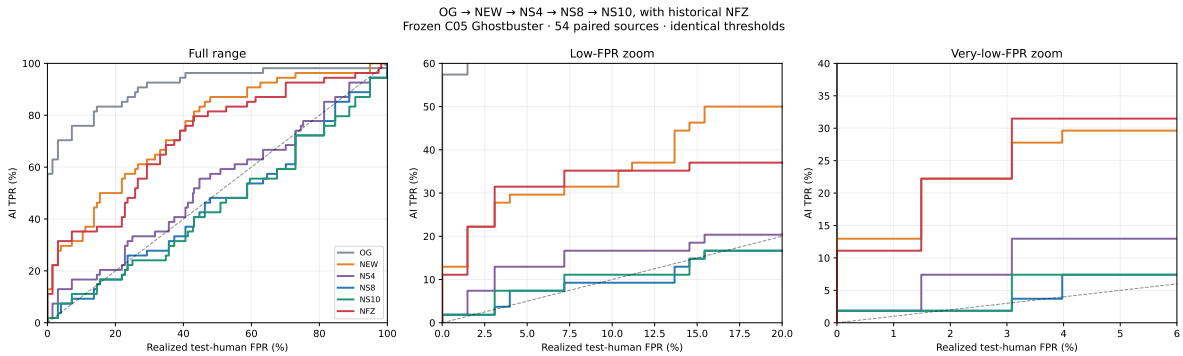


Figure 33: Frozen Ghostbuster staircase on Original AI, NEW, NS4, NS8, NS10, and historical NFZ. All curves use the same 54 sources and fixed detector, with full-range and low-FPR panels.

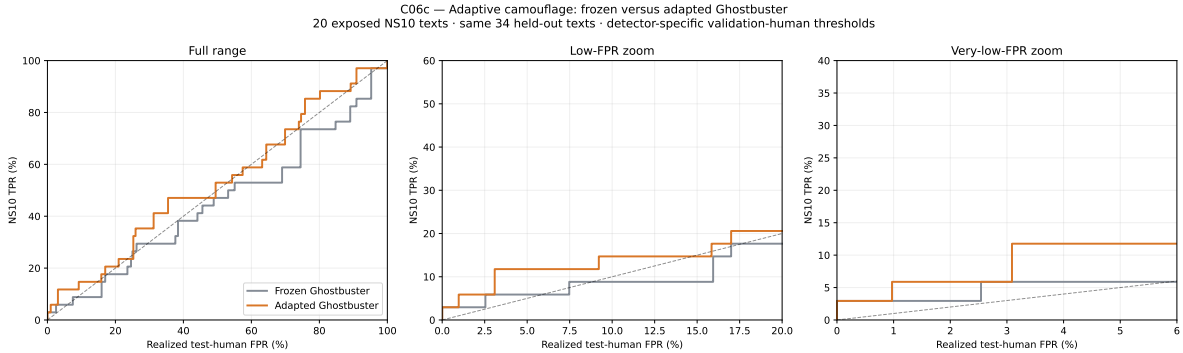


Figure 34: C06c: frozen and adapted Ghostbuster on the same 34 SHADE sources. Thresholds are detector-specific and selected from validation humans; axes show realized test-human FPR and SHADE TPR. Exposure to twenty SHADE examples yields partial recovery.

Intervention endpoint and next phase. C06 closes the current intervention sequence with strong attenuation of the frozen Ghostbuster reproduction and limited recovery under the tested twenty-example adaptation. The adaptive result is one random split without a reported uncertainty interval or repeated exposure sets, and it holds the representation fixed. It does not establish resistance to feature reselection, larger adaptation sets, or other detector families. Earlier author inspection covered the first twenty M75 outputs; a complete blinded content/fidelity evaluation of all final SHADE texts has not been established. At the C06 endpoint, the 54 saved SHADE texts were fixed for subsequent cross-detector stress testing. C07 below reports the first such test on WaveDetect; broad or universal transfer is not established by C06 or by that single additional detector.

F.26 C07: Zero-Shot Transfer to RAID-Trained WaveDetect

Unseen-detector evaluation. We freeze the intervention configuration and all 54 final SHADE texts before evaluating them with WaveDetect (Liu et al., 2026c). No WaveDetect scores or evaluations of these attacked texts were observed during intervention development, and neither the intervention parameters nor the outputs were selected or revised using WaveDetect feedback. C07 is therefore a *zero-shot attack-transfer evaluation with respect to WaveDetect*. Here zero-shot describes the attack's lack of target-detector feedback, not the defender's training: the WaveDetect defender is supervised on regular RAID data. Earlier observational and complementarity experiments used WaveDetect on other corpora; they did not evaluate or optimize the fixed SHADE attack endpoint. We distinguish this initial transfer test from the subsequent twenty-example defender-adaptation experiment.

RAID retraining and evaluation protocol. We reuse the exact RAID document membership and source-group train/validation/test assignments used by C05, checking document keys and split integrity. Unlike the earlier notebooks' public pretrained WaveDetect scorer, C07 retrains the architecture from a pretrained Qwen2.5-0.5B base LM (Qwen et al., 2024; Qwen Team, 2024b) and newly initialized ResNet-18/classifier weights (He et al., 2016). The implementation follows the pinned official WaveDetect code (Liu et al., 2026b): next-token probabilities, a Morlet transform with 16 scales and kernel size 128, and the convolutional classifier. Stage 1 freezes the LM and trains the CNN/head for three epochs at initial learning rate 10^{-3}; stage 2 jointly trains all parameters for three epochs at 10^{-5}. Both use AdamW and linear learning-rate decay. The selected checkpoint minimizes original RAID validation loss among stage-2 epochs; SHADE never enters training, validation selection, or threshold calibration.

For the local single-GPU implementation, microbatch size is one and logical batch size is eight, with gradient checkpointing and chunked vocabulary projection. We preserve the released class-weighted loss over each logical batch, visit all training rows each epoch instead of applying the example script's class-ratio undersampling, and update BatchNorm on microbatches. All inputs use a fixed padded/truncated width of 1024 tokens at both training and evaluation. Thus this is an architecture-matched RAID retraining, not an exact reproduction of the authors' multi-GPU training run or published checkpoint. The implementation records model revision, input hashes, truncation, and resumable checkpoints.

All 54 original AI/SHADE pairs are scored against the same held-out human pool, with human genre weights matching the AI cohort. Thresholds come only from validation-human scores; the staircase reports realized test-human FPR. AUROC uses full score rankings rather than the area under that calibrated staircase. Human reference prose is used for defender training/evaluation, never supplied to the attack generation or editing pipeline.

Transfer results. AUROC decreases from .797419 on original AI to .548795 on SHADE (Table 26), a paired change of $-.248624$ with 95% case-bootstrap interval $[-.354292, -.145500]$. Mean AI score decreases by .218798 (interval $[-.313675, -.125285]$), with decreases in 39 of 54 texts. At the threshold targeting 5% validation-human FPR, realized test-human FPR is 7.2381% for both arms; TPR decreases from 48.1481% to 24.0741%. These results establish substantial transfer to a detector that was not an intervention optimization target. Residual low-FPR detection remains, so near-chance aggregate AUROC does not imply complete neutralization or human/AI distributional identity.

Text	AUROC	Mean AI score	Median AI score
Original RAID AI	.797419	.706225	.768298
SHADE	.548795	.487428	.427104

Table 26: C07 zero-shot attack transfer: the same RAID-trained WaveDetect evaluates all 54 paired sources. Its softmax AI scores are not independently calibrated posterior probabilities.

Twenty-example adaptation. We subsequently sample twenty SHADE cases without replacement from sorted case identifiers using NumPy seed 20260923, save that assignment, and reserve the remaining 34 for evaluation. Starting from the selected RAID-trained WaveDetect, the defender jointly fine-tunes all parameters for three additional epochs on the original RAID training set plus the twenty SHADE positives, using initial learning rate 10^{-5} and the same loss and batching protocol. Checkpoint selection again minimizes original RAID validation loss. The held-out 34 SHADE texts are excluded from fitting and selection. Each detector receives its own validation-human thresholds, and both use the same test-human rows weighted to the 34-case genre mixture. This adaptation test is *not* zero-shot: it explicitly gives the defender twenty attacked examples.

Adaptation does not produce an observed AUROC rebound: the paired change is $-.023619$, with 95% case-bootstrap interval $[-.062273, .008026]$ (Table 27). The interval includes zero, so this is evidence of no detectable recovery under the tested protocol, not established deterioration. Mean AI score rises from .538597 to .598111 without improved relative

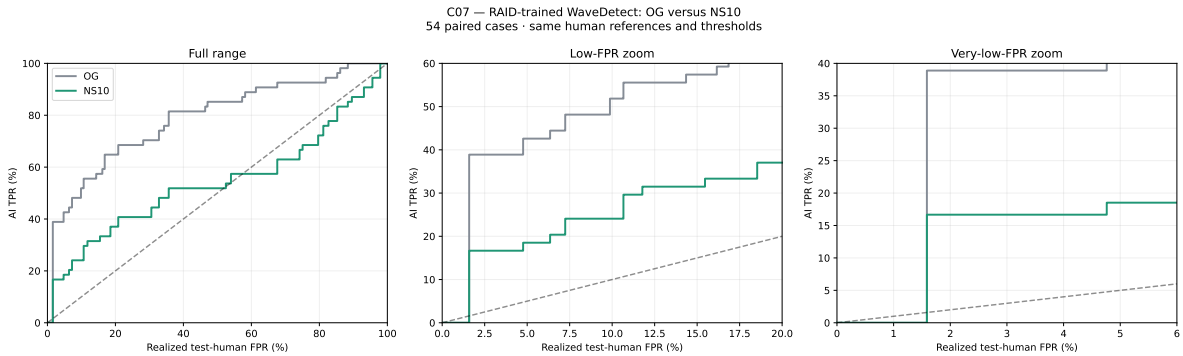


Figure 35: Zero-shot transfer to RAID-trained WaveDetect. The fixed original AI and SHADE texts share 54 sources, human references, and thresholds. No WaveDetect evaluation of SHADE was observed before the attack endpoint was fixed.

Detector	AUROC	Mean AI score	Median AI score
Frozen	.608108	.538597	.493935
Adapted	.584489	.598111	.630554

Table 27: WaveDetect adaptation on the same 34 held-out SHADE cases. The subset and human genre weights differ from the 54-case transfer evaluation; the appropriate baseline is the frozen row within this table.

ranking against humans. At the 5% validation-target operating point, frozen/adapted TPR is 26.4706%/23.5294%, with realized test-human FPR 8.6667%/7.5766%; these are separately calibrated operating points, not an exact matched-test-FPR comparison.

Scope and next tests. C07 supplies zero-shot cross-detector transfer evidence and resistance to the specified limited-exposure adaptation procedure. Both intervals resample paired AI cases 10,000 times while holding the human reference and fitted detectors fixed; they exclude training-seed, exposure-split, and human-reference uncertainty. The 54-source cohort remains part of intervention development, even though WaveDetect feedback was withheld; the 34-case split is held out from defender adaptation, not from all prior development. One exposure split and a checkpoint criterion based on clean RAID validation do not establish resistance to larger exposure sets or adaptation tuned on attack-specific validation. Binoculars NF4 is evaluated next in Appendix G; ProSSD remains pending. C07 does not establish universal detector evasion or isolate which component of the combined intervention causes transfer.

G C07a: Transfer to Binoculars NF4 and Limited-Exposure Feature Adaptation

Detector and precision. We next evaluate the already fixed SHADE endpoint using Binoculars (Hans et al., 2024), a training-free detector contrasting two pretrained language models. We use Falcon-7B as observer and Falcon-7B-Instruct as performer (Almazrouei et al., 2023). Hardware constraints require four-bit NF4 weights with double quantization (Dettmers et al., 2023) and bfloat16 computation, loading one model at a time. We therefore call this configuration *Binoculars NF4*; these results are not a full-precision replication. We retain the published default mean-NLL/cross-entropy ratio and score the first 512 model tokens. The raw ratio is lower for more AI-like text; we negate it for AI-oriented ranking. It is not a probability. The implementation follows the metric definitions at official repository commit `c8ae2f90d50ee696418bc71d8d9e5020e5f9d7b8`, with CPU metric reductions. Prefix truncation affects 25/54 SHADE texts, 1/54 originals, and 11/273 validation/test human references; this limits claims about full-document detection.

Fixed-attack transfer protocol. All 54 original RAID AI texts (OG) and their fixed SHADE counterparts are scored without further attack tuning. No Binoculars feedback was used to produce these outputs, so this is zero-shot *attack transfer* to a new target configuration. Unlike the supervised WaveDetect experiment, native Binoculars has no RAID

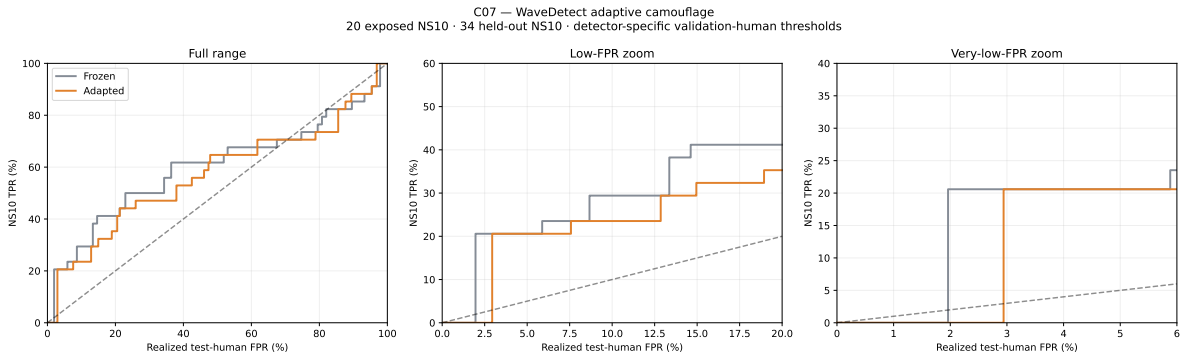


Figure 36: Frozen versus adapted WaveDetect on 34 adaptation-held-out SHADE texts after exposure to twenty other SHADE examples. Thresholds are detector-specific; the curves show no broad recovery of discrimination under this adaptation procedure.

Measure	OG	SHADE
Cases	54	54
AUROC	.905163	.721606
TPR at 1% validation target	.740741	.370370
TPR at 5% validation target	.814815	.481481

Table 28: C07a Binoculars NF4 transfer. Realized test-human FPRs are 1.6708% and 4.8670% at the respective 1% and 5% validation targets.

Defender on 34 held-out SHADE texts	AUROC
Frozen Binoculars NF4 ratio	.684260
RAID-only two-feature classifier	.672084
Adapted two-feature classifier	.678159

Table 29: C07a limited-exposure adaptation. The frozen 34-case baseline differs from the 54-case result because the cases and human genre weights differ.

training stage. Original RAID validation humans determine conservative thresholds, while the original test humans, weighted to the evaluated AI genre mixture, determine realized false-positive rates (FPRs). Each paired comparison shares thresholds and human references. Published BF16 thresholds are retained only as uncalibrated diagnostics for NF4, not as guarantees of their original FPRs.

Transfer results. Table 28 shows a paired AUROC change of $-.183557$ (95% case-bootstrap interval $[-.283240, -.084525]$). The mean negated-ratio change is $-.154232$ ($[-.194652, -.110421]$), with lower AI-oriented scores for 44/54 cases and higher scores for ten. At the 5% validation-FPR target, TPR decreases by 33.33 percentage points at the same realized 4.87% test FPR. This is substantial partial transfer, but Binoculars NF4 retains meaningful discrimination; the experiment does not support near-chance detection for this target.

Adaptation protocol and control. Since changing only a ratio threshold cannot change its ranking or AUROC, we test an explicitly different defender: a logistic classifier on the two Binoculars components, performer mean NLL and observer–performer cross-entropy. Using seed 20260923, we sample twenty SHADE case identifiers without replacement from the sorted 54 and reserve the other 34. Both feature classifiers use $C = 1$, balanced class weights, the LBFGS solver, and a common standardizer fitted only on the original RAID training set. The *RAID-only* classifier fits the 1,272 original training documents; the *adapted* classifier fits those documents plus the twenty SHADE positives. The Falcon models remain frozen. No held-out SHADE examples tune the classifiers or thresholds. Detector-specific thresholds use the original validation humans; all three detectors share test humans reweighted to the 34-case genre mixture. This control separates changing the scoring rule from exposing that rule to attacked examples.

Adaptation results. The clean exposure contrast is adapted minus RAID-only: $+.006075$ AUROC (95% interval $[+.001869, +.011251]$), a small but resolved conditional increase of 0.61 percentage points. Adapted minus frozen ratio is $-.006100$ ($[-.017103, +.003447]$); RAID-only minus frozen ratio is $-.012175$ ($[-.021088, -.000823]$).

At the 5% validation target, RAID-only and adapted classifiers share a realized test FPR of 6.7201%, with TPRs 47.0588% and 44.1176%, respectively. Thus exposure yields little overall recovery under this particular adaptation procedure, despite a small positive AUROC shift. Remaining distinguishability is substantial for all three rules.

Scope and reproducibility. Intervals use 10,000 paired-case bootstrap draws with the fitted models and human references fixed. They exclude exposure-split, human-sampling, and training uncertainty. The 34 cases are held out from defender adaptation, not from the broader intervention-development process. Twenty examples are added once to the original training set, without attack-specific oversampling or hyperparameter tuning. These findings support limited sensitivity to this two-feature update, not resistance to arbitrary adaptive defenders or Falcon fine-tuning. Scores, component values, model revisions, runtime versions, split membership, fitted coefficients, calibration thresholds, and figures are saved. Batched observer/performer passes retain per-document checkpoints and delete intermediate logits only after final scores are saved. Together with C07, this extends evidence of attack transfer while identifying Binoculars NF4 as a target retaining appreciable residual detection power.

H RoBERTa Adaptation and Local-Attack Compatibility (C07b/R01/R11)

Module selection and pipeline position. The residual module must operate as a suffix on an existing text, so our candidate pool comprises post-generation editors rather than humanizers in general. A generator-based humanizer such as StealthRL, used as a fresh generation stage, would replace the earlier output rather than preserve its attained features; it therefore does not occupy this suffix position in the evaluated pipeline. In our preliminary standalone RoBERTa screening, RAFT was clearly the strongest editor considered and was the only candidate composed with SHADE; the weaker standalone candidates were not carried forward. Standalone strength does not establish composability, as the StealthRL contrast in Table 4 demonstrates. The experiment establishes that SHADE admits a complementary module, not that RAFT is optimal. This restriction concerns pipeline position rather than method quality: the staged design makes a preservation-aware suffix position available, a point generalized in Section 6.

Scope and naming. C07a above denotes Binoculars; the RoBERTa experiment is stored in C07b, with local-attack comparisons in R01 and their composition in R11. SHADE remains the primary method; SHADE+RAFT denotes RAFT applied to its frozen output, an optional compatibility experiment rather than a replacement endpoint satisfying all our neutralization criteria.

Matched evaluation and adaptation. We evaluate a RAID-trained RoBERTa classifier (Liu et al., 2019) using the original source-group data split. One saved random split (seed 20260923) assigns twenty of the 54 intervention cases to defender adaptation and 34 to evaluation. We use the same case identifiers for every attack variant. Each adapted defender starts from the same frozen RAID checkpoint and trains on the original RAID training documents plus twenty AI-labeled examples of its designated variant: OG-to-RAFT, SHADE, or SHADE+RAFT. The runs reuse the saved SHADE adaptation hyperparameters, retain original RAID validation-loss checkpoint selection, and exclude the remaining 34 cases from adaptation. This is matched limited-exposure fine-tuning, not attack-specific validation or unrestricted adversarial training. Scoring uses the first 512 tokenizer tokens. All comparisons use the same original test-human references, weighted to the held-out AI genre mixture, and equal weighting of AI cases. Human texts serve defender training/evaluation; they are not supplied as subject-specific material to the attack.

Local attack and composition. Our RAFT implementation follows the released local ranking, candidate substitution, and POS-filtering procedure (Wang et al., 2024a), substituting Qwen2.5-3B-Instruct (Qwen Team, 2024d) with NF4 weights as the candidate generator. It uses up to fifteen parsed candidate words per position and a successful-substitution budget of ten percent of whitespace words. Candidates are selected using

frozen RoBERTa feedback; neither the adapted models nor their held-out scores guide substitutions. The source-style selection chooses the lowest-scoring eligible alternative without requiring improvement over the unchanged word. Candidate-list early termination and a finite generation ceiling are implementation accommodations. These results describe our implementation, not an exact reproduction of the original paper's model configuration or quality findings. The 54 OG-to-RAFT and 54 SHADE+RAFT outputs are saved before the corresponding adaptation comparisons. Unlike SHADE's initial unseen-detector transfer tests, RAFT and SHADE+RAFT are explicitly detector-guided.

Frozen compatibility. Relative to OG, RAFT removes 39.40 AUROC percentage points and SHADE removes 26.26; SHADE+RAFT removes 71.70, exceeding their summed reductions by 6.04 points (Table 30). This is a descriptive positive interaction on this AUROC scale, not a significance test or proof of independent mechanisms. Against the RAFT-adapted and SHADE-adapted checkpoints, SHADE+RAFT also has lower AUROC than either standalone intervention. Frozen SHADE+RAFT AUROC below .5 indicates reversed ranking under the original score orientation; it does not establish distributional indistinguishability or attainment of human feature bands.

Recovery under direct adaptation. SHADE-specific adaptation raises SHADE AUROC from .7291 to .9657. RAFT-specific adaptation raises OG-to-RAFT from .5977 to .8534, while SHADE+RAFT-specific adaptation raises SHADE+RAFT from .2747 to .8836. Thus the large frozen compatibility gain does not yield adaptation-proof behavior. Under the SHADE+RAFT-adapted checkpoint, SHADE+RAFT and OG-to-RAFT score .8836 and .8938, respectively; no uncertainty interval is available to resolve this small difference. A previously evaluated RAID-only continuation control yielded .7705 on SHADE, compared with .7291 frozen and .9657 after SHADE exposure, supporting a substantial exposure-associated recovery beyond that continuation control.

Quality and mechanism limits. Manual comparison of one displayed case reveals additional SHADE+RAFT damage: a named location becomes a generic word, a butler becomes a prefect, and substitutions introduce agreement and selectional errors. The SHADE input itself also contains awkward or semantically defective passages. This observation is not a corpus-wide quality estimate, but it prevents treating detector-score reduction as sufficient evidence of successful substantive neutralization. Attribution and intermediate-stage inspections motivate the hypothesis that surface edits leave recoverable lexical/conjunction signatures; they do not causally identify the features responsible for recovery. The contrast with limited recovery in the specified WaveDetect adaptation is consistent with transfer through shared statistical features, but detector architecture, representation, and adaptation procedure also differ. We therefore do not infer a general ranking of attack philosophies or claim that the source RAFT method intrinsically fails to preserve quality.

Interpretation and reproducibility. This experiment demonstrates compatibility in lowering existing RoBERTa scores, with substantial recovery after direct adaptation and observed quality costs. Preservation of SHADE's context-utility, recurrence, surface-feature bands, and performance against other detectors after adding RAFT remains untested. SHADE stays the primary endpoint; SHADE+RAFT is an exploratory extension. Results use one exposure split and training configuration without repeated-seed or paired uncertainty estimates. The 34 cases are held out from adaptation, not all intervention development. Outputs are retained under `results/causal/R11_NS10_RAFT_v1/`: `NS11_texts.csv`, per-case generation records, `adaptation20_34/`, and `evaluation34/`. The latter contains scores, AUROC summaries, and TPR staircase exports. No new generation or training was performed for this manuscript update.

I C07c: GLTR-Feature Transfer and Limited-Exposure Adaptation

Detector definition and data. GLTR was introduced as a statistical visualization tool

RoBERTa checkpoint	OG	OG-to-RAFT	SHADE	SHADE+RAFT
Frozen	.9917	.5977	.7291	.2747
RAFT-adapted	.9949	.8534	.8860	.6764
SHADE-adapted	.9951	.8082	.9657	.7356
SHADE+RAFT-adapted	.9925	.8938	.9349	.8836

Table 30: AUROC on the same 34 cases against genre-weighted test humans. Each adapted checkpoint sees the same twenty source cases in its designated attack form. Lower AUROC means weaker ranking in the original AI-score orientation, not necessarily human-distribution alignment. Values are transcribed from the saved evaluation display, rounded to four decimals.

supporting human judgments, rather than a released binary classifier (Gehrmann et al., 2019). We therefore explicitly evaluate our *GLTR-feature classifier*: logistic regression on four token-rank fractions under frozen GPT-2 (124M), for ranks 1–10, 11–100, 101–1000, and above 1000. This uses the rank histogram, not every statistic available in GLTR or its human-in-the-loop interface. GPT-2 runs in FP32 and scores the first 1024 tokenizer tokens, excluding the first token and adding no artificial beginning-of-sequence token. Rank is one plus the number of vocabulary logits strictly exceeding the observed token's logit. The fractions use scored-token count as denominator. Per-document token counts and prefix-truncation flags are saved; numerical truncation totals were not provided with the reported results.

We reuse the exact C05 RAID membership and original source-group train/validation/test assignments. The standardizer is fitted on original RAID training features only. Logistic regression uses fixed $C = 1$, balanced class weights, and the LBFGS solver; no SHADE result selects features or hyperparameters. The original RAID validation humans calibrate thresholds, while original test humans are weighted to the evaluated AI genre mixture. All paired comparisons share human references and equal AI-case weighting. The saved 54 SHADE texts are evaluated without modification or GLTR-guided attack tuning. This is transfer of the fixed attack to a newly fitted rank-feature detector, not a claim that token-rank statistics themselves were unknown during the research.

All-54 transfer results. Frozen AUROC decreases from .8160 on OG to .5062 on SHADE.

Classifier / evaluation set	Cases	OG	SHADE
Frozen / full cohort	54	.8160	.5062
Frozen / adaptation holdout	34	.8178	.5083
SHADE-adapted / holdout	34	.8145	.5099

Table 31: C07c AUROC against genre-weighted test humans. Frozen and RAID-only are the same deterministic classifier. Values are transcribed to the precision shown in the evaluation displays.

The saved paired estimate is $-.3097$ (95% interval $[-.3861, -.2344]$); minor differences from subtracting displayed means reflect rounding. At the 5% validation-FPR target, realized test-human FPR is 3.17%: OG TPR is 27.78% and SHADE TPR is 0%. At the 10% validation target, realized FPR is 8.13%, with TPRs 59.26% and 5.56%, respectively. Thus rank-feature discrimination approaches chance on SHADE while remaining appreciable on the matched originals.

Adaptation protocol and outcome. Twenty case identifiers are reserved for SHADE exposure and the other 34 for evaluation. The notebook reuses the saved RoBERTa 20/34 split when available; otherwise it samples and saves one split with seed 20260923. The adapted logistic head is refitted on original RAID training examples plus twenty SHADE positives, with the original standardizer, GPT-2 model, feature definition, and hyperparameters fixed. A deterministic RAID-only refit reproduces the frozen head, so the exposure contrast requires no separate continuation-training correction. No held-out SHADE features enter fitting or threshold calibration.

On the 34 held-out cases, SHADE AUROC changes from .5083 to .5099, an approximately

+.0016 increase (0.16 percentage points from displayed values). OG AUROC changes from .8178 to .8145. The persistence of OG discrimination argues against a wholesale loss of classifier functionality as the explanation for near-chance SHADE ranking. No adaptation confidence interval was supplied in the result displays, so we report this rebound descriptively rather than asserting statistical equivalence or a resolved nonzero effect.

Interpretation and limits. The result extends the observed transfer to a coarse token-rank representation and supports resistance to this specified twenty-example feature-level adaptation. Together with the WaveDetect and RoBERTa contrast, it is consistent with the hypothesis that feature-directed interventions transfer to detectors using related statistical signals. It does not establish that the four-dimensional feature distributions are identical, isolate which SHADE step causes transfer, or establish resistance to nonlinear heads, more exposure, or language-model fine-tuning. The transfer interval uses 5000 paired AI-case bootstrap draws with human references and fitted models fixed; it excludes human-sampling, training, and exposure-split uncertainty. The 34 cases are held out from adaptation, not all earlier intervention development. This detector evaluation supplies no new text-quality assessment.

Artifacts. The notebook is `C07c_GLTR_RAID_OG_NS10.ipynb`; outputs are saved under `results/causal/C07c_GLTR_RAID_OG_NS10_v1/`. The directory retains pinned model/extraction settings, per-document cached features, original data manifests, the adaptation split, fitted classifiers, per-arm scores, operating points, paired changes, and staircase figures. Original 54-case and held-out 34-case comparisons remain distinct. C07d is reserved for a subsequent detector and has no reported outcome here.

J C08: Competitor Humanizers and Own-Attack Adaptation

Comparison scope. We compare SHADE with six completed competitor configurations on the same 54 original RAID machine texts (OG), using the unchanged C05 GHOSTBUSTER classifier and reference pools. The methods are HIP (Xu et al., 2026), MASH (Gu et al., 2026), two UTS-inspired procedures (Galat and Rizoiu, 2026), and StealthRL (Ranganath and Ramesh, 2026). Four configurations are *limited reproductions*: RAID-trained HIP (0.6B), RAID-trained MASH with Ghostbuster feedback, and the UTS_SYNTH and UTS_STREAM prompt/procedure reconstructions. These are not exact reproductions of the publications' training conditions or competition submissions. The two *released-checkpoint* configurations use the authors' HIP and StealthRL 4B adapters. "Released" identifies weight provenance, not the strongest available model or a quality guarantee. The separate locally trained StealthRL run remains incomplete and contributes no results.

Released-checkpoint inference. We load `YixuanEvenXu/Qwen3-4B-HIP-adapter` and `suraj-ranganath/StealthRL`, recording their revisions and declared base models. Hardware constraints require NF4 four-bit loading and batch size one. HIP applies ten successive paraphrases using the official runner defaults: 1024-token input and output limits, temperature 1, and top-$p = .95$. We did not locate a 4B-specific inference configuration; these are runner defaults rather than a verified match to every published experiment. StealthRL uses its released local evaluation procedure: one candidate, 1024-token input and 512-token output limits, temperature .9, and top-$p = .95$. Neither released arm uses Ghostbuster feedback for inference-time selection. All 54 outputs complete for each arm with zero recorded input truncations or generation caps, including intermediate HIP rounds. Quantization and independent per-case seeds prevent a claim of exact numerical replication. HIP training-data overlap with this cohort has not been audited; StealthRL's training detector families also preclude describing every future cross-detector evaluation as unseen.

Frozen-detector results. SHADE has lower AUROC than every evaluated competitor (Table 32). Relative to SHADE, released HIP increases detector AUROC by .394 (95% paired case-bootstrap interval [.304, .483]), and released StealthRL by

Attack / text	Ghostbuster	Binoculars NF4	RoBERTa	GLTR	WaveDetect
Original AI	.917 [.865,.959]	.905 [.834,.963]	.992 [.984,.997]	.816 [.744,.880]	.797 [.726,.863]
SHADE	.459 [.380,.538]	.722 [.628,.809]	.733 [.657,.806]	.506 [.431,.581]	.549 [.453,.647]
HIP, limited	.931 [.890,.964]	.892 [.818,.951]	.755 [.688,.818]	.775 [.689,.854]	.699 [.619,.777]
MASH, limited	.935 [.872,.985]	.851 [.798,.901]	.842 [.789,.888]	.945 [.899,.978]	.738 [.670,.801]
UTS-SYNTH, limited	.796 [.736,.851]	.879 [.822,.930]	.979 [.965,.990]	.612 [.539,.682]	.682 [.602,.758]
UTS-STREAM, limited	.856 [.791,.912]	.965 [.918,.998]	.984 [.972,.993]	.673 [.621,.723]	.742 [.663,.817]
HIP, released 4B	.853 [.806,.899]	.918 [.871,.957]	.894 [.856,.927]	.856 [.795,.909]	.671 [.590,.750]
StealthRL, released 4B	.737 [.639,.826]	.382 [.272,.496]	.964 [.949,.976]	.309 [.214,.409]	.775 [.733,.813]
Original → RAFT	.256 [.190,.329]	.662 [.572,.746]	.591 [.498,.686]	.320 [.256,.389]	.587 [.503,.668]
SHADE +RAFT (SHADE+RAFT)	.148 [.096,.204]	.535 [.451,.619]	.259 [.190,.336]	.097 [.069,.127]	.253 [.181,.332]
StealthRL → RAFT	.495 [.373,.617]	.156 [.084,.237]	.699 [.612,.781]	.099 [.048,.158]	.693 [.633,.747]

Table 2: **Frozen AUROC with conditional 95% intervals on the original 54-source cohort.** Entries show AUROC above its interval. Intervals use 10,000 paired case-bootstrap resamples; their cohort-only scope is defined in Section 4. Lower values mean weaker discrimination in the original AI-positive orientation; below .5 can indicate reversal. Limited/released labels identify provenance. SHADE+RAFT uses RoBERTa-guided RAFT; these arms were not rerun on the 74-source complement.

Configuration	AUROC	Mean AI score
OG	.917	.738
SHADE	**.459**	**.217**
HIP, limited	.931	.750
MASH, limited	.935	.839
UTS_SYNTH, limited	.796	.483
UTS_STREAM, limited	.856	.584
HIP 4B, released	.853	.552
StealthRL 4B, released	.737	.589

Table 32: C08 frozen Ghostbuster comparison on all 54 matched sources. Lower AUROC means lower discrimination in the fixed AI-positive orientation. Limited reproductions and released checkpoints remain distinct. Human references are genre weighted; capped outputs are retained.

.278 ([.167, .396]). Their AI scores exceed SHADE's on 49/54 and 42/54 sources, respectively. The four limited arms also have positive paired AUROC differences from SHADE: HIP .472 ([.393, .554]), MASH .476 ([.375, .571]), UTS_SYNTH .337 ([.249, .429]), and UTS_STREAM .397 ([.289, .495]). These are pointwise, unadjusted intervals from 5000 paired AI-case bootstrap draws with human references and fitted detectors fixed, not uncertainty across training or generation seeds. Overall AUROC and strict-threshold performance need not rank competitors identically: released HIP outperforms released StealthRL at the strictest displayed thresholds despite having higher overall AUROC.

Recorded truncation sensitivity. Limited HIP has 14 input-truncation and 13 output-cap flags; limited MASH has one and seventeen, respectively. UTS_SYNTH has one output-cap flag; UTS_STREAM has none. These counts can overlap within a document. Restricting OG, SHADE, and the four limited arms to their shared subset without recorded truncation/cap flags leaves 25 sources. AUROC is .911 for OG, .450 for SHADE, .917 for HIP, .956 for MASH, .783 for UTS_SYNTH, and .906 for UTS_STREAM. This sensitivity analysis retains the ordering relative to SHADE but changes the cohort: no abstracts or reviews remain. It is not a replacement for the all-attempt comparison or an audit of unrecorded intermediate truncation.

Own-attack adaptation protocol. We select the two competitors with full-cohort frozen AUROC below .8: limited UTS_SYNTH and released StealthRL. A shared random source split supplies twenty adaptation examples and thirty-four held-out examples for each arm. Each adapted classifier is cloned from the frozen classifier and fitted on the original RAID training set plus twenty AI-positive examples from

Attack	Frozen	Adapted	Δ (pp)
UTS_SYNTH, limited	.833	.842	+.84
StealthRL 4B, released	.796	.910	+11.41

Table 33: C08 own-attack adaptation: AUROC on the same 34 held-out sources per arm after exposure to twenty examples. These frozen scores differ from the 54-source scores because the evaluation subset and human-domain weights change.

its own attack, with unit weights. UTS examples never train the StealthRL-adapted detector or vice versa. The C05 feature features and normalization remain frozen; there is no new symbolic-feature search. Each classifier obtains its own thresholds from the same validation-human pool; test-human weights match the held-out domain mixture. Held-out attack texts do not enter fitting or threshold selection. Thus this measures adaptation of the decision rule within the frozen representation, not unrestricted defender retraining.

Adaptation outcome and interpretation. StealthRL loses substantial evasion effectiveness: AUROC rises from .796 to .910, while median AI score rises from .875 to .991. UTS_SYNTH shows only a small rebound, .833 to .842; its median score changes from .557 to .594. StealthRL's frozen AUROC advantage over UTS therefore reverses after own-attack adaptation. A small rebound alone is not successful neutralization: UTS remains substantially distinguishable both before and after adaptation. These results support separately reporting absolute adapted discrimination and recovery magnitude. They describe one twenty-example exposure split; no repeated-split uncertainty or mechanism identifying newly learned traces is established. We do not substitute the earlier SHADE adaptation result as a same-split numerical control without verifying source-ID agreement.

Limits and artifacts. C08 establishes a frozen-Ghostbuster advantage over these six evaluated configurations, not universal superiority over their method families or larger checkpoints. Competitor fidelity has not received a matched blinded assessment, and inference/training budgets are not equalized. The following C08a–d sections extend the comparison to other detectors; larger adaptation budgets remain separate tests. Results are transcribed at the precision in the saved evaluation displays. The base output directory is `results/causal/C08_Ghostbuster_only_RAID54_v1/`; released-checkpoint outputs occupy `released_checkpoints_v1/`, and the adaptation split, fitted models, scores, operating points and PNG/PDF staircases occupy `adaptive_competitors_20_34_v1/`. No additional generation is required to reproduce the saved-score plots.

K C08a–d: Cross-Detector Comparison and Modular Composition

Scope and provenance. We close the C-series with Binoculars NF4, RAID-trained RoBERTa, a GLTR-feature classifier, and RAID-trained WaveDetect, extending the Ghostbuster comparison above. All frozen comparisons use the same 54 source cases; human references are weighted to their domain mixture. We retain all outputs, including capped or detector-prefix-truncated texts. Four competitor configurations are limited reproductions; HIP and StealthRL additionally have released 4B configurations. These labels describe provenance, not an assumed performance ordering: limited HIP outperforms released HIP on several detectors. They do not establish implementation equivalence or performance of the complete method families. SHADE remains the core endpoint; SHADE+RAFT is the exploratory composition RAFT(SHADE), compared with RAFT(OG). RAFT used frozen RoBERTa feedback, so its RoBERTa evaluation is detector-aware; its Binoculars, GLTR and WaveDetect evaluations test transfer without those detectors guiding the saved rewrites. Released-model training may have involved related detector families and cannot be described as uniformly detector-unseen. GPTZero and Sapling are not expanded to the full competitor matrix because commercial access/credit limits preclude that experiment; earlier partial commercial observations are not pooled into this comparison.

Frozen transfer and ranking reversals. Table 34 precludes a universal-best claim. SHADE has lower frozen AUROC than each evaluated HIP, MASH and UTS configuration

Arm	Ghostbuster	Binoculars NF4	RoBERTa	GLTR features	WaveDetect
OG	.917	.905	.992	.816	.797
SHADE	.459	.722	.733	.506	.549
HIP (limited)	.931	.892	.755	.775	.699
MASH (limited)	.935	.851	.842	.945	.738
UTS SYNTH (limited)	.796	.880	.979	.612	.682
UTS STREAM (limited)	.856	.965	.984	.673	.742
HIP (released 4B)	.853	.918	.894	.856	.671
StealthRL (released 4B)	.737	.382	.964	.309	.775
OG → RAFT	.256	.662	.591	.320	.587
SHADE+RAFT	.148	.535	.259	.097	.253

Table 34: Frozen-detector AUROC on 54 paired sources. Lower denotes weaker discrimination in the fixed AI-positive orientation, not necessarily closer human-distribution matching. Ghostbuster entries for OG→RAFT and SHADE+RAFT are completed by C08e. Detector implementations and precision differ as specified in their respective sections.

across all five detectors. StealthRL is substantially stronger against Binoculars and GLTR, but weaker against Ghostbuster, RoBERTa and WaveDetect. SHADE+RAFT lowers AUROC relative to SHADE by 18.69, 47.39, 40.96 and 29.54 percentage points on Binoculars, RoBERTa, GLTR and WaveDetect, respectively. It also outperforms OG→RAFT on each of those four comparisons. It has the lowest frozen AUROC of the evaluated arms on the latter three detectors, whereas StealthRL retains the lowest Binoculars AUROC. These are observed paired-cohort rankings; large numerical gaps are not substituted for paired significance tests. Lower overall AUROC also need not imply uniformly lower TPR at every operating point.

Exposure protocol and controls. For each detector we select non-OG arms with full-cohort frozen AUROC strictly below .8, expose a separate defender to twenty examples from its own arm, and evaluate the remaining 34 on a common saved source split. Because selection used all 54 cases, these are exploratory selected-arm comparisons, not independent post-selection tests. Original RAID training data remain in every adapted fit. GLTR refits its logistic classifier with the original train-only scaler and frozen four-rank representation. Binoculars keeps the Falcon pair frozen and fits a two-component logistic readout; its RAID-only readout controls for changing the decision rule. RoBERTa and WaveDetect perform full neural continuation from their respective frozen checkpoints and include RAID-only continuation controls. C08d WaveDetect uses three epochs at learning rate 10^{-5} with linear decay, logical batch eight/microbatch one, the original class-weighted loss, and checkpoint selection by original RAID validation loss. These are newly recorded C08 runs, not interchangeable with earlier C07 adaptation results. The same held-out human references are rescored under each neural model; validation humans determine thresholds, and test-human FPR is reported. Intervals, where reported, resample cases conditional on the chosen split, fitted models and human references, excluding training-seed uncertainty.

Binoculars: composition helps, StealthRL remains strongest by AUROC. SHADE+RAFT moves frozen AUROC from SHADE's .722 to .535, while StealthRL reaches .382. On the held-out 34, SHADE+RAFT's adapted AUROC is .517, only .0084 above the frozen ratio (95% interval $[-.012, .028]$). However, adaptation adds .074 over the RAID-only readout (interval $[.058, .095]$). Thus attack exposure recovers discrimination relative to the proper readout control even though the final AUROC stays near chance. StealthRL's corresponding exposure increment is .0030 (interval $[-.0002, .0068]$). SHADE+RAFT has favorable low-FPR behavior in the frozen staircase, but we do not infer uniform operating-point dominance from AUROC.

RoBERTa: a substantive adaptation failure remains. All four selected arms rebound. SHADE rises from .729 to .960, while SHADE+RAFT rises from .275 to .803. The latter remains the lowest adapted AUROC among these arms but is substantially

Detector	Arm	Frozen (34)	RAID-only control	Own-arm adapted
Binoculars NF4	SHADE	.684	.672	.678
	StealthRL released	.390	.372	.375
	OG → RAFT	.713	.706	.715
	SHADE+RAFT	.509	.443	.517
RoBERTa	SHADE	.729	.754	.960
	HIP limited	.758	.768	.889
	OG → RAFT	.598	.654	.892
	SHADE+RAFT	.275	.348	.803
GLTR features	SHADE	.508	—	.510
	HIP limited	.796	—	.796
	UTS SYNTH limited	.614	—	.618
	UTS STREAM limited	.711	—	.714
	StealthRL released	.312	—	.295
	OG → RAFT	.356	—	.408
	SHADE+RAFT	.104	—	.154
WaveDetect	SHADE	.569	.578	.539
	HIP limited	.715	.724	.690
	MASH limited	.740	.745	.728
	UTS SYNTH limited	.656	.712	.690
	UTS STREAM limited	.734	.760	.744
	HIP released	.681	.704	.614
	StealthRL released	.790	.849	.815
	OG → RAFT	.616	.648	.635
	SHADE+RAFT	.282	.281	.353

Table 35: C08 own-arm adaptation on 34 held-out sources after twenty-example exposure. Controls are a RAID-only readout for Binoculars and neural continuation for RoBERTa/WaveDetect. GLTR's frozen classifier already supplies the RAID-only reference. Frozen54 and frozen34 scores must not be directly differenced as adaptation effects. Each adapted row uses its own trained defender.

distinguishable. Relative to RAID-only continuation, exposure increases SHADE AUROC by .206 (95% interval $[.130, .291]$) and SHADE+RAFT by .454 ($[.354, .547]$); comparable increases are .121 for limited HIP and .238 for OG→RAFT. The frozen improvement therefore does not establish durable neutralization of RoBERTa's learned representation. Earlier local-attribution observations motivate a residual syntactic/lexical-trace hypothesis, not a complete causal identification of the recovered signal.

GLTR: stable SHADE, stronger score suppression after composition. SHADE changes from .508 to .510 (change .0016, interval $[-.0026, .0058]$). SHADE+RAFT retains the lowest AUROC, .104 frozen and .154 adapted, despite a .050 rebound ($[.035, .064]$); OG→RAFT rises by .052 ($[.035, .070]$). StealthRL changes from .312 to .295, with an interval spanning zero for the change. SHADE supplies a near-chance ranking result, whereas SHADE+RAFT and StealthRL exhibit reversed rankings under the original score orientation. Neither below-chance AUROC nor a reduced mean score proves distributional equality.

WaveDetect: composition and near-chance SHADE survive this exposure budget. SHADE+RAFT retains the lowest adapted AUROC, .353, followed by SHADE at .539; the lowest competitor is released HIP at .614. SHADE+RAFT rebounds by .071 relative to frozen and .072 relative to RAID-only continuation. SHADE instead decreases by .030 relative to frozen and .039 relative to continuation. These changes are descriptive here: the supplied result displays do not establish their paired uncertainty. SHADE's mean AI score nevertheless rises from .502 to .631, illustrating why absolute score shifts and discrimination against rescored human references must be reported separately. This neural adaptation result supports robustness at the tested budget without implying resistance to arbitrary retraining.

Artifacts and limits. The reproducible workflows are `C08a_Binoculars_Competitors_Adaptive.ipynb` with

`C08a_Append_RAFT_NS11.ipynb`, `C08b_RoBERTa_Competitors_RAFT_NS11.ipynb`, `C08c_GLTR_Competitors_RAFT_NS11.ipynb`, and `C08d_WaveDetect_Competitors_RAFT_NS11.ipynb`. Their result folders preserve split roles, input identities, scores, models, tables and combined staircase exports. Numbers above are transcribed from the completed evaluation displays, not recomputed from unavailable raw score files. The same 54-source development cohort, a single exposure split, unmatched compute budgets, reproduction differences and incomplete blinded quality assessment limit generalization. The intervention sequence and RAFT composition were selected during this research; detector transfer does not make the source cohort an untouched final test set.

L C-Series Synthesis: Feature-Directed Neutralization and Composition

An operational feature taxonomy. We organize interventions into overlapping families. Structural/contextual statistics include surprisal distributions, context utility and recurrence organization. Surface statistics include lexical ratios, punctuation, selected POS/NER statistics and sentence-length measures. Residual learned signals denote discriminative patterns not explicitly targeted or adequately measured here; they are not assumed to form one coherent or orthogonal feature family. RoBERTa can use signals in all three groups. Positional embeddings and classification pooling are architectural components, not identified residual linguistic features. $U(k)$ is a scorer-relative probe of contextual predictive benefit, used as a heuristic for related contextual statistics; correlated measurements and transfer do not prove that the whole family has been matched. F remains informative about source/generation differences but is not a universal human–AI discriminator in our tests; heterogeneity motivates caution rather than a universal negative claim.

Pipeline and evidence. SHADE combines controlled generation targeting human-referenced Unique and surprisal/context behavior with local surface operations. The measurements document selected feature movements; they do not establish that every targeted feature or joint distribution reaches a human band. A final optional module applies a compatible local attack, instantiated by RAFT to form SHADE+RAFT. The observed improvements over both constituent routes on all five detectors support operational complementarity by point estimate in this tested order (Section M). They do not establish commutativity, statistical superadditivity, or independence of the affected features. RoBERTa's adaptation rebound exposes a boundary; the more limited recovery under several probability-based detectors is consistent with the feature-directed interpretation but does not uniquely prove it.

Why joint feature matching would limit retraining. For a fixed observable representation $Z = \phi(X)$, write P_H^ϕ and P_A^ϕ for human and intervened-AI feature distributions. For any measurable decision region B,

$$\left| P_A^\phi(B) - P_H^\phi(B) \right| \le \mathrm{TV}(P_A^\phi, P_H^\phi). \qquad (33)$$

Hence equality of the joint feature distributions makes TPR equal FPR for every classifier restricted to ϕ, including a retrained classifier. This is a standard distributional argument, related to existing detection limits (Sadasivan et al., 2025), not a new impossibility theorem. Matching individual means, marginal bands or one detector score does not ensure joint equality: tails, dependencies, conditional patterns and newly learned representations may remain discriminative. The experiments test finite-budget adaptation; they do not establish the equality premise or force future defenders to abandon their existing features.

Compatibility and quality. SHADE+RAFT demonstrates that substantial additional evasion can coexist with the SHADE pipeline across the evaluated detectors. Author inspection also identified coherence decay after RAFT. We therefore keep SHADE as the primary endpoint and SHADE+RAFT as a compatibility experiment, not an unqualified quality improvement. A more faithful local module could occupy the same position in future work, but that prospect requires direct content-preservation evaluation. C08e directly tests StealthRL with the same local RAFT procedure: it beats both constituents on GLTR and Binoculars, whereas

SHADE+RAFT does so by point estimate on all five detectors (Section M). StealthRL's superior Binoculars composition remains a counterexample to universal superiority, and these results do not establish exclusive compatibility of our pipeline.

Novelty boundary and contribution. Feature-directed obfuscation and modular style intervention have substantial precedent. Anonymouth identifies stylometric features and suggested changes (McDonald et al., 2012); StyleRemix controls fine-grained style axes with LoRA modules (Fisher et al., 2024). TempParaphraser exploits sampling-related distributional weaknesses (Huang et al., 2025); MASH is explicitly multi-stage (Gu et al., 2026); RAFT studies local attacks and adversarial training (Wang et al., 2024a); StealthRL studies ensemble-guided paraphrasing and transfer (Ranganath and Ramesh, 2026); detector-guided paraphrasing and detector-reward optimization also appear in Adversarial Paraphrasing, the work of Nicks et al., and AuthorMist (Cheng et al., 2025; Nicks et al., 2024; David and Gervais, 2025). Accordingly, we do not claim the first feature-based, modular, multi-stage, transferable, or adaptation-tested humanizer. Our contribution is an operational combination of human-reference feature controls, generation and local editing, feature-attainment audits, own-attack exposure tests, and empirical composition with a local attack on a matched cohort. This is evidence for a modular research workflow: identify a measurable discrepancy, construct an intervention, audit target attainment and collateral changes, then test quality, transfer, composition and adaptation. A targeted literature review did not establish priority for this exact combination; we therefore use "we develop and evaluate" rather than "first" or "pioneered."

M C08e: operational complementarity of local attack compositions

We apply the same project implementation of RAFT (Wang et al., 2024a) to all 54 saved outputs of the released 4B StealthRL checkpoint (Ranganath and Ramesh, 2026). The local stage uses the frozen C07b RoBERTa detector for both attention-mask proxy ranking and candidate selection, as in OG→RAFT and SHADE+RAFT. The pinned Qwen candidate generator, POS constraint, fifteen proposed candidates, ten-percent word-substitution budget, sequential updates and candidate tie rule are retained. This standardizes the local procedure across starting texts; it does not assert exact equivalence to every configuration in the original RAFT paper or matched realized edit counts. Ghostbuster, WaveDetect, Binoculars and GLTR are evaluation-only for this local search; RoBERTa is detector-aware. The released parent's detector-training exposure prevents an unrestricted claim of detector-unseen lineage. No C08e detector adaptation is performed.

Success requires beating both standalone methods. Let $a_d(A)$ be frozen-detector AUROC after applying method A to the common OG cohort, with the original AI-score orientation. For the ordered composition $B \circ A$, define

$$G_d(A, B) = \min\{a_d(A), a_d(B)\} - a_d(B \circ A). \tag{34}$$

We call the composition operationally complementary only when it improves on both standalone routes. Helping the weaker parent alone does not meet this criterion. Positive estimated G_d records the point-estimate criterion; a statistically supported claim additionally requires paired evidence of improvement over each constituent. A minimum practically meaningful gain, if desired, must be specified separately; none was prespecified here. This is an experiment-specific operational definition, not a claim of additive gains, causal feature independence, or mathematical superadditivity.

Broad SHADE complementarity and detector-dependent StealthRL complementarity. SHADE+RAFT satisfies the point-estimate criterion on all five detectors; StealthRL→RAFT satisfies it on GLTR and Binoculars only. RAFT nevertheless lowers StealthRL's AUROC on every detector: decreases are .241, .082, .264, .210 and .226 for Ghostbuster, WaveDetect, RoBERTa, GLTR and Binoculars, respectively, with all displayed paired 95% case-bootstrap intervals excluding zero. Improvement over the parent therefore must not be conflated with complementarity.

Detector	SHADE	StealthRL	OG→RAFT	SHADE+RAFT	StealthRL→RAFT	G_d(SHADE,RAFT)	G_d(StealthRL,RAFT)
Ghostbuster	.459	.737	.256	.148	.495	.108	-.239
WaveDetect	.549	.775	.587	.253	.693	.295	-.106
RoBERTa	.733	.964	.591	.259	.699	.332	-.108
GLTR	.506	.309	.320	.097	.099	.223	.210
Binoculars NF4	.722	.382	.662	.535	.156	.127	.226

Table 36: C08e frozen comparison on 54 matched sources. Lower AUROC denotes greater evasion in the fixed scoring direction. SHADE+RAFT is RAFT(SHADE). Positive G_d means beating both standalone routes by point estimate; differences use displayed rounded inputs. Values below .5 do not establish equality with the human distribution.

StealthRL→RAFT is worse than OG→RAFT on Ghostbuster by .239 (95% interval [.110, .373]), WaveDetect by .106 ([.0038, .207]), and RoBERTa by .108 ([.039, .177]). It is also worse than SHADE+RAFT on these three detectors, with all three intervals excluding zero. On GLTR, the two compositions are nearly tied: StealthRL→RAFT minus SHADE+RAFT is .0025 ([−.052, .065]). On Binoculars, StealthRL→RAFT is better by .379 ([−.482, −.279]). Thus SHADE supplies a consistently beneficial starting point for RAFT by observed AUROC, while StealthRL's benefit is detector-dependent. StealthRL's two successful compositions improve significantly over both constituent routes under the reported intervals. The available C08e contrast tables do not contain every SHADE+RAFT-versus-constituent interval, so the five-of-five SHADE+RAFT statement is a point-estimate result rather than five newly established significance claims.

Interpretation and limits. The pattern is consistent with SHADE leaving useful additional scope for RAFT across a broader set of detectors, whereas StealthRL's composition preserves weaknesses on several detectors where its standalone output was already comparatively detectable. It does not identify the responsible features or prove that the interventions act on disjoint feature sets. Overall AUROC rankings need not hold at every FPR. Coherence loss observed after RAFT remains relevant, and neither composition has a matched blinded quality evaluation here. These frozen results establish neither C08e adaptation resistance nor universal superiority. The definition was formalized after inspecting the results; the shared development cohort, single generation realization and conditional case-bootstrap intervals further limit confirmatory interpretation.

The workflow is `C08e_StealthRL_RAFT_Five_Detectors.ipynb`, with outputs under `C08e_StealthRL_RAFT_RoBERTaGuide_v1`. It saves parent-linked texts, per-edit traces, five detector score tables, paired contrasts and staircase plots. Figures compare the new composition with OG, SHADE, OG→RAFT and SHADE+RAFT; StealthRL alone remains the parent control in numerical tables. Reported values are transcribed from completed displays, without claiming a fresh raw-score recomputation. Human weighting, threshold calibration and conditional bootstrap conventions follow C08a–d.

N C09a: RoBERTa partition sensitivity and recovery normalization

Recovery relative to the opened gap. For detector d and budget b, define

$$\mathrm{RF}_{d,b} = \frac{A^{\mathrm{adapt}}_{d,b}(\mathrm{SHADE}) - A^{\mathrm{frozen}}_{d,b}(\mathrm{SHADE})}{A^{\mathrm{frozen}}_{d,b}(\mathrm{OG}) - A^{\mathrm{frozen}}_{d,b}(\mathrm{SHADE})}. \tag{35}$$

All terms use the same held-out source identities from the 74-source complement, rather than mixing the full cohort with budget-specific subsets. A value of one restores the original-AI AUROC; zero leaves the attacked AUROC unchanged. The normalization compares recovery against the detector-specific gap the attack opened. At twenty examples, recovered fractions are 8.6% for Ghostbuster, 0.4% for GLTR, −39.6% for WaveDetect, and 68.0% for RoBERTa. At thirty/forty examples they are 17.7/28.9%, 0.3/0.6%, −21.9/ − 31.2%, and 1.0/80.8%, respectively.

For Binoculars, the saved frozen-ratio gap on the twenty-example holdout is .90772 − .70456 = .20316, comparable to RoBERTa's .98566 − .73990 = .24576. Readout recovery

is only .00441, or 2.2% of that ratio-defined reference gap (4.2/6.4% at thirty/forty examples). This comparison shows that the small Binoculars update is not simply paired with a negligible benchmark attack gap. The exact readout-relative RF requires original-AI scores under the RAID-only readout, which are not present in the saved exports; Table 3 explicitly distinguishes this reference normalization from the matched fraction. Original-AI scores are taken from the saved frozen evaluation; WaveDetect's separately exported frozen SHADE AUROC differs on the first two holdouts by less than .0001, which does not change the displayed RF precision materially.

WaveDetect exhibits ranking reversal after adaptation. WaveDetect's adapted AUROCs of .414/.456/.399 are below .5 at all three budgets: held-out SHADE is more often ranked below than above the weighted human reference pool. The negative recovery therefore reflects a shift toward reversed ranking, not proof that every exploitable signal has disappeared. Reversing the score would yield AUROC .586/.544/.601 on these same comparisons (an algebraic diagnostic, not a separately selected defender). We interpret this as failure of the specified continuation procedure to recover the original AI-positive ordering under the attacked distribution, consistent with a representation or training-objective mismatch. The data do not isolate that mechanism from optimization effects. This is the concrete ranking-reversal case anticipated in table 2.

The anomalous 30/44 result is partition-sensitive. C09a repeats the thirty-example RoBERTa exposure using five partition seeds while keeping the training seed and adaptation settings fixed. On each partition's own forty-four held-out SHADE texts, adapted-minus-frozen AUROC is $+.00233, +.03251, +.08554, +.06618, +.04380$ (mean $+.04607$, sample SD .03187). The original partition has the smallest recovery. Thus its near-zero rebound is not representative of all tested partitions, although this sweep does not resolve the non-monotonic exposure–response pattern across budgets. Both exposure and held-out composition change, and the five held-out sets overlap; this is a descriptive sensitivity check, not five independent replications or a training-seed uncertainty estimate. Full results and protocol limits appear in Appendix N.

Split seed	Frozen	Adapted	Δ (pp)
20260923	.742044	.744376	0.2333
20260924	.764189	.796702	3.2514
20260925	.729038	.814574	8.5536
20260926	.761845	.828023	6.6178
20260927	.716430	.760225	4.3795

Table 37: C09a RoBERTa sensitivity analysis: thirty exposed and forty-four held-out SHADE sources per partition, with a fixed training seed. Recovery uses the reported underlying differences; rounding the displayed AUROCs can change the last digit.

C09 extends the SHADE pipeline to the remaining 74 sources and evaluates frozen detectors and nested adaptation exposures of 20, 30 and 40 examples. Following unexpectedly small RoBERTa recovery at the 30/44 partition, C09a repeats that exposure size across five partition seeds (20260923–20260927), holding the training seed fixed at 20260923. Each run starts from the same frozen RoBERTa checkpoint and adds thirty SHADE examples, labeled AI, to the original RAID training data; the other forty-four SHADE examples are held out. Training retains three epochs, learning rate 10^{-5}, unweighted cross-entropy and checkpoint selection using the unchanged original RAID validation set. No SHADE texts are regenerated.

For each partition, frozen and adapted AUROC are evaluated on exactly the same forty-four attack texts. Human-reference weights match that partition's held-out domain mixture, and threshold calibration uses validation humans separately from test humans. Thus the frozen reference changes with the partition rather than reusing the original partition's AUROC. Define recovery as adapted minus frozen AUROC, so a positive value denotes improved detector discrimination.

Results. Recovery is positive in all five partitions (Table 37), with mean 4.6071 percentage points, median 4.3795, sample standard deviation 3.1874, and range 0.2333–8.5536. The original partition has the smallest recovery. Its near-zero aggregate change therefore does not generalize across these alternative partitions:

adaptation generally recovers some discrimination, with appreciable variation in magnitude. The staircase curves also show that a small aggregate AUROC change can coexist with improved detection at some low-FPR operating points and losses elsewhere.

Scope of inference. This is a post hoc partition-sensitivity analysis, not a training-seed sweep. Both exposure membership and its held-out complement vary, so the differences cannot be attributed solely to which examples the detector sees during adaptation. The five held-out sets overlap; their sample standard deviation is descriptive and is not a standard error from independent replications. No significance or equivalence claim follows from the five positive point estimates alone. The results qualify a claim of negligible recovery on the original split, but do not certify an error-free implementation or establish complete adaptation resistance. Without a matched RAID-only continuation control in this sweep, recovery also includes effects of additional training beyond exposure to SHADE itself.

Reproducibility. The workflow is `C09a_RoBERTa_30of74_SplitSweep.ipynb`, with results under `C09a_RoBERTa_30of74_split_sweep_v1`. It retains split manifests, matched frozen/adapted scores, selected checkpoints, per-split staircase exports, `split_sweep_summary.csv` and `recovery_stats.csv`. Numerical values here are transcribed from the completed run displays rather than recomputed from unavailable raw scores.

O C09b: Cross-Attack Adaptation Transfer

Exposure-specific cross-adaptation. On the common 20/34 split, RAFT exposure raises SHADE+RAFT AUROC more than SHADE exposure on Ghostbuster, Binoculars and GLTR. This comparison already fixes the evaluated arm and its frozen baseline. To distinguish preferential transfer from the general tendency for SHADE+RAFT to recover more, we additionally contrast the exposure difference on SHADE+RAFT with that on SHADE:

$$I_d = [A_{R,11} - A_{N,11}] - [A_{R,10} - A_{N,10}], \tag{36}$$

where R/N denote RAFT/SHADE exposure and 10/11 index the evaluated configurations NS10 (SHADE) and NS11 (SHADE+RAFT). The equivalent difference of frozen-relative recoveries cancels the frozen terms. I_d is +.07495 for Ghostbuster (95% CI $[.05320, .09821]$) and +.02845 for GLTR ($[.01679, .04060]$); Binoculars and WaveDetect also have positive conditional intervals, while RoBERTa's includes zero (Appendix O). This interaction tests relative exposure specificity; it does not rescale the bounded AUROC headroom or establish causal attribution. Together with the feature audit, it supports preserved surface gains, partial contextual overlap, and detectable transfer of shared signals.

Question and protocol. Does exposure to RAFT outputs transfer preferentially to the composition, and does exposure to SHADE+RAFT transfer back to its constituents? C09b returns to the original 54-source composition cohort, not the disjoint 74-source C09 replication. One shared random split (seed 20260923) assigns twenty source identities to exposure and thirty-four to evaluation. Each detector sees either SHADE, RAFT(OG), or SHADE+RAFT versions of the same twenty sources, labeled AI and added to the original RAID training data. All three arms are evaluated on the same thirty-four held-out identities. Frozen and RAID-only fits provide controls. The five focal exposure→evaluation pairs are SHADE →SHADE+RAFT, RAFT→SHADE+RAFT, SHADE+RAFT→SHADE+RAFT, SHADE+RAFT→SHADE, and SHADE+RAFT→RAFT; the complete matrix is retained in table 38.

Compatible saved checkpoints are reused after checking their protocol and split. Ghostbuster refits its calibrated classifier over frozen features; Binoculars adapts a two-feature readout with frozen Falcon components; GLTR refits the rank-feature classifier. RoBERTa and WaveDetect continue training from their frozen checkpoints, with original RAID validation for selection. RAID-only controls include equivalent continuation where applicable; the saved Ghostbuster and GLTR RAID-only predictions equal their frozen controls. Human pools are evaluated under the relevant detector

and weighted to the held-out genre mixture. Paired intervals use 5,000 case resamples conditional on fitted models and human references, not repeated training seeds or exposure partitions.

Cross-adaptation reveals shared attack-sensitive signals. The exposure-by-evaluation interaction in equation (36) additionally subtracts the SHADE-evaluation exposure effect; its conditional intervals are reported in Appendix Q.

On SHADE+RAFT, replacing SHADE exposure with RAFT exposure changes AUROC by +.05429 for Ghostbuster (95% CI [.03286, .07711]), +.00478 for Binoculars ([.00242, .00754]), and +.02120 for GLTR ([.01394, .02913]). The corresponding differences are −.01695 for RoBERTa ([−.06357, .02713]) and +.00239 for WaveDetect ([−.00648, .01073]). Thus RAFT exposure transfers more strongly to SHADE+RAFT in three tested representations, but there is no uniform five-detector ordering. In the reverse direction, SHADE+RAFT exposure produces a reliably larger frozen-relative recovery on RAFT than on SHADE only for GLTR among these five conditional contrasts (+.10620, CI [.07187, .13924]). Own-SHADE+RAFT exposure gives the strongest SHADE+RAFT point-estimate discrimination for Ghostbuster, Binoculars, GLTR and RoBERTa; WaveDetect differs negligibly across exposure arms.

The positive transfer contrasts provide evidence that RAFT and SHADE+RAFT share signals usable by several adapted representations. The complete matrix also locates detector-specific differences, including strong RoBERTa recovery after SHADE exposure. C09c follows this behavioral evidence with direct measurements of the constituent and composed texts.

P C09c: Feature-Level Evidence for Compatibility

Weighted relevance and full-feature coverage. The intervention objective selects features with residual AI-ward discrepancy and weights their relevance (equation (2)); unweighted coverage of every extracted feature measures a different quantity. As a transparent retrospective diagnostic, fix $w_j \propto \max(0, \bar{c}_j^{OG} - \bar{c}_j^{H})$ using the original frozen-FZ301 mean contribution gaps, normalize the weights to sum to one, and apply them unchanged to all arms. Weighted central-80% coverage is 77.33% (OG), 78.31% (SHADE), 77.81% (RAFT), and 77.76% (SHADE+RAFT), compared with unweighted coverage of 84.67/81.01/84.86/80.60%. The weighted gain is modest; the stronger detector-relevant result is the human-adjusted score-excess reduction of 80.8% for SHADE and 89.2% for SHADE+RAFT. These fixed post hoc weights are an audit of relevance, not a reconstruction of historical stage weights or a measurement of the entire distance loss.

Utility differences and absolute-surprisal exits. On 39 jointly eligible sources, $U(k) = L_8 - L_k$ and the extra context benefit $E = L_{32} - L_{128}$ are loss differences; L_k measures absolute surprisal. SHADE →SHADE+RAFT increases U32/U128/E band attainment from 64.1/71.8/66.7% to 71.8/76.9/79.5%, leaving 28/30/31 of 39 cases in-band. At the same time, L32/L128 attainment falls from 87.2/74.4% to 51.3/48.7%. Thus the improvement in utility differences and the absolute-surprisal exits are simultaneous, distinct outcomes. Relative to OG's mean outside-band distance under the same case-specific bands, SHADE+RAFT's ratios are 1.924 for U32, .998 for U128 and .656 for E, but 3.901 for L32 and 1.696 for L128. The denominator includes all 39 cases, assigning zero distance to in-band cases. These results support retained context-utility attainment alongside an absolute-surprisal composition cost; they do not show that SHADE+RAFT is mostly outside U bands or that every exit is smaller than the original-AI deviation.

The audit identifies a concrete preservation pattern: SHADE removes much of the dominant FZ301 AI-score excess that standalone RAFT leaves intact, and SHADE+RAFT retains and extends that reduction. Context measurements then identify partial overlap and distinguish preserved utility differences from absolute-surprisal costs.

Surface-feature audit. We extract the project's C04 FZ301 representation for OG, SHADE, RAFT(OG), and SHADE+RAFT on all 54 sources. This implementation includes inherited FZ118, POS, named-entity and entity-grid features, coreference proxies, lexi-

Detector	Exposure	SHADE	RAFT	SHADE+RAFT
Ghostbuster	Frozen	0.4961	0.2817	0.1562
Ghostbuster	RAID only	0.4961	0.2817	0.1562
Ghostbuster	SHADE	0.5741	0.3345	0.2460
Ghostbuster	RAFT	0.5535	0.3999	0.3003
Ghostbuster	SHADE+RAFT	0.6136	0.4412	0.4077
Binoculars NF4	Frozen	0.6843	0.7129	0.5087
Binoculars NF4	RAID only	0.6721	0.7056	0.4433
Binoculars NF4	SHADE	0.6782	0.7134	0.4587
Binoculars NF4	RAFT	0.6791	0.7151	0.4635
Binoculars NF4	SHADE+RAFT	0.6946	0.7359	0.5171
GLTR	Frozen	0.5083	0.3557	0.1039
GLTR	RAID only	0.5083	0.3557	0.1039
GLTR	SHADE	0.5099	0.3692	0.1085
GLTR	RAFT	0.5027	0.4082	0.1297
GLTR	SHADE+RAFT	0.4984	0.4520	0.1536
RoBERTa	Frozen	0.7291	0.5977	0.2747
RoBERTa	RAID only	0.7545	0.6545	0.3484
RoBERTa	SHADE	0.9602	0.8220	0.7406
RoBERTa	RAFT	0.9206	0.8922	0.7237
RoBERTa	SHADE+RAFT	0.8911	0.8329	0.8025
WaveDetect	Frozen	0.5692	0.6159	0.2824
WaveDetect	RAID only	0.5782	0.6477	0.2810
WaveDetect	SHADE	0.5392	0.6357	0.3514
WaveDetect	RAFT	0.5302	0.6352	0.3538
WaveDetect	SHADE+RAFT	0.5338	0.6320	0.3533

Table 38: C09b AUROC on the same 34 held-out sources. Exposure denotes the twenty-example attack arm added to RAID training. RAFT means RAFT(OG). Readout adaptation for Binoculars is not Falcon fine-tuning.

cal/semantic summaries, Yule's statistic and topic redundancy; it is not asserted to be an exact reconstruction of every original FZ feature. Diagnostic central 80% intervals use up to forty same-genre RAID training-human texts nearest to each OG document in log word length, fixed across its four versions; at least ten references are required. Genre-only and central 90% sensitivity exports are retained. These post hoc intervals differ from the heterogeneous running bands and stage-specific targets used during intervention development.

Across 16,254 case–feature pairs, in-band fractions are 84.675% (OG), 81.014% (SHADE), 84.859% (RAFT), and 80.602% (SHADE+RAFT). OG→SHADE gains 1,543 pairs but loses 2,138; SHADE →SHADE+RAFT gains 394 and loses 461, preserving 12,707 of 13,168 previously in-band pairs (96.5%). Hence unweighted attainment over all 301 features does not improve overall under SHADE. There are no strict standalone RAFT crossings from the training-AI side to the opposite side of the entire human interval in this surface audit. Neither fact rules out smaller systematic shifts or joint-distribution changes.

Detector-relevant surface contributions. We additionally decompose the frozen C04 FZ301 forest's AI probability into its mean root probability and feature-assigned parent-to-child probability changes along each decision path, following the path-contribution approach described by Saabas (2014). Contributions sum exactly to the model prediction; we subtract fixed same-genre historical test-human mean contributions. These model-specific attributions are not causal effects or SHAP values and can change when another feature changes the path.

Mean AI scores are .59942 (OG), .44207 (SHADE), .57833 (RAFT), and .42584 (SHADE+RAFT); corresponding human-adjusted excesses are .19470, .03735, .17361, and .02112. SHADE suppresses dominant original contributions involving POS-X, colons, NUM, CARDINAL and ORG, while RAFT alone leaves much of this signal intact and SHADE+RAFT largely preserves its suppression. Long-word and some punctuation/readability signals remain. This gives direct support for surface-feature compatibility: the composition retains reductions in dominant signals that standalone RAFT largely leaves

intact. The full-feature and attribution results distinguish broad marginal coverage from preservation of detector-relevant gains.

Context utility and absolute surprisal. Using frozen Qwen3-0.6B-Base (Yang et al., 2025; Qwen Team, 2025a), we score eight fixed zero-indexed targets evenly spaced from 128 through 255, at contexts 8, 32 and 128. Let L_k be their mean negative log probability, $U(k) = L_8 - L_k$, and $E = L_{32} - L_{128}$. All four versions must contain at least 256 scorer tokens: 39 of 54 sources qualify, with the other fifteen excluded by this shared-support rule. This diagnostic uses eight targets and a 256-token prefix, not a whole-text evaluation or an interchangeable estimate of all earlier context protocols.

Both SHADE and RAFT increase absolute surprisal relative to OG at each context, and RAFT further increases it after SHADE; all nine corresponding paired-change intervals exclude zero. U128 changes are −.18350 (SHADE minus OG), −.25260 (RAFT minus OG), and −.08533 (SHADE+RAFT minus SHADE), each with an interval including zero. Per-case SHADE and RAFT U128 changes correlate at .4101 and share a direction in 25/39 cases; shared OG subtraction can contribute to that correlation. These measurements suggest partial overlap in context-utility effects; composition-interaction intervals also include zero.

Human-reference attainment. For this scorer audit, each source uses up to forty same-genre training-human documents with at least 256 tokens, nearest in log scorer-token length to OG, fixed across arms (minimum ten). These references differ from the word-length-matched FZ audit. Table 39 reports the resulting central 80% band fractions and mean distance outside the interval, in nats. Reference-mean plot intervals resample cases within genre and condition on the selected human documents; they do not quantify reference-pool uncertainty. No held-out-human coverage calibration is claimed.

SHADE improves L32 and L128 attainment from 66.7% to 87.2% and from 41.0% to 74.4%, respectively. After RAFT, SHADE+RAFT falls to 51.3% and 48.7%; mean outside-band distances rise from .1083 to .5433 and from .1493 to .5047 nats. Specifically, among features brought in-band by SHADE from outside in OG, RAFT subsequently moves seven cases out for L8, six for L32 and six for L128 (these counts can overlap across features). Aggregate means show upward surprisal displacement, while band-exit counts alone do not establish every exit's direction.

SHADE →SHADE+RAFT improves attainment and reduces outside-band distance for U32, U128 and E. Since $U(k)$ is a difference of losses, this can coexist with increased absolute surprisal. The result makes compatibility more specific: SHADE+RAFT retains dominant surface-score gains and improves utility-band attainment relative to SHADE, while the audit localizes interference to absolute-surprisal features. SHADE itself has lower U-band coverage than OG on all three measured features despite closer aggregate means. These distinct outcomes motivate preservation constraints that track both utility differences and loss levels when adding a residual module.

Reproducibility and scope. C09b uses `experiments/C09b_Cross_Attack_Adaptation.ipynb` and `results/causal/C09b_Cross_Attack_Adaptation_20_34_v1/`; the complete AUROC matrix and paired contrasts are saved. C09c uses `experiments/C09c_FZ301_Band_Attainment.ipynb` and `results/causal/C09c_FZ301_band_attainment_v1/`, including reference membership, per-feature transitions, `frozen_FZ301_contributions/`, and `U_overlap_v1/human_bands/`. Reported numbers are checked against the committed result CSVs.

Scope of the mechanism evidence. These post hoc diagnostics reuse the development cohort and fixed fitted models, with conditional intervals. Contribution changes and cross-adaptation transfer establish measured associations; they do not isolate a causal RAFT fingerprint or show that the absence of band targeting causes rebound. Marginal attainment is not joint distributional equality. Explaining StealthRL's different composition behavior remains open.

Attribution boundary and stopping point. The current detector panel cannot cleanly separate SHADE+RAFT's residual recoverability into renewed detectability of context-related

	In-band fraction (%)				Mean outside-band distance (nats)			
Feature	OG	SHADE	RAFT	SHADE+RAFT	OG	SHADE	RAFT	SHADE+RAFT
L8	61.5	69.2	82.1	53.8	0.1060	0.1962	0.3545	0.4426
L32	66.7	87.2	71.8	51.3	0.1393	0.1083	0.3387	0.5433
L128	41.0	74.4	64.1	48.7	0.2976	0.1493	0.3933	0.5047
U32	79.5	64.1	79.5	71.8	0.0492	0.1390	0.0822	0.0946
U128	76.9	71.8	82.1	76.9	0.1038	0.1607	0.2858	0.1036
E	79.5	66.7	64.1	79.5	0.1149	0.1149	0.2137	0.0754

Table 39: C09c human-band diagnostic on 39 shared sources. Attainment and outside-band distance can disagree because a few large deviations affect distance. Both quantities are reported to distinguish coverage from the magnitude of residual deviations.

structure and a RAFT-specific signature: the detectors expose different representations, and the measured surface features do not account for RAFT's full effect. Although U makes part of the non-surface overlap observable, it does not identify every signal available to an adapted detector. We therefore stop at demonstrated cross-adaptation transfer and feature-level preservation, overlap and interference; a finer attribution would require controlled feature-restricted defenders or interventions that isolate those candidate signals.

Q Additional uncertainty and weighting diagnostics

Conditional inference. Tables 2 and 4 use 10,000 paired source resamples (seed 20261005), recomputing the better-constituent minimum inside each replicate for composition. Per-source AUROC contributions compare the AI score with the fixed saved human test scores using the original genre weights and half credit for ties. Their means reproduce the saved AUROCs. The human pool and fitted detectors are held fixed; intervals are pointwise. This analysis reruns no generation or model training.

Exposure interaction. The C09b exposure-by-evaluation DiD in equation (36) uses all four paired per-source AUROC contributions in each resample. Unlike the earlier single-exposure contrast of SHADE+RAFT versus SHADE recovery, it subtracts the corresponding differential under the other exposure arm. Its values (95% pointwise intervals) are Ghostbuster .07495 ([.05320, .09821]), Binoculars .00387 ([.00137, .00664]), RoBERTa .02261 ([−.02588, .06873]), GLTR .02845 ([.01679, .04060]), and WaveDetect .01143 ([.00076, .02175]). These are conditional interactions on the AUROC scale, not ceiling-normalized recovery fractions.

Retrospective weighting and distance ratios. The retrospective weights in Appendix P use positive original-AI mean contribution gaps from the frozen FZ301 forest, fixed before comparing transformed-arm coverage. Weighted coverage is $\sum_j w_j n^{-1} \sum_i \mathbf{1}\{\phi_j(x_i) \in B_{ij}\}$. This binary diagnostic is distinct from the weighted, scaled outside-band distance in equation (2), and from the historical feature-selection and target rules. Outside-band ratios divide mean SHADE+RAFT distance by mean OG distance using identical case-specific intervals and all 39 eligible cases. They are ratios of means, not means of potentially undefined individual ratios. For completeness, the corresponding L8 ratio is 4.175. Neither these ratios nor increased coverage establish equality of distributions.

Detector	Original AI	SHADE	Δ AUROC	95% paired CI
Ghostbuster	.91321	.47607	-.43713	$[-.49762, -.37585]$
Binoculars NF4	.90549	.71277	-.19272	$[-.25754, -.12954]$
RoBERTa	.98792	.73273	-.25519	$[-.30365, -.20712]$
GLTR	.82727	.52466	-.30261	$[-.35596, -.25092]$
WaveDetect	.78455	.54981	-.23474	$[-.30277, -.16714]$

Table 40: **Secondary frozen evaluation pooled over 128 matched RAID sources.** The 128 sources are the original 54-source development/evaluation cohort plus the disjoint 74-source complement generated after the SHADE pipeline was frozen. Δ is SHADE minus original-AI AUROC; all five paired case-bootstrap intervals exclude zero. Human references are genre-weighted to the pooled 128-source mixture.